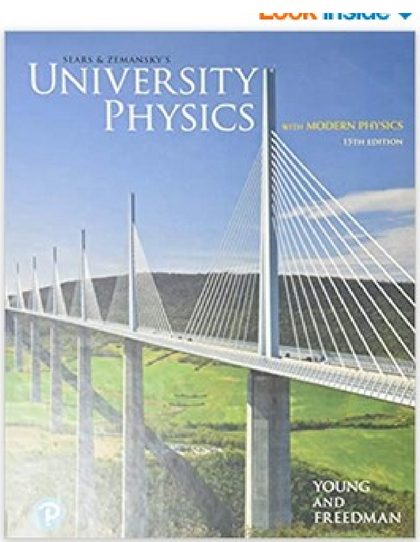
Physics W calculus – Spring 2026

Electrostatics/Electric circuits / Electromagnetism/Optics

See canvas and syllabus

University Physics with
Modern Physics

by Hugh Young, Roger Freedman



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Relative weight will be assigned as follow:

Homework 18 %

tests (test1, test2, final) 60%

pop-up quizzes 12%

take home test (test3) 10%

Attendance 5%

Tentative exams:
Test1: Feb 5th, Test2: March 12th, test3: April 9th

Unit 1.1 Electric Forces and Electric Fields

→ historical notes

→atomic structure. Conductor, insulator, semi conduction

→ charging by induction or contact

→ coulombs law, applications

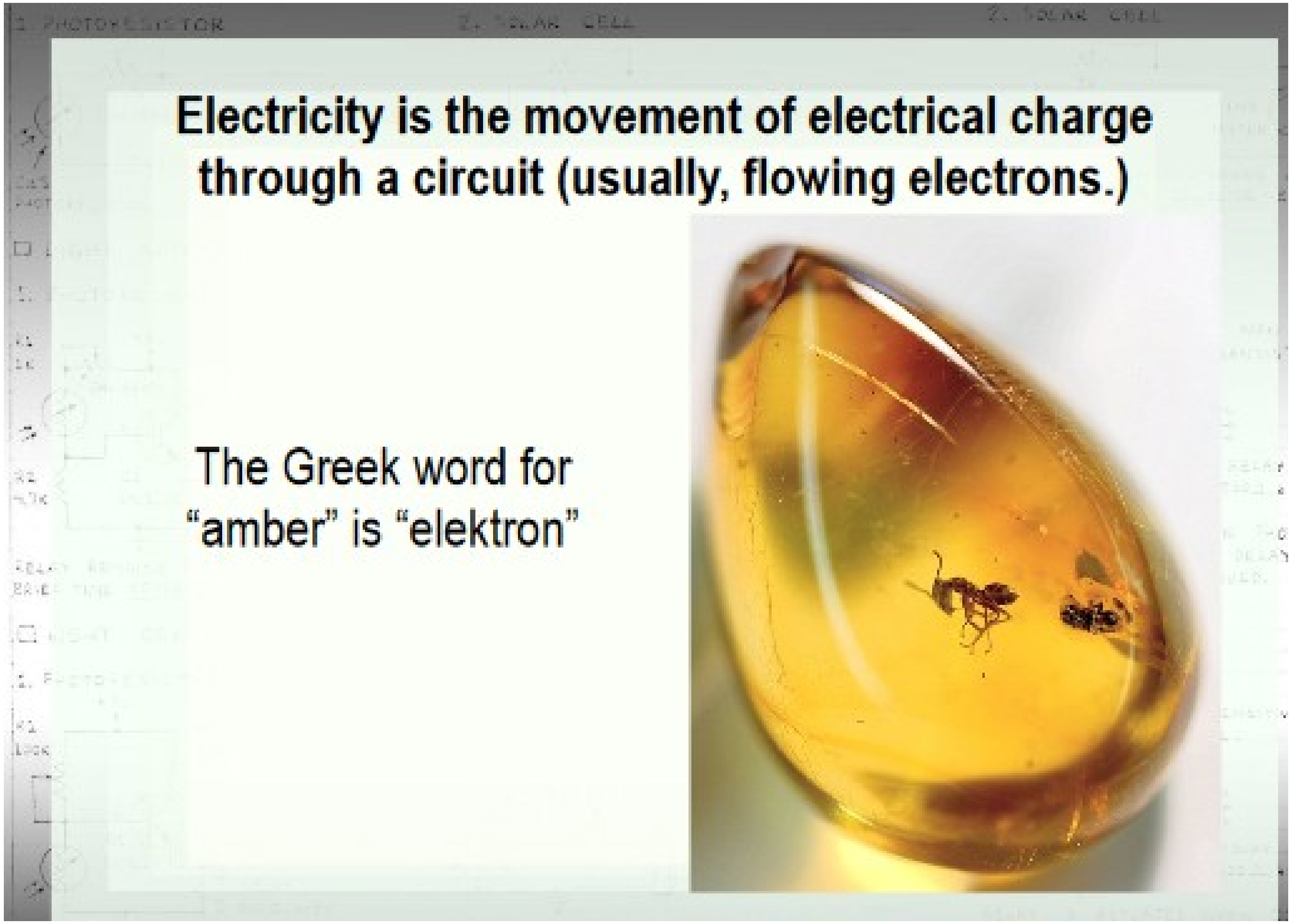
→ vector nature of the electric force

→ electric field, calculus part

→ motion in an electric field

→ electric dipole, electric moment, torque, PE of a dipole

history



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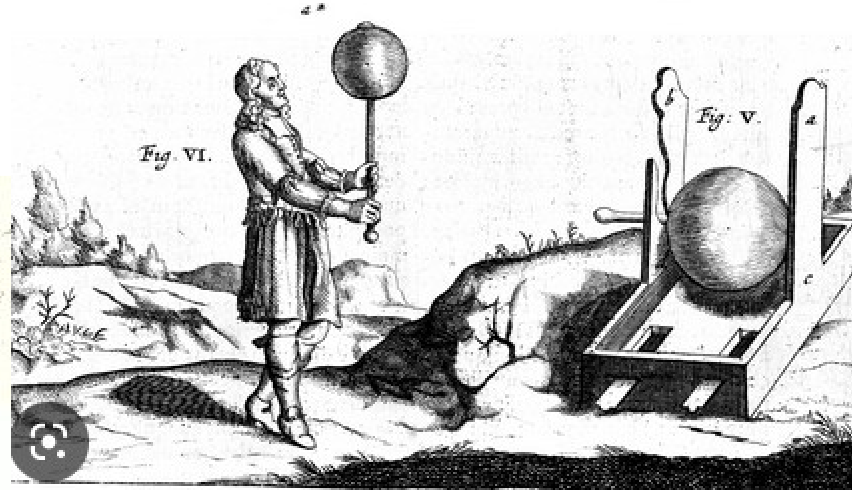
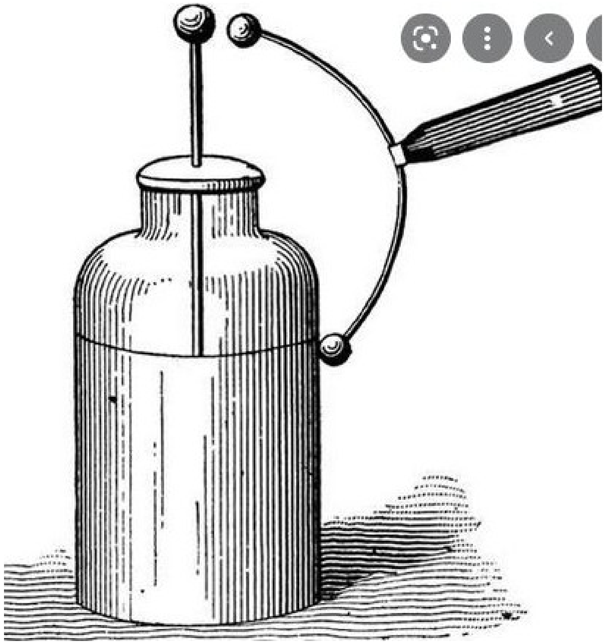
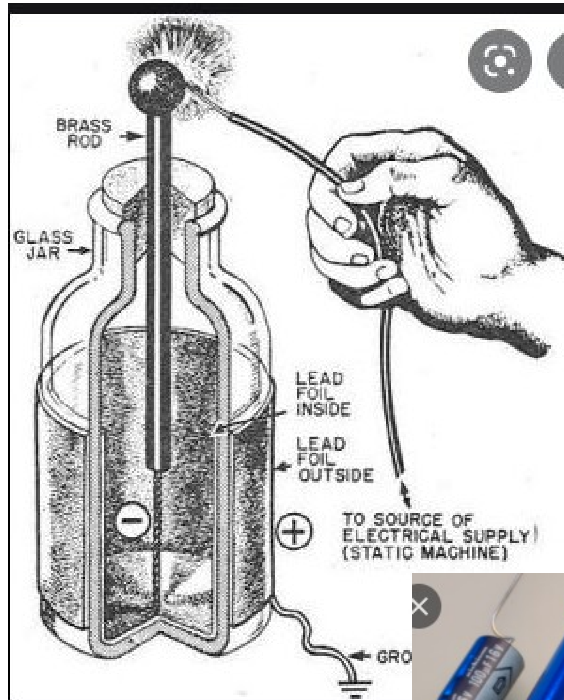
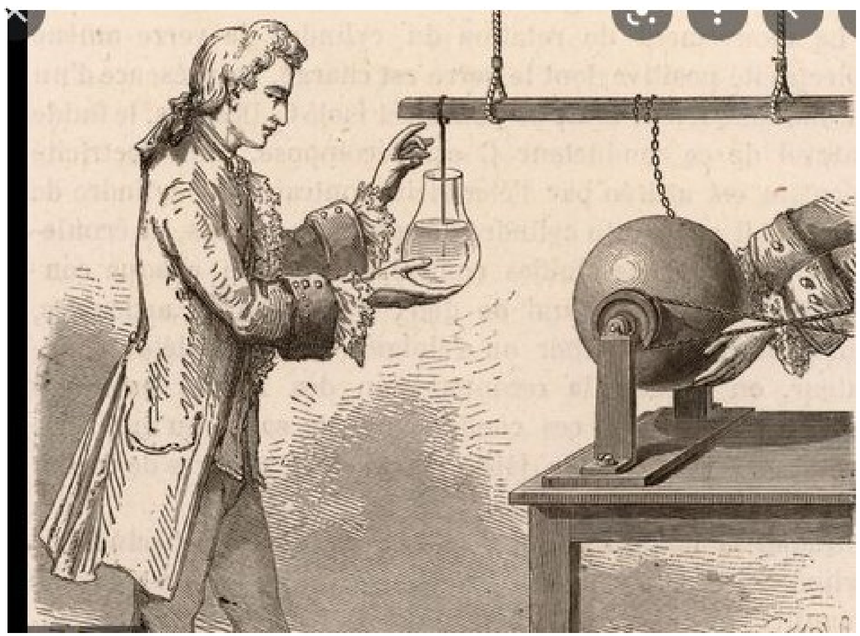
Greek philosophers noticed that **when a piece of amber was rubbed with cloth, it would attract pieces of straw**. They recorded the first references to electrical effects, such as static electricity and lightning, over 2,500 years ago.

In the mid seventeenth century, Otto von Guericke of Germany invented one of the first devices capable of generating electricity for research. Basically it consisted of a ball of sulfur mounted in a sort of wooden cradle that he manually rotated against another object to produce a charge. Mar 13, 2015





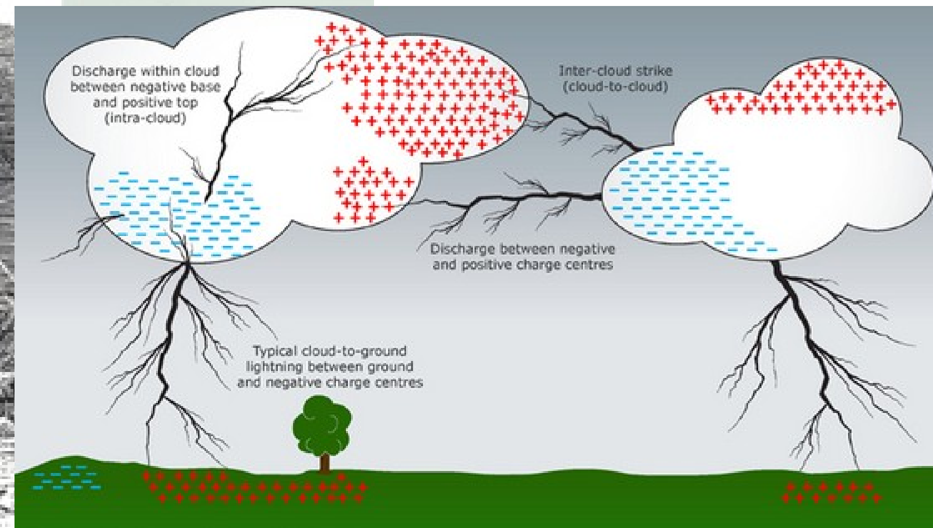




1745

In **1745** a cheap and convenient source of electric sparks was invented by Pieter van Musschenbroek, a physicist and mathematician in Leiden, Netherlands. Later called the Leyden jar, it was the first device that could store large amounts of electric charge.

It was initially believed that the charge was stored in the water in early Leyden jars. In the 1700s American statesman and scientist Benjamin Franklin performed extensive investigations of both water-filled and foil Leyden jars, which led him to conclude that the charge was stored in the glass, not in the water.

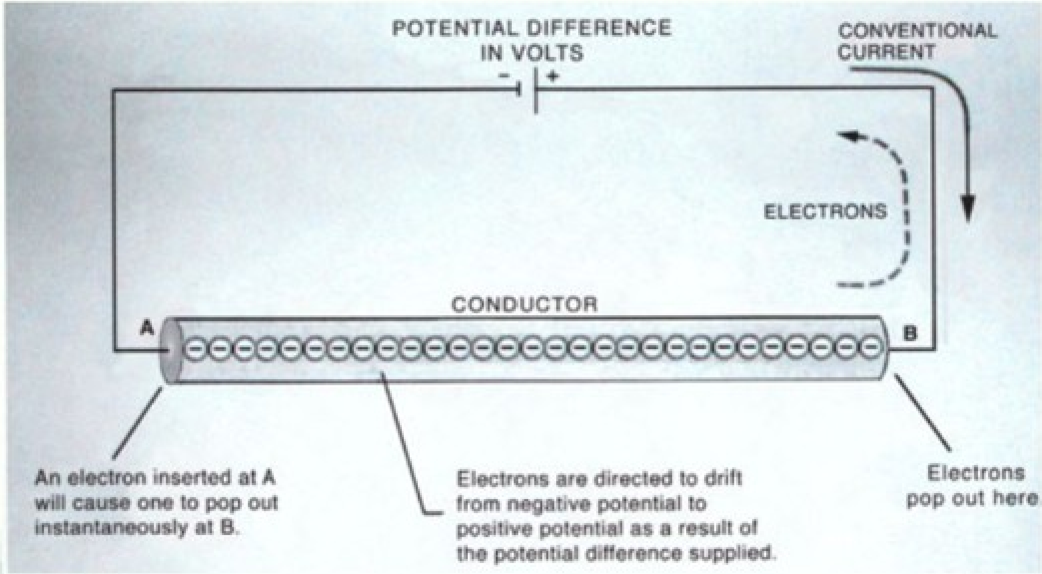


Benjamin Franklin (1705-1790) had ideas about electricity:

1. That Lightning was electricity.
2. The two types of electrical charge (at the time, referred to as two "fluids": "resinous" and "vitreous") were in fact the result of the relative presence or absence of a "single fluid", which he referred to as "positive electricity" and "negative electricity", respectively.

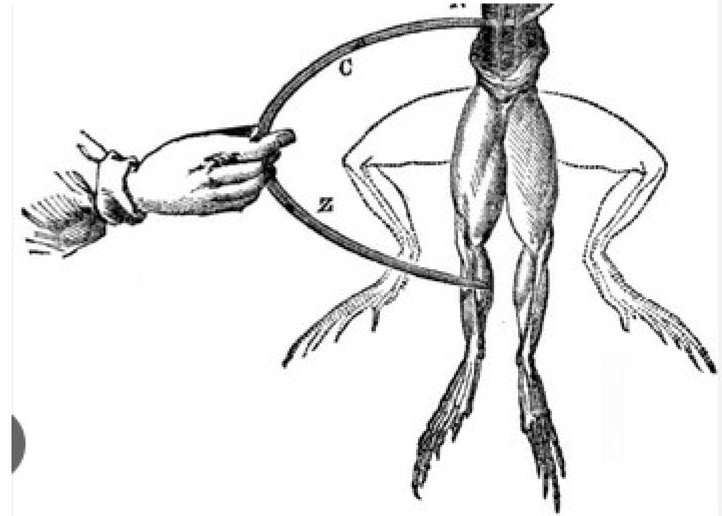
Franklin had a 50/50 chance... Ooops!

Franklin postulated that the two types of observed charge were actually the result of the relative presence or absence of a single “fluid”. He labeled its presence in "vitreous electricity" (electrons) as positive, and its absence in "resinous electricity" as negative. When the electron was discovered, it turned out that Franklin got it backwards. To this day, (and in this class) we refer to “conventional current” flow, described as flowing from positive to negative, even though electrons (negatively charged particles) flow toward a positive pole.



In the 1780s, the Bolognese physician Luigi Galvani (1737-1798) conducted a vast range of experiments on electricity's effect on "prepared" frog specimens - that is, frog legs severed at the base of the spine, with nerves exposed.

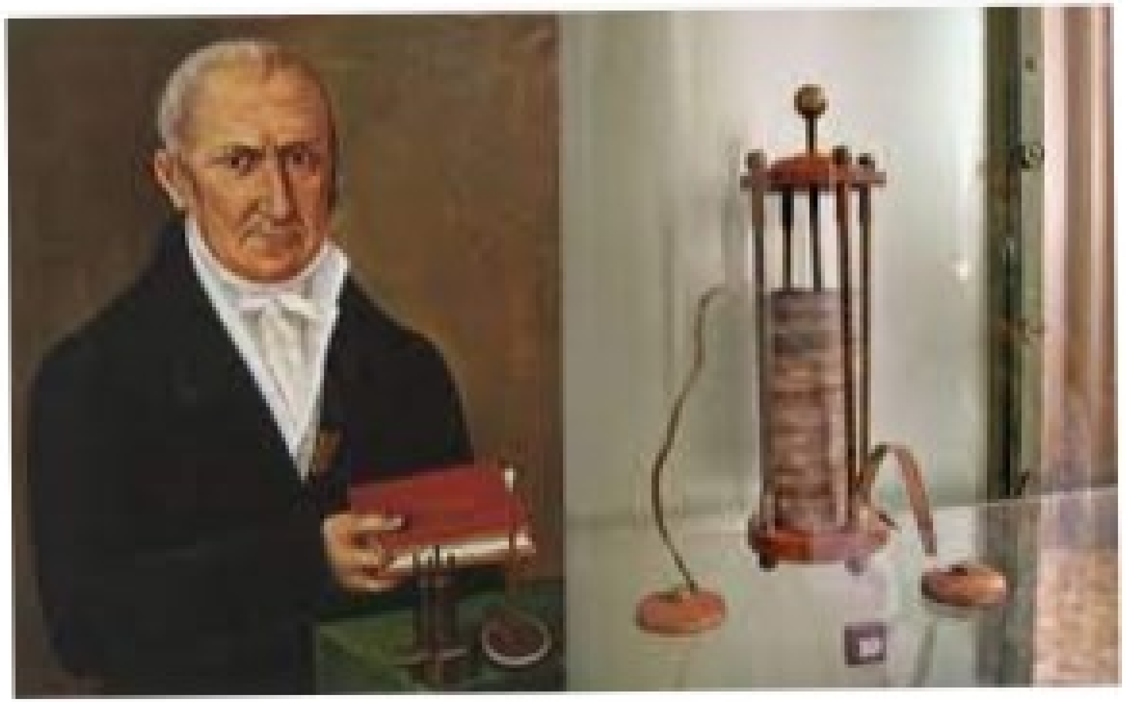
Anatomist Luigi Galvani's interpretaion of animal electricity led to the invention of the battery, a reliable source of electricity



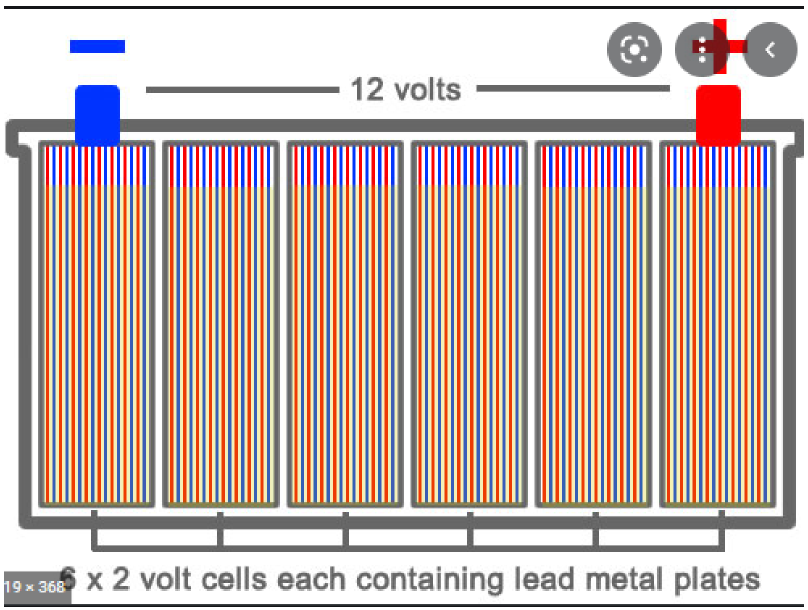
Mary Shelley created the story on a rainy afternoon in 1816 in Geneva

"It's Pronounced Fronkensteen"; A Look Inside the Making of Mel Brooks' Hilarious Satire Young Frankenstein

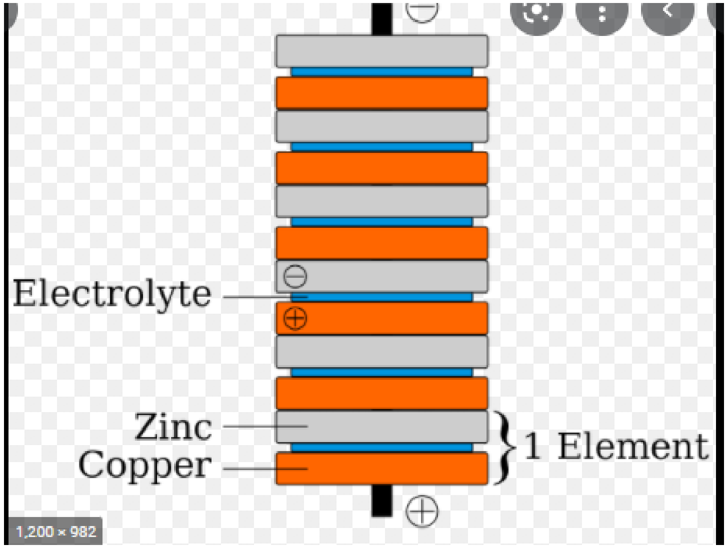
Director Mel Brooks, right, on the set of his 1974 classic "Young Frankenstein" with a prone Peter Boyle, Teri Garr, Gene Wilder and Marty Feldman. (Los Angeles Times)



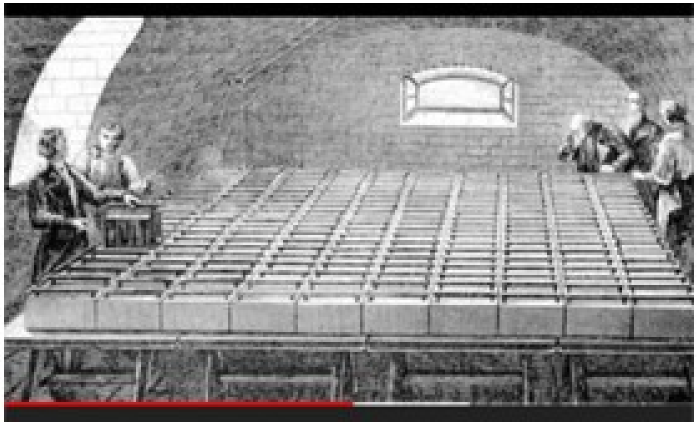
In 1800, at the time Alessandro Volta invented the battery, he was a professor of physics at the University of Pavia, in Italy



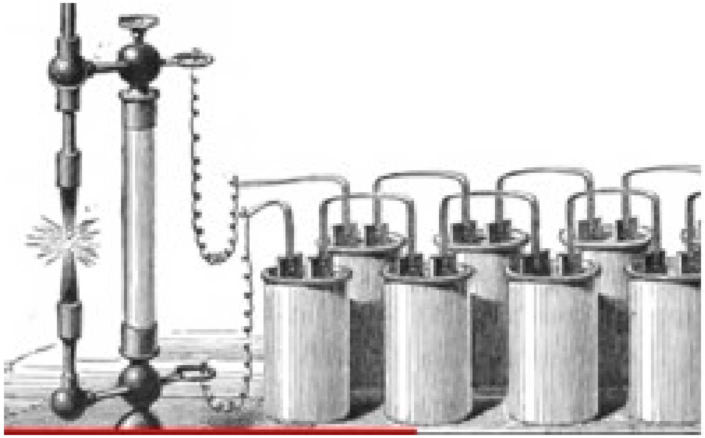
2.1V x 6 = 12.6V

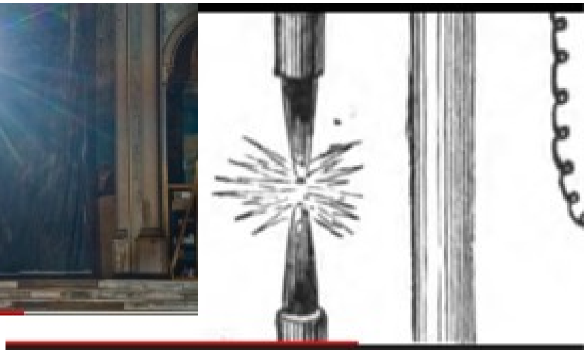


Humphry Davy

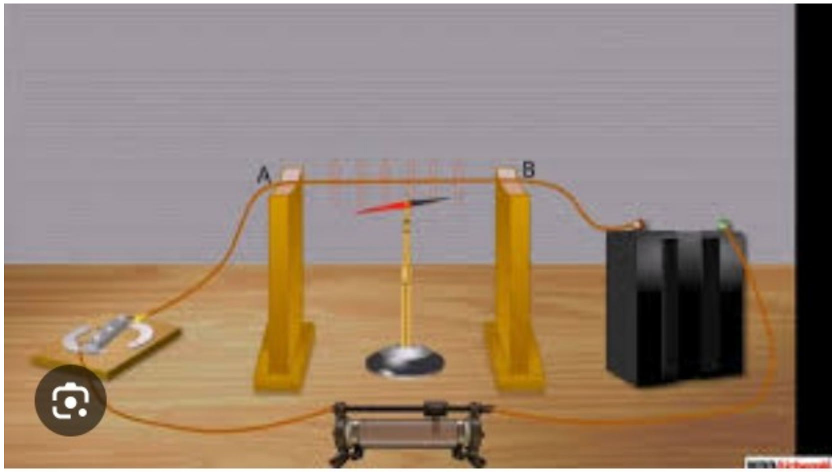


Sir Humphry Davy constructed the first arc lamp (**1807**), using a battery of 2,000 cells to create a 100-millimetre (4-inch) arc between two charcoal sticks. When suitable electric generators became available in the late 1870s, the practical use of arc lamps began.

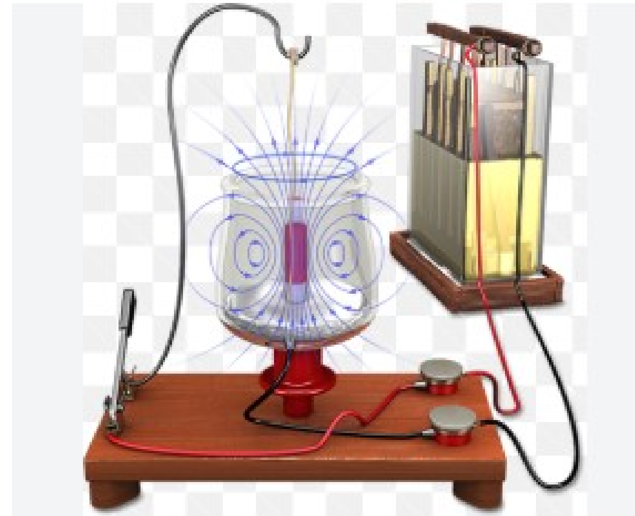




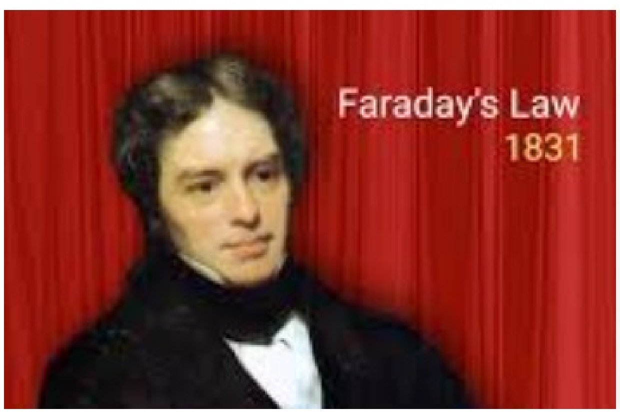
Davy is the best known for isolating, by using electrolysis, many elements for the first time: **potassium (K) and sodium (Na) in 1807 and calcium (Ca), strontium (Sr), barium (Ba), and magnesium (Mg) in 1808**. Actually, Davy was one of the originators of electrochemistry as a subarea of chemistry. May 4, 2021



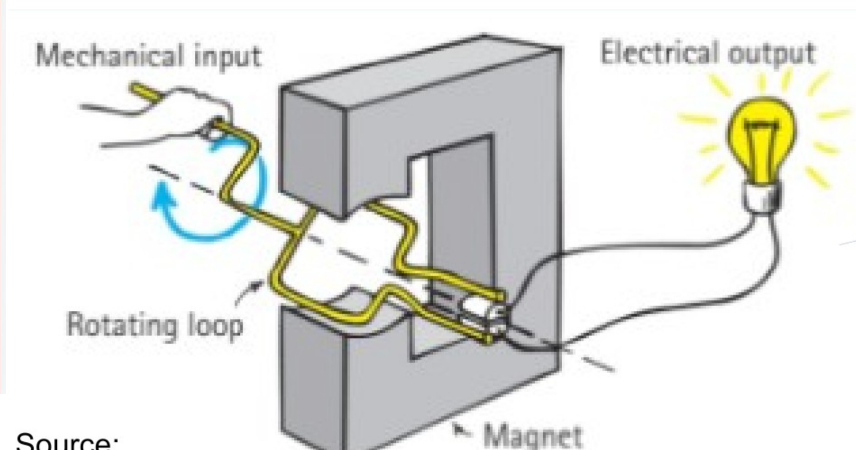
1810 Oersted experiment
Electricity → magentic field



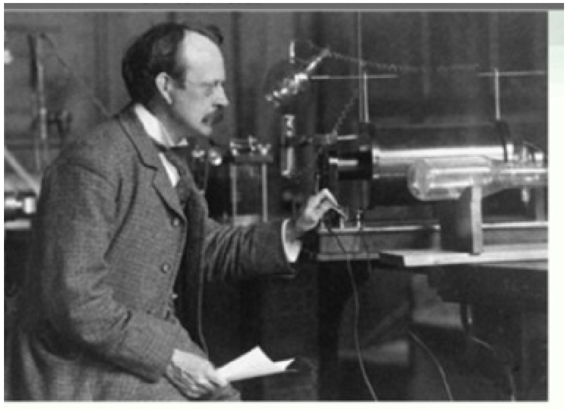
1821 Michael Faraday built the First motor.

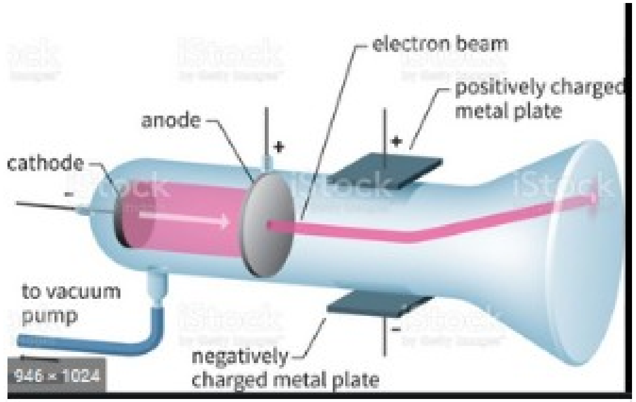


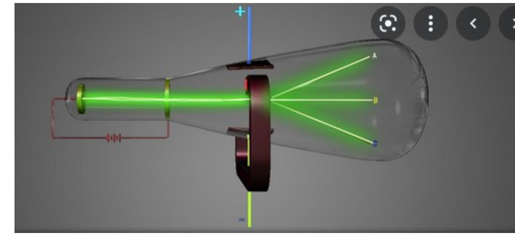
1831 law of induction



Source:
Paul Hewitt. Conceptual physics;



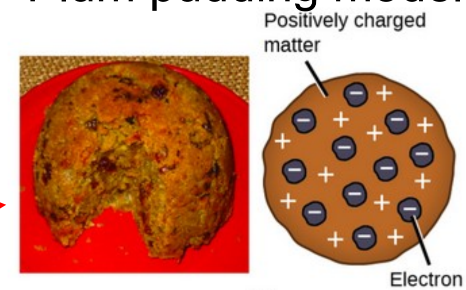




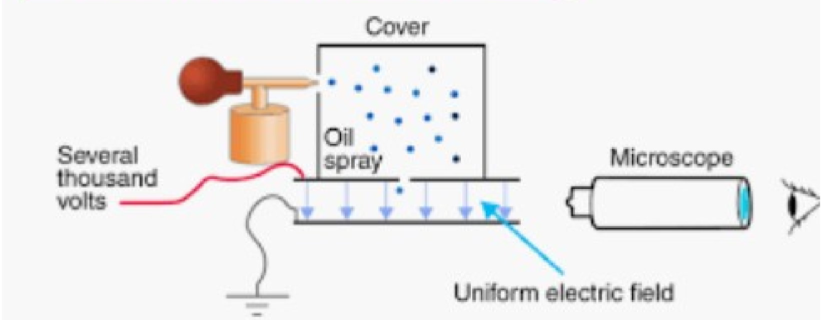
In 1897 Thomson discovered the electron and then went on to propose a model for the structure of the atom. His work also led to the invention of the mass spectrograph. Jan 9, 2018

J. J. Thomson → ratio mass versus charge for electrons.

Plum pudding model

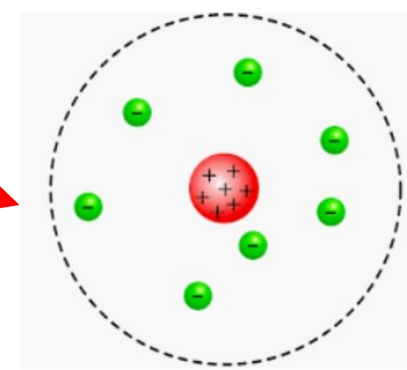


1909: Robert Millikan



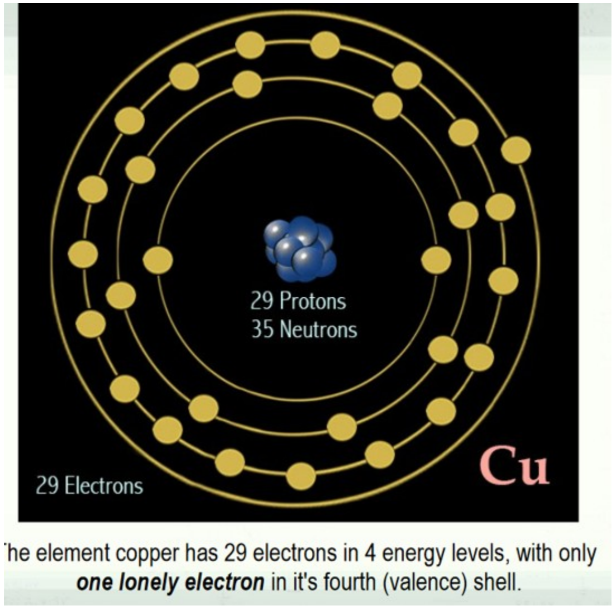
E. Rutherford → protons (structure atoms) – 1911

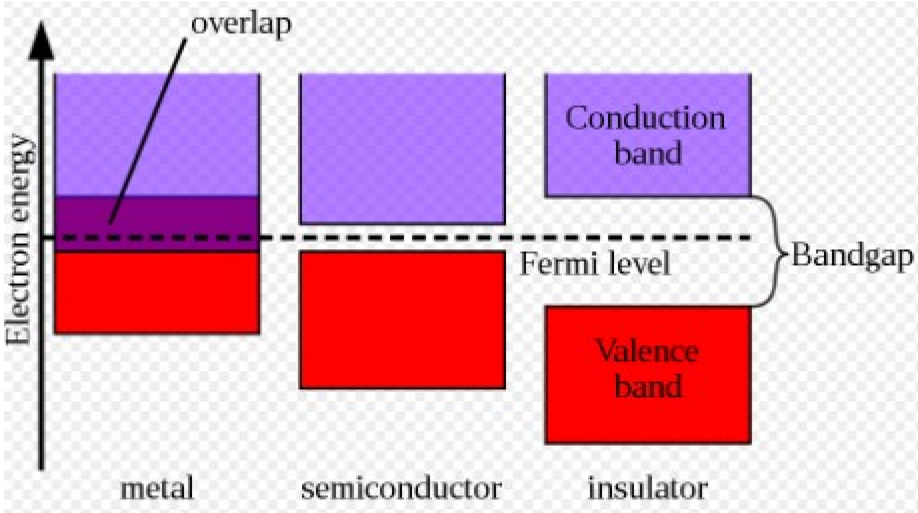
proved Thomson's plum pudding structure incorrect. … Rutherford concluded that the atom consisted of a small, dense, positively charged nucleus in the center of the atom with negatively charged electrons surrounding it.



https://www.youtube.com/watch?v=HVxBdMxgVX0 for Physics majors

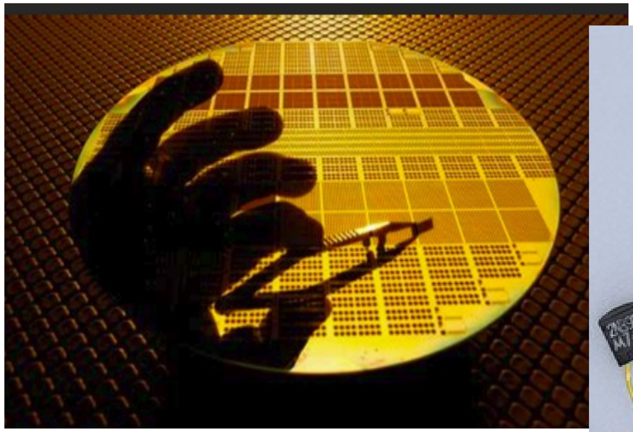
N. Bohr → model of the atom 1913
J. Chadwick → neutrons 1932

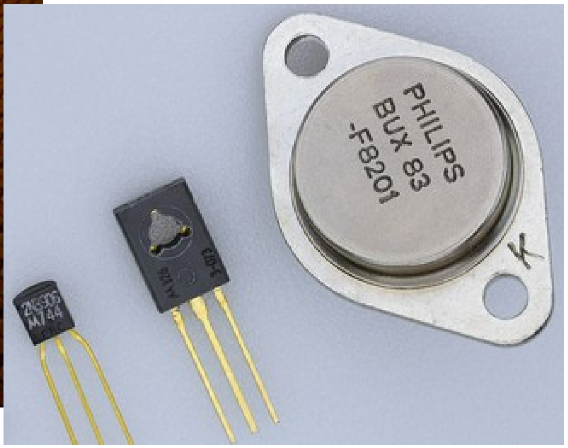


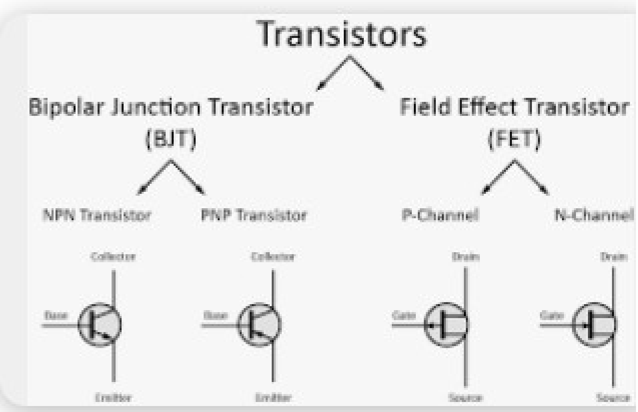


Bardeen and Brattain at Bell Laboratories in the US invented the point-contact transistor in 1947, and Shockley invented the junction transistor in 1948

Semiconductors

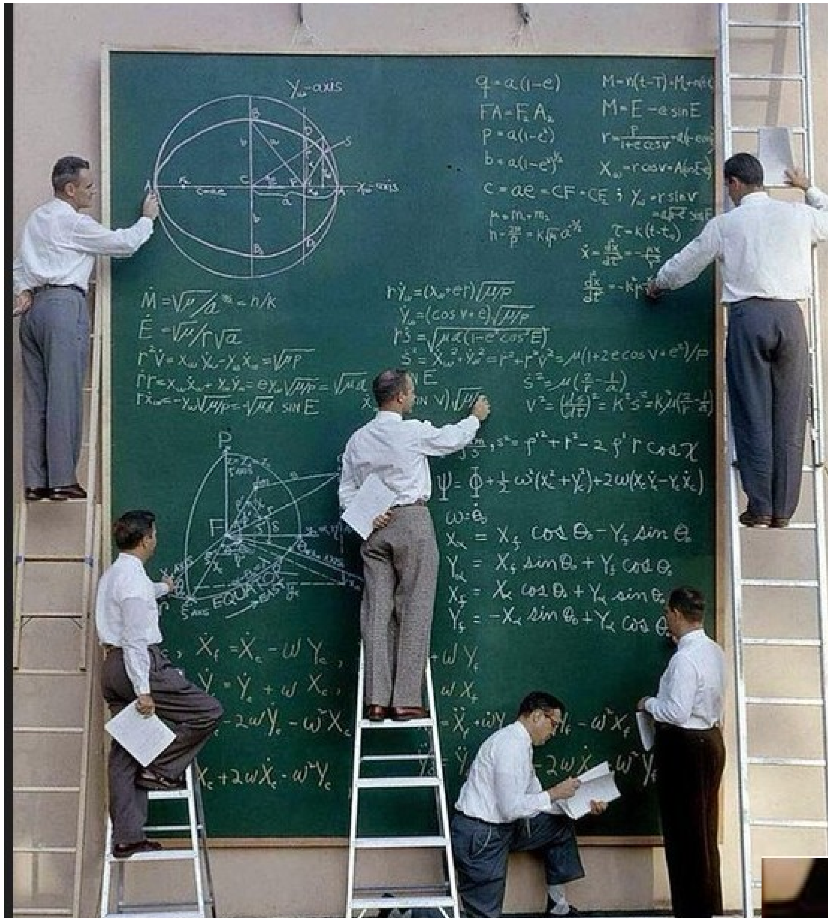


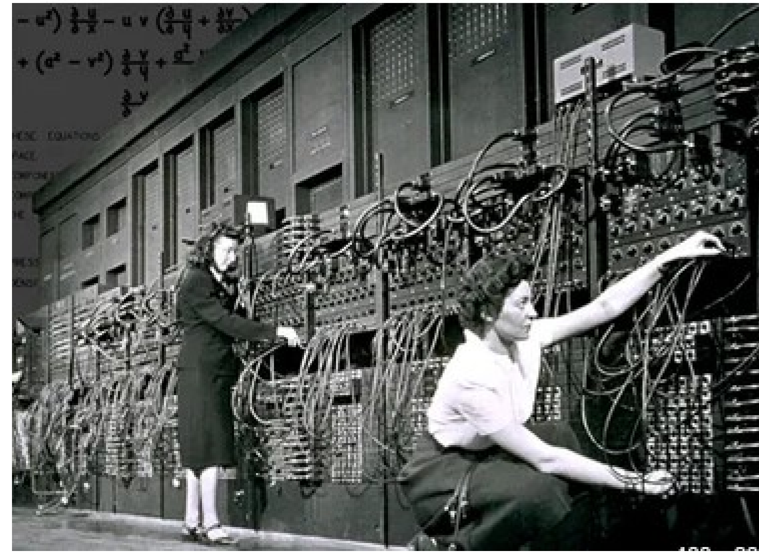


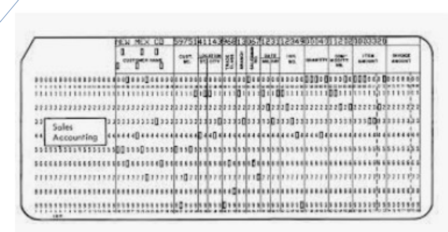


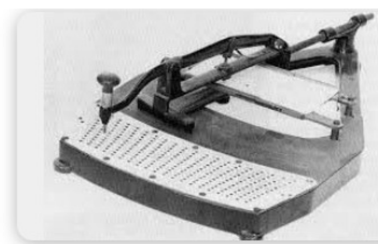
https://www.hitachi-hightech.com/global/en/knowledge/semiconductor/room/about/history.html

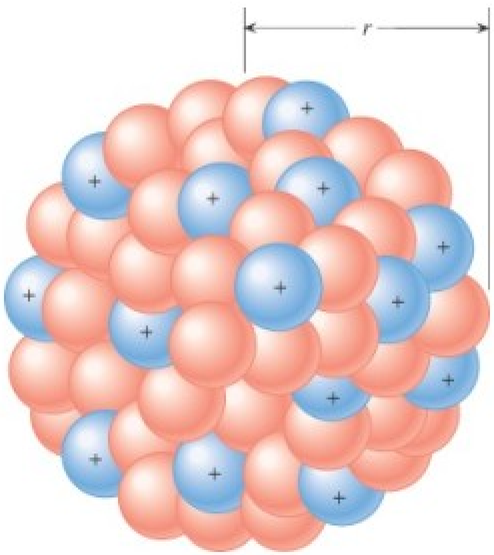
Before LAPTOPS

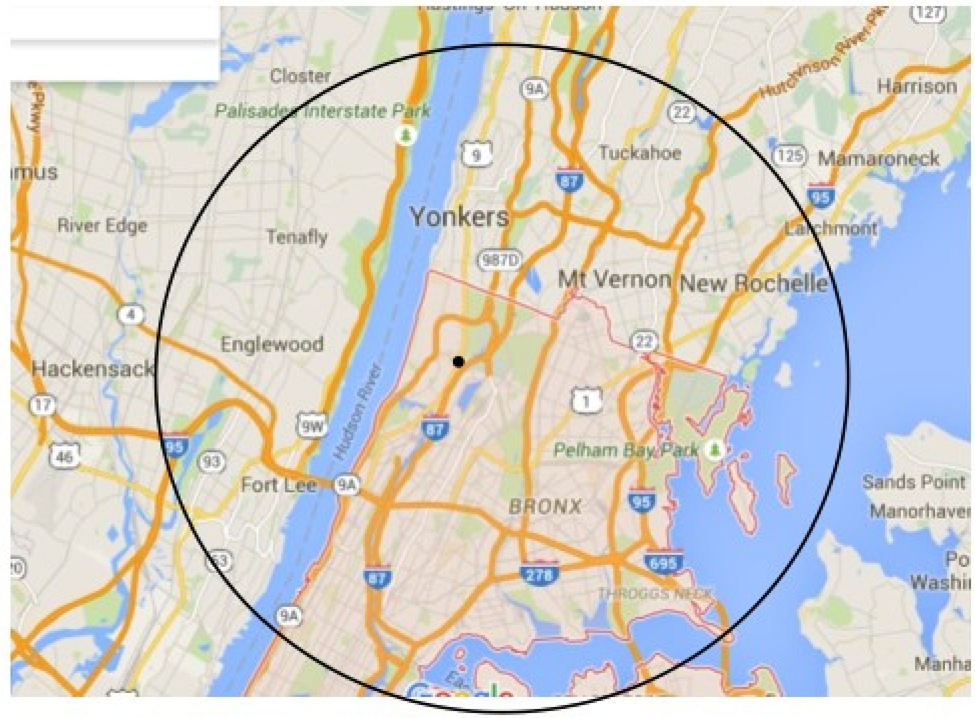












The atom has a size of about 10^{-10} m.

If the size of the nucleus is about 10^{-15}m

If the nucleus was a basketball at Manhattan College
The electrons are 7.5 miles away.

What is the ratio ?

It the nucleus is 1mm (ant) then the size of the atom is about _______ mm = ____m

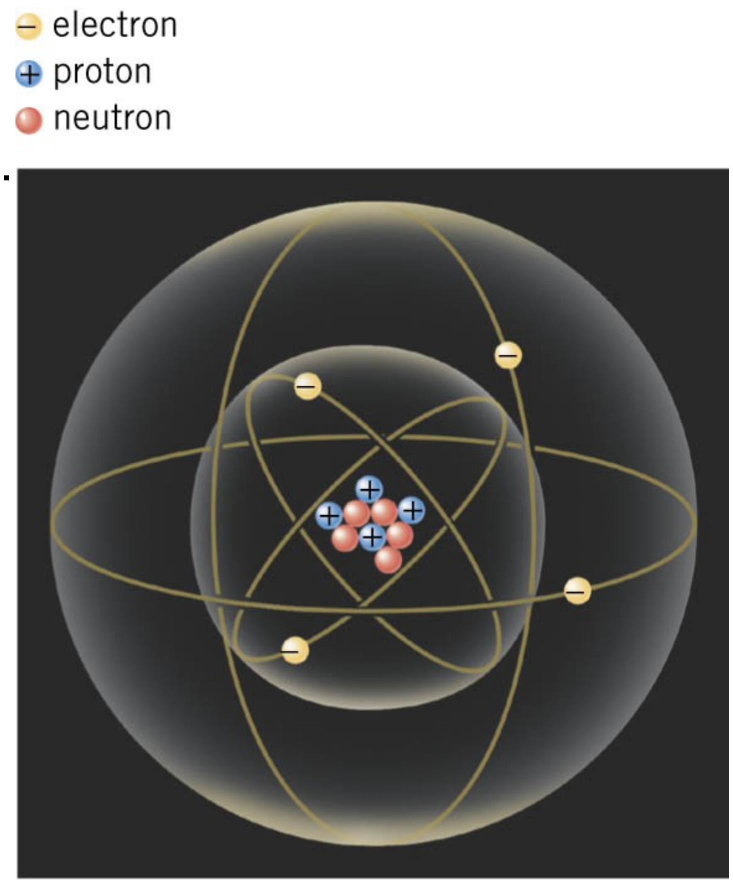
(divide by 1000. 1m = 1 yard about). It's the ratio between a grape seed and

A baseball stadium !!!

A lot of space. (the space is filled by electron waves)

18.1 *The Origin of Electricity*
Bohr model

The electrical nature of matter is inherent in atomic structure. Charge is a fundamental Property of nature like mass. Unit is the coulomb (C).



$$m_p = 1.673\times10^{-27}\,\text{kg}$$

$$m_n = 1.675\times10^{-27}\,\text{kg}$$

$$m_e = 9.11\times10^{-31}\,\text{kg}$$

$$e = 1.60\times10^{-19}\,\text{C}$$

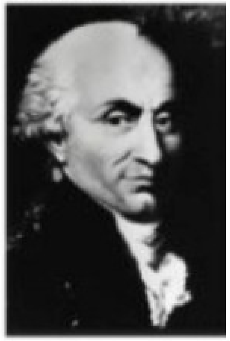
coulombs

The main difference between Bohr model and Rutherford model (nuclear model) is that in Rutherford model, electrons can revolve in any orbit around the nucleus, whereas in Bohr model, electrons can revolve in a definite shelf.

Properties of Electric Charges

- Electric charge is always conserved (not created, only exchanged) in an isolated system
- Objects become charged because negative charge (electrons) is transferred from one object to another
- Charge is quantized (a multiple of a fundamental unit of charge, e): electrons have a charge of – e and protons have a charge of + e
- The SI unit of charge is the Coulomb (C)

$$\boxed{e = 1.6 \times 10^{-19}\ C}$$



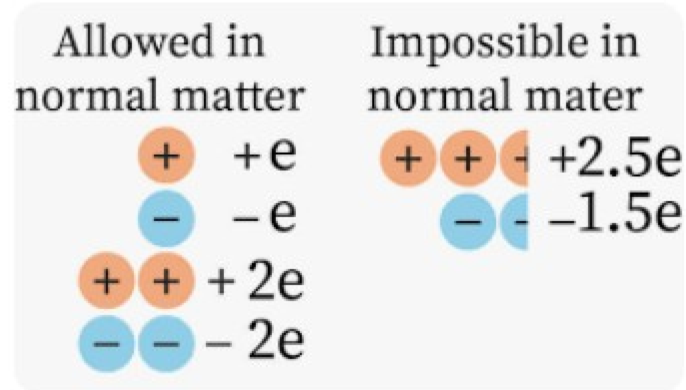
Charles Coulomb
1736 – 1806

18.1 The Origin of Electricity

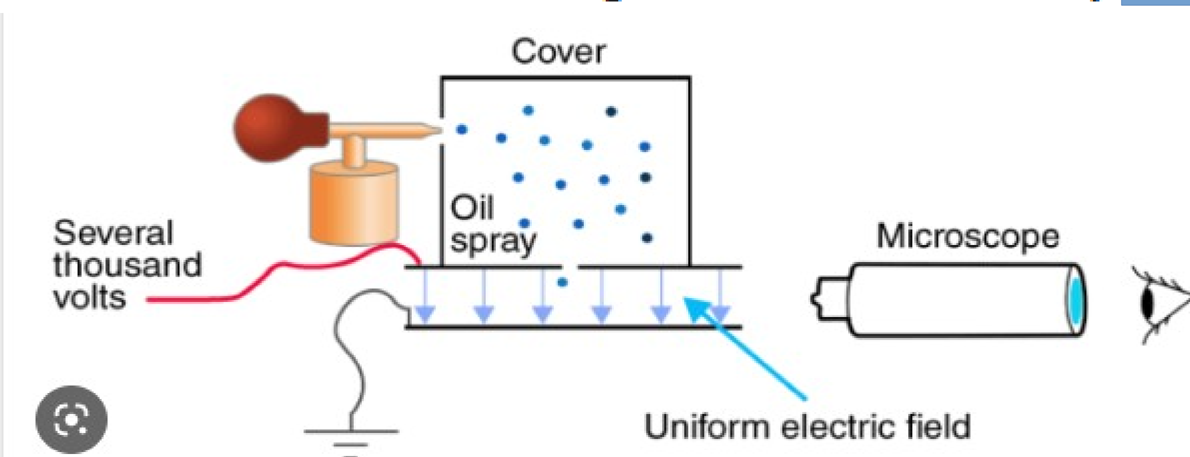
In nature, atoms are normally found with equal numbers of protons and electrons, so they are electrically neutral.

By adding or removing electrons from matter it will acquire a net electric charge with magnitude equal to *e* times the number of electrons added or removed. *N*.

$$q = Ne$$



Charge is quantized; it comes in integer multiples of **individual small units called the elementary charge, e**, about 1.602×10^{-19} C, which is the smallest charge that can exist freely



Millikan experiment
(oil drop experiment)
→ charge of an electron
Mass is also known
Thomson found earlier
The ratio e/m with his
CRT experiment.

Example 1 A Lot of Electrons

How many electrons are there in one coulomb of negative charge?

$$q = Ne$$

Charge is a property of matter. It is quantized
Mass is a property of matter it is not quantized.

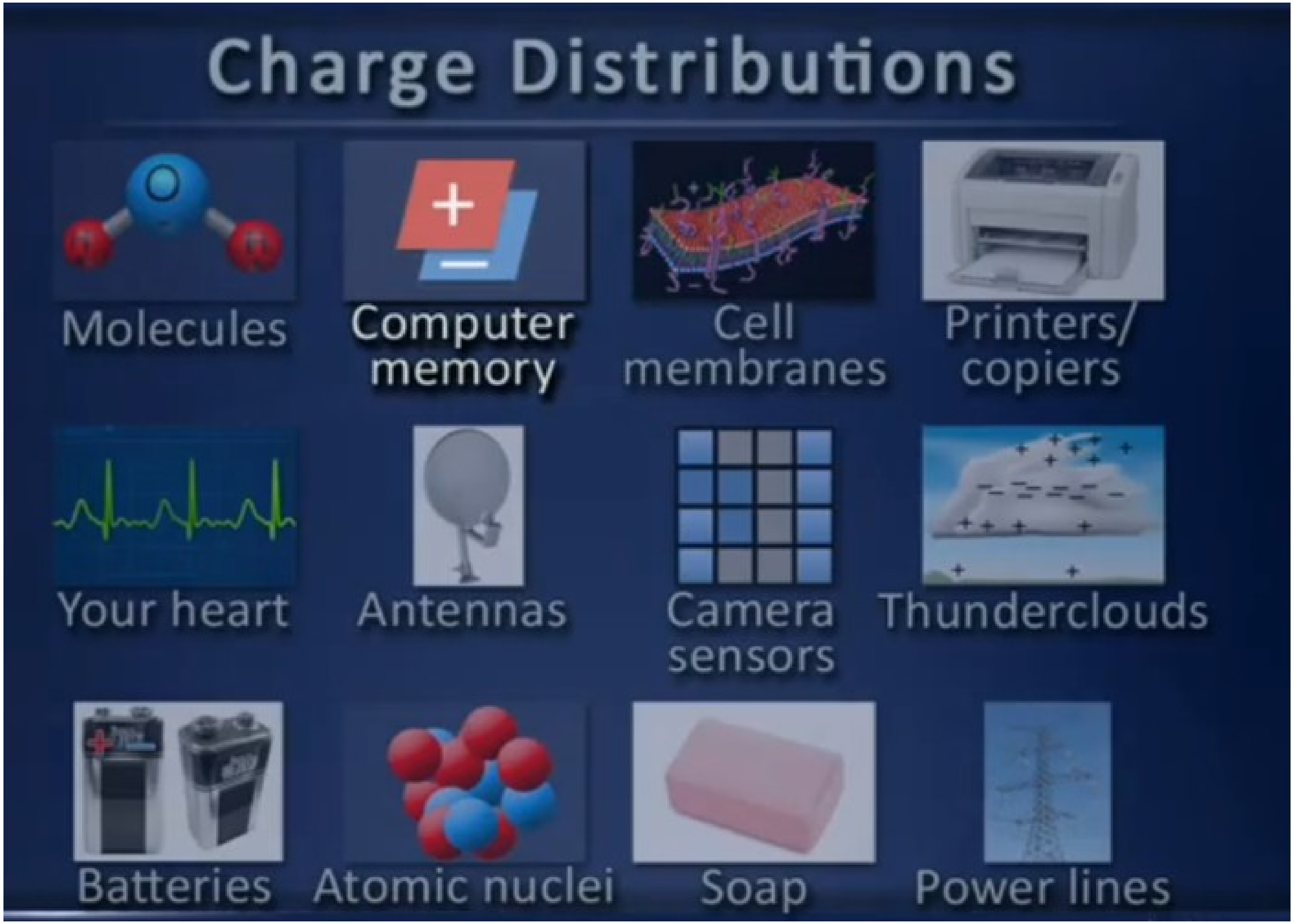
Example 1 **A Lot of Electrons**

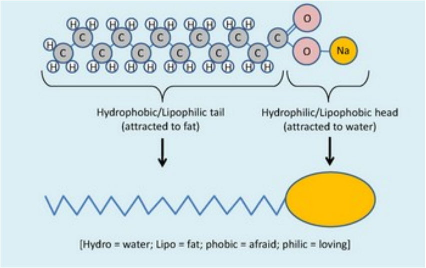
How many electrons are there in one coulomb of negative charge?

$$q = Ne$$

$$N = \frac{q}{e} = \frac{1.00\,\mathrm{C}}{1.60\times10^{-19}\,\mathrm{C}} = 6.25\times10^{18}$$

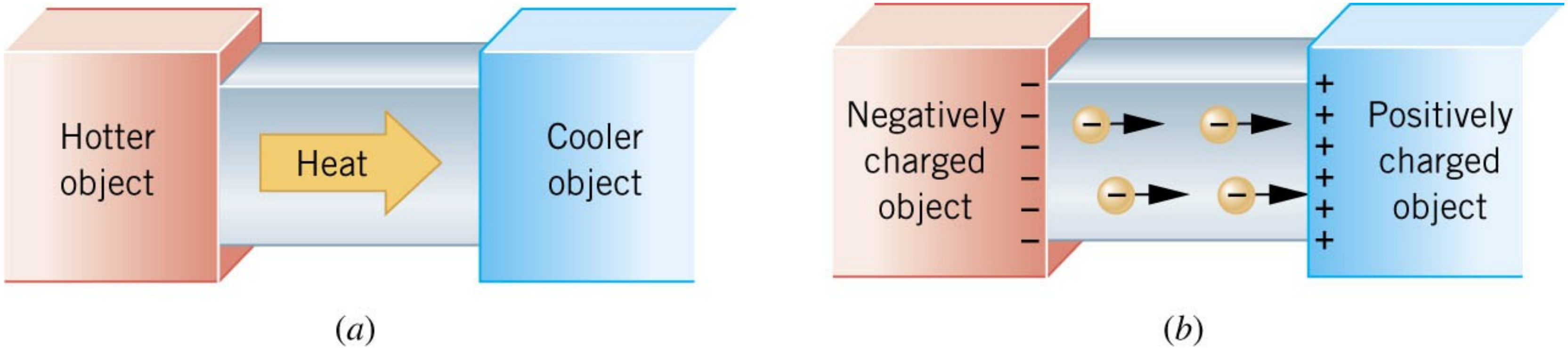
Charge is a property of matter. It is quantized
Mass is a property of matter it is not quantized.





https://www.radiantvisionsystems.com/blog/ccd-sensors-albert-einstein-and-photoelectric-effect

18.3 Conductors and Insulators

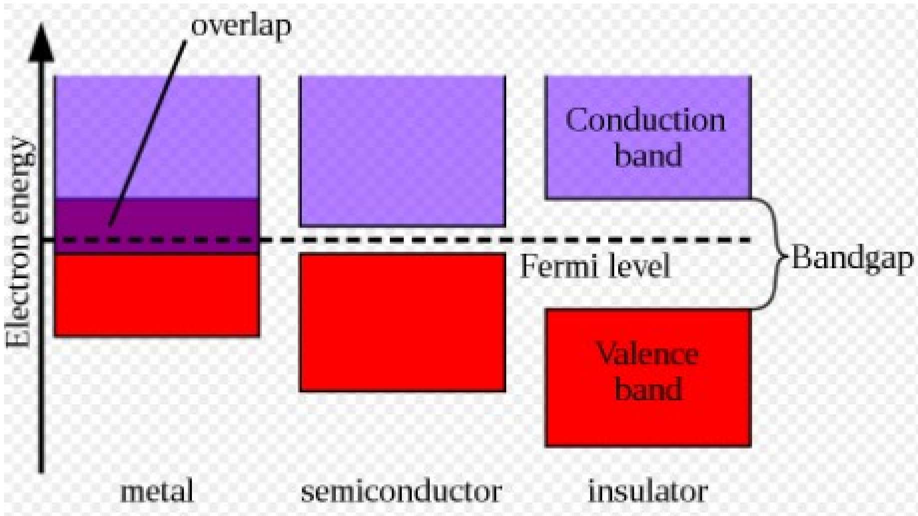


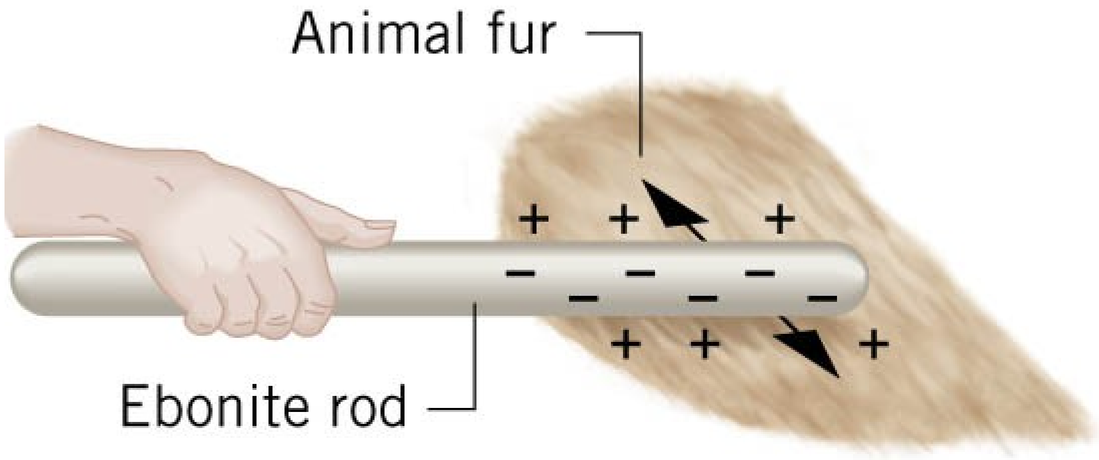
(a) (b)

Not only can electric charge exist *on an object,* but it can also move *through and object*.

Substances that readily conduct electric charge are called ***electrical conductors***.

Materials that conduct electric charge poorly are called ***electrical insulators.***





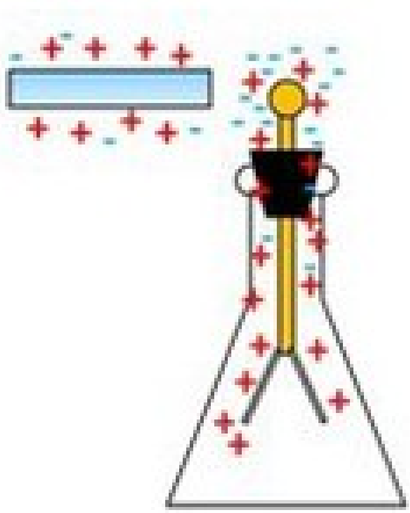
It is possible to transfer electric charge from one object to another.

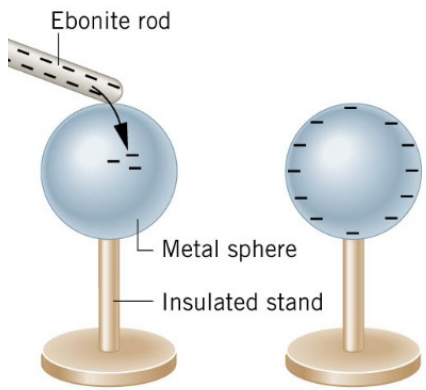
The body that loses electrons has an excess of positive charge, while the body that gains electrons has an excess of negative charge.

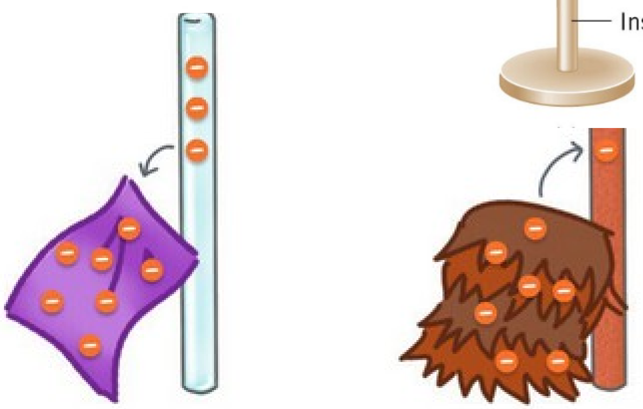
LAW OF CONSERVATION OF ELECTRIC CHARGE

During any process, the net electric charge of an isolated system remains constant (is conserved).

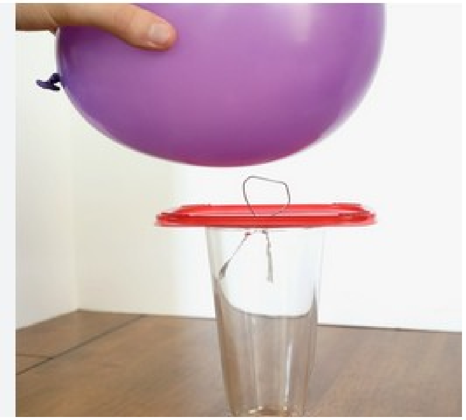
Demo electroscope!!
Get electroscope, silk, glass, fur





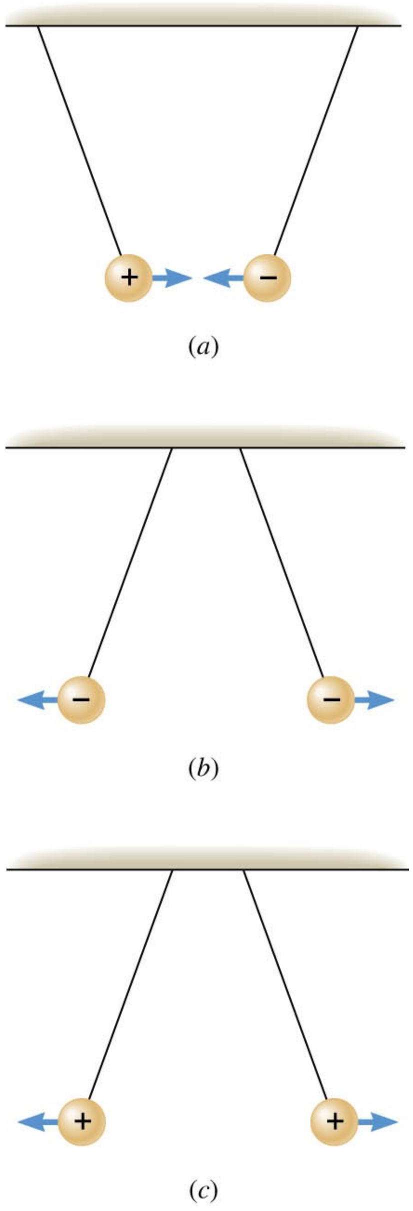


Silk removes electrons from the glass rod and fur adds electrons to the rubber.



https://phet.colorado.edu/sims/html/balloons-and-static-electricity/latest/balloons-and-static-electricity_en.html

18.2 Charged Objects and the Electric Force



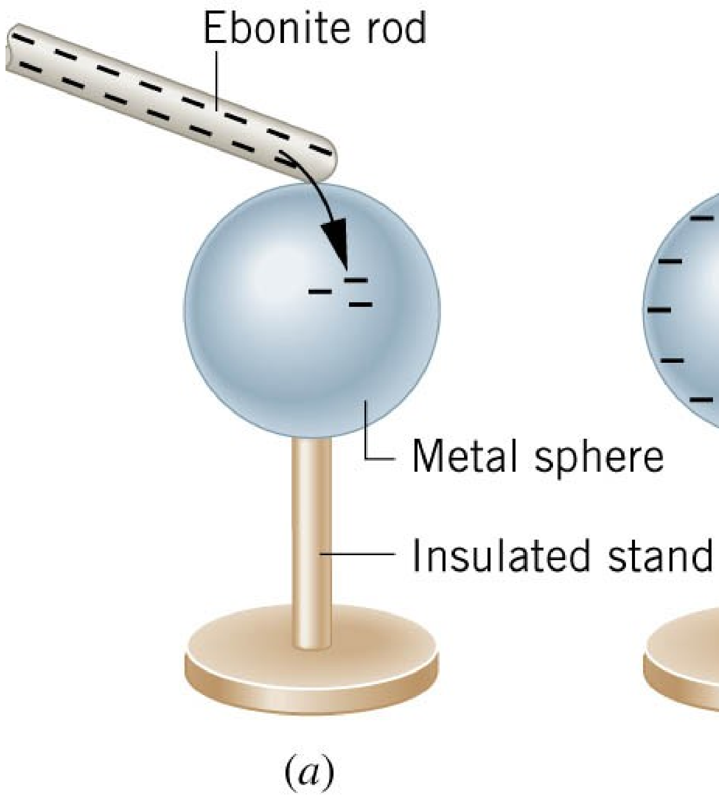
Like charges repel and unlike charges attract each other.

Demo attract – repel

- Consider a plastic rod that is being charged up by rubbing a wool cloth. It is then brought to an initially neutral metallic sphere, and allowed to touch the sphere for a few seconds. The rod is then separated from the sphere by a small distance. After becoming separated, the rod

  A. is repelled by the sphere.
  B. is attracted to the sphere.
  C. experiences no force due to the sphere.

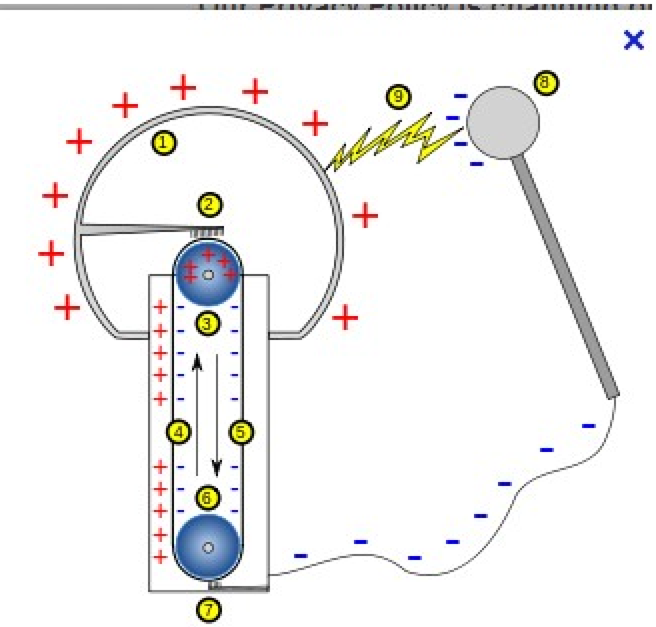
18.4 Charging by Contact and by Induction



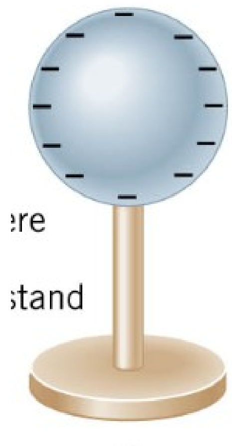
Charging by contact.

https://phet.colorado.edu/en/simulations/john-travoltage

Van graaf generator -

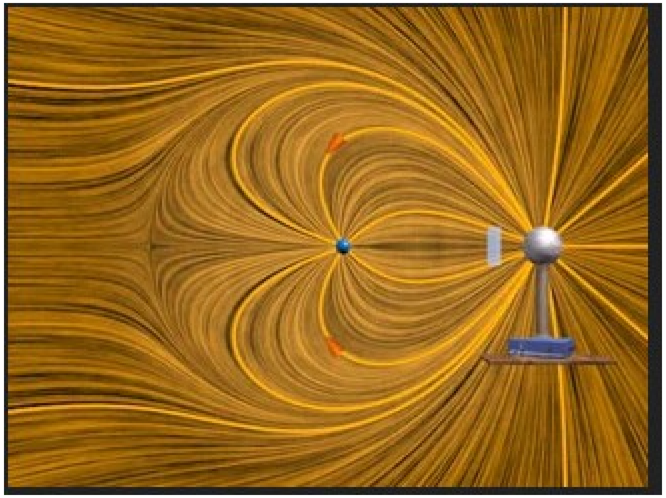


The charges placed
On a conductor
Move on its surface.



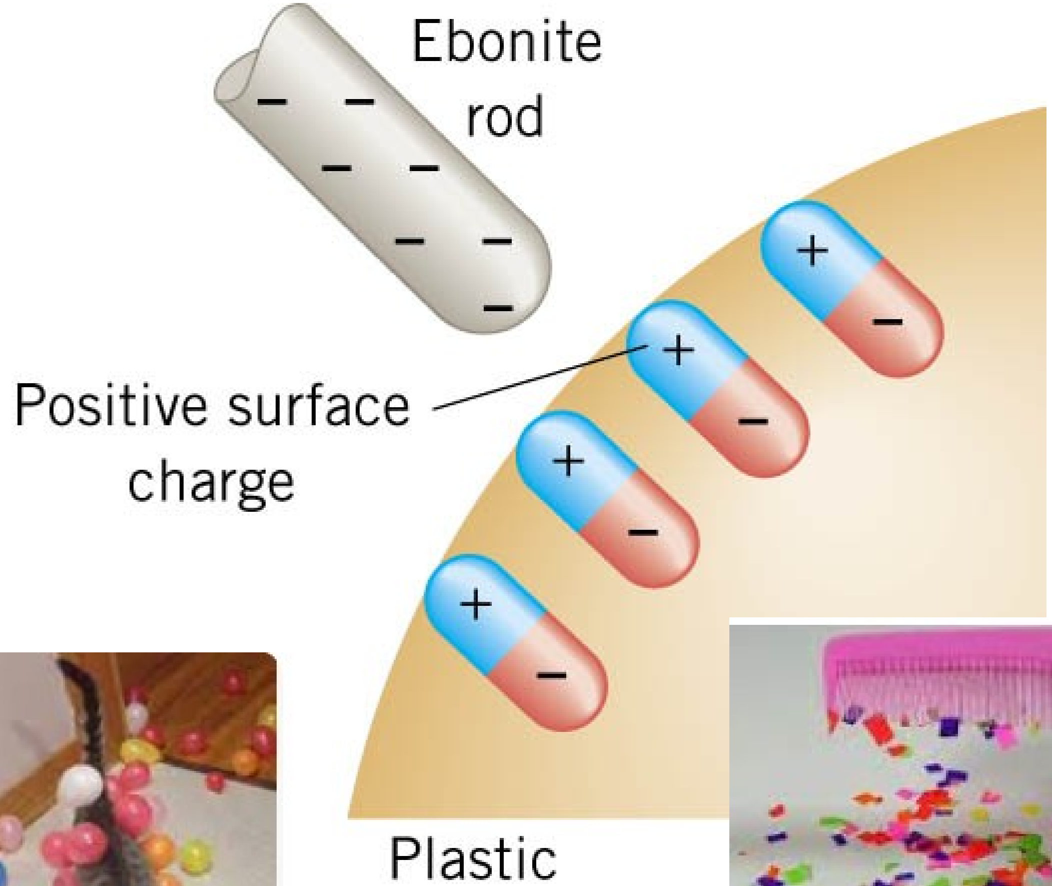
There is no charge inside.

Demo mini Van Graaf. Charge electroscope
Show the butterfly repelled.
Show the aluminum plate charged → spark



https://phet.colorado.edu/en/simulations/john-travoltage

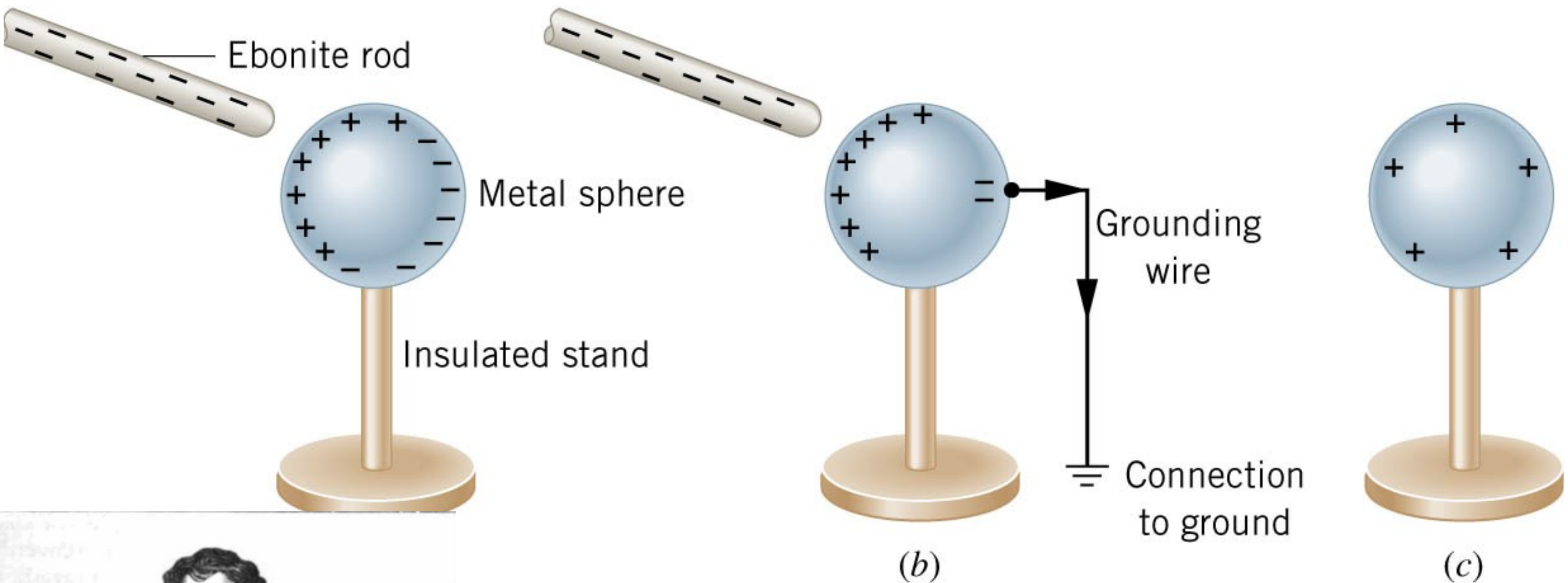
18.4 Charging by Contact and by Induction



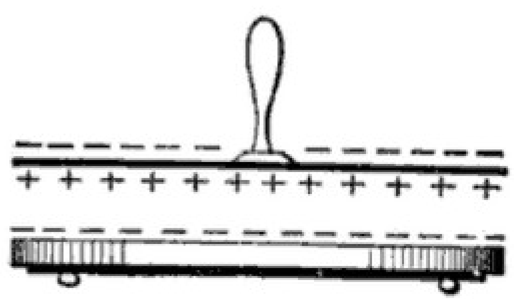
The negatively charged rod induces a slight positive surface charge
→ polarization
Demo feathers , balloon and hair

18.4 Charging by Contact and by Induction

Charging by induction.



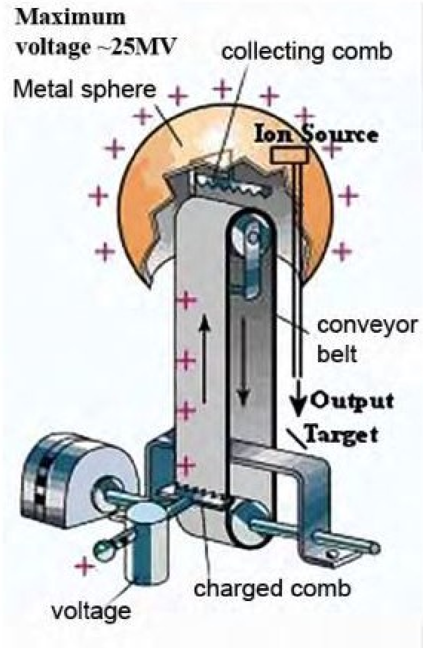
electrophorus



https://www.youtube.com/watch?v=zrvIXhmDA7E&t=31s

Van graaf generator

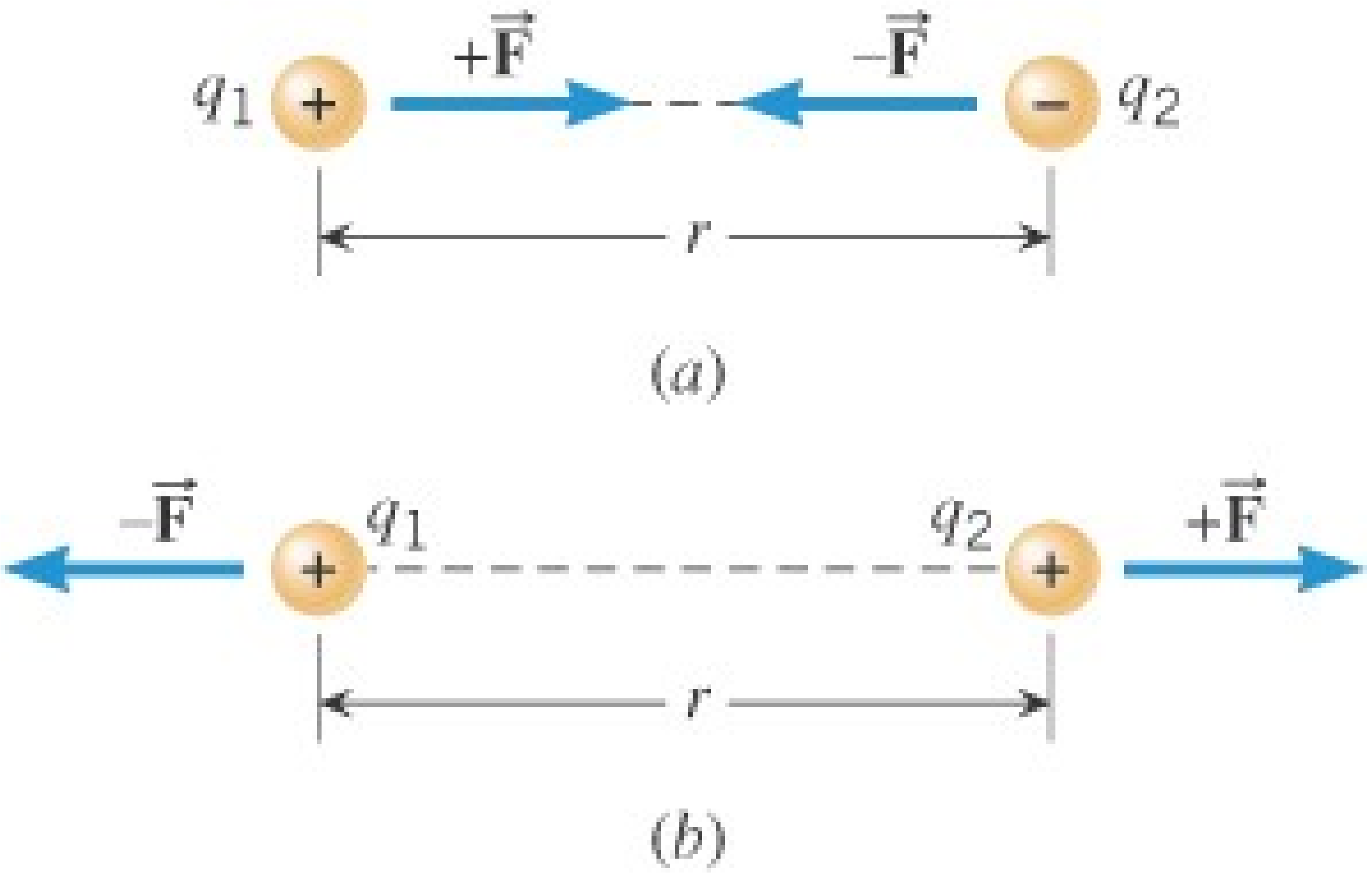
Built in the 30s (MIT has the patent). Used to generate 2MV (max 5MV) → particle Accelerator. Then if placed in pressurized container → 25MV
For demo in classes (hair due) → 100,000 V. Safe. Not enough charges.
Still used in nuclear medicine (Xray) or research. Video Walter Lewin

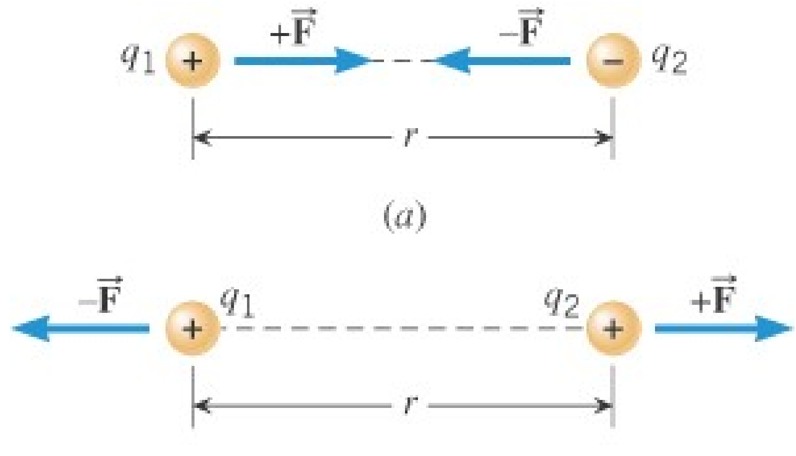


Van de Graaff's generator a Round Hill MA

https://oncologymedicalphysics.com/van-de-graaff-accelerators/

18.5 Coulomb's Law





https://ux1.eiu.edu/~addavis/1360/23EFields/Coulomb.html

COULOMB'S LAW

The magnitude of the electrostatic force exerted by one point charge on another point charge is directly proportional to the magnitude of the charges and inversely proportional to the square of the distance between them.

$$F = k\frac{|q_1||q_2|}{r^2}$$

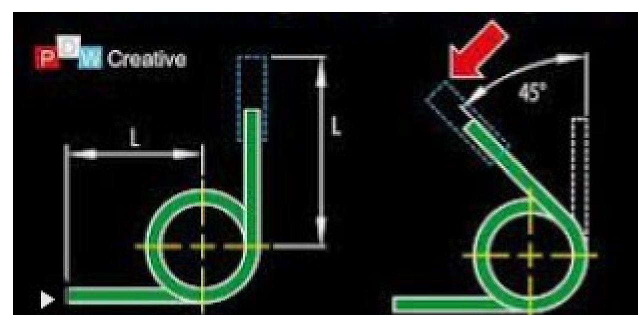
$$\varepsilon_o = 8.85\times10^{-12}\ \mathrm{C^2/(N\cdot m^2)}$$

Vacuum permittivity, commonly denoted ε0

$$k = 1/(4\pi\varepsilon_o) = 8.99\times10^{9}\ \mathrm{N\cdot m^2/C^2}$$

Coulomb's constant was discovered and named after Charles-Augustin de Coulomb (French!). He determined the strength of the electric force by measuring the force between charged objects using a torsion balance
Late 18th century.

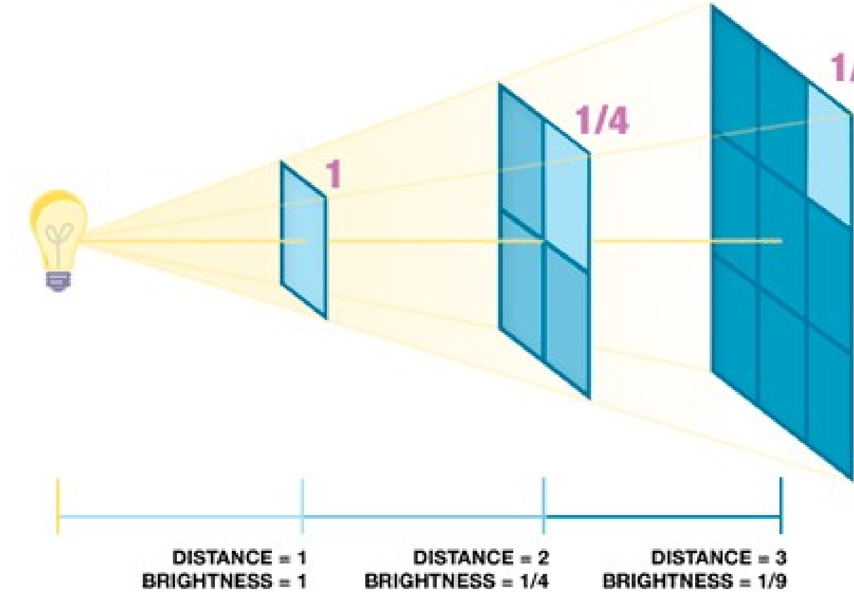
$$k = 1/(4\pi\varepsilon_o) = 8.99 \times 10^9\, N \cdot m^2/C^2$$

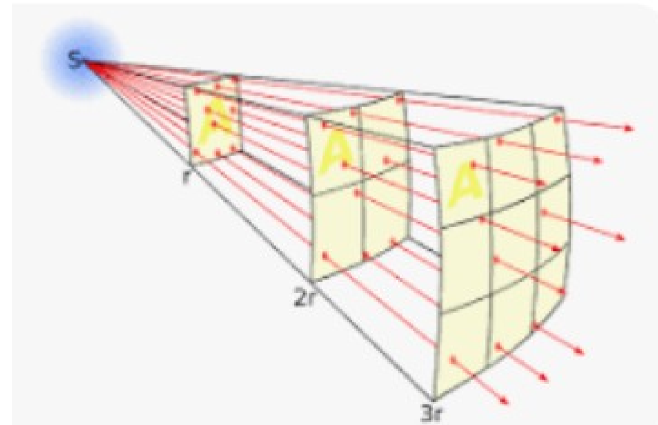


The wire (fiber) twist and the angle is proportional
To the force. (like Hookes law)

Experiment similar to Cavendish experiment

2 charges q1 and q2 are repelling each other with a force of 100 N (they are pinned)
If you half the distance between them, what's the new force?



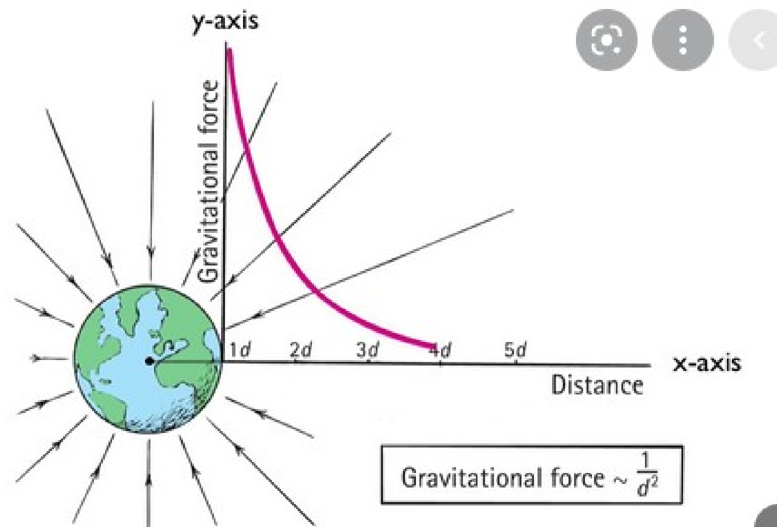


Inverse square law for radiation. Number of lines (intensity of light) Going through a square unit decreases with distance to the source.

Inverse square law for gravity.

$$F = \frac{GM_1M_2}{r^2}.$$

$$G = 6.67430 \times 10^{-11} \frac{N \cdot m^2}{kg^2}$$



A point charge produces an electric field that decreases with the square of the distance

- Two charged particles attract each other with a force F. If the charges of both particles are doubled, and the separation between them is also doubled, the force between them will be

A. $F/8$.

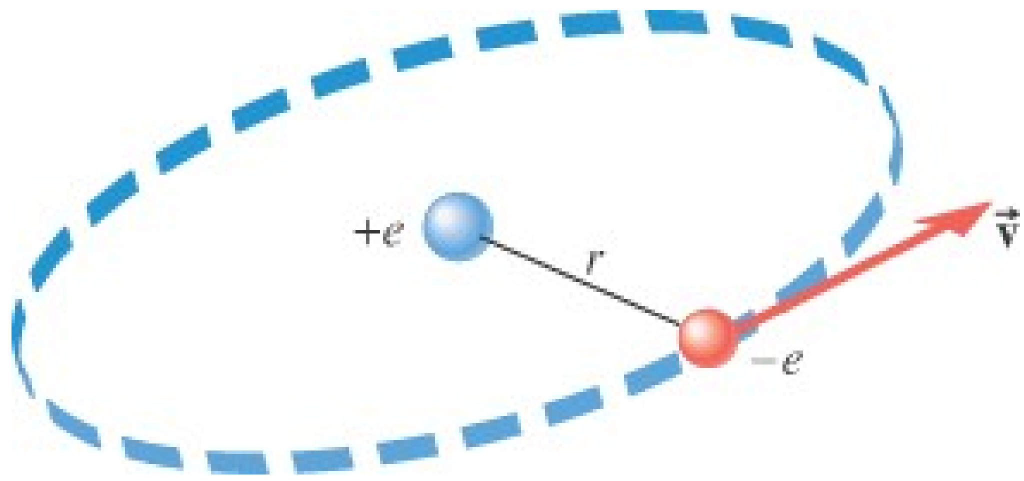
B. $F/2$.

C. F.

D. $2F$.

E. $8F$.

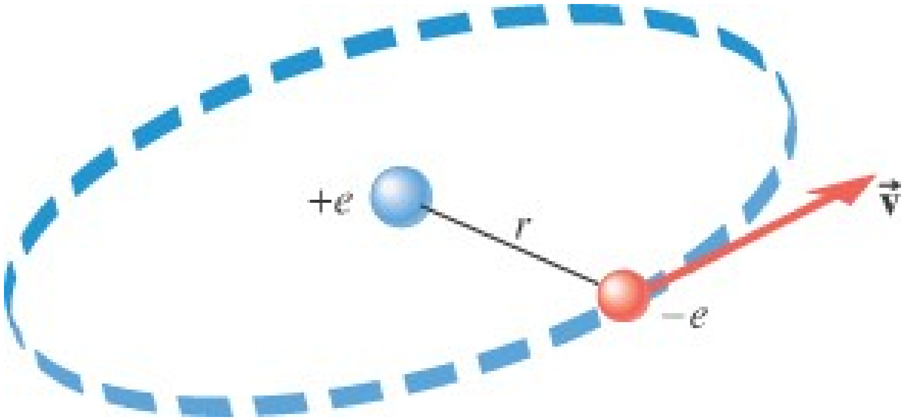
Mass e = 9 10^{-31}kg
Charge e = 1.9 10^{-19}C
Ke= 9 10^{9}



***Example 3* A Model of the Hydrogen Atom**

In the Bohr model of the hydrogen atom, the electron is in orbit about the nuclear proton at a radius of $5.29\text{x}10^{-11}$m. Determine the speed of the electron, assuming the orbit to be circular.

$$F = k\frac{|q_1||q_2|}{r^2}$$



$$F = k\frac{|q_1||q_2|}{r^2} = \frac{\left(8.99\times10^{9}\ \mathrm{N\cdot m^2/C^2}\right)\left(1.60\times10^{-19}\ \mathrm{C}\right)^2}{\left(5.29\times10^{-11}\ \mathrm{m}\right)^2} = 8.22\times10^{-8}\ \mathrm{N}$$

$$F = ma_c = mv^2/r$$

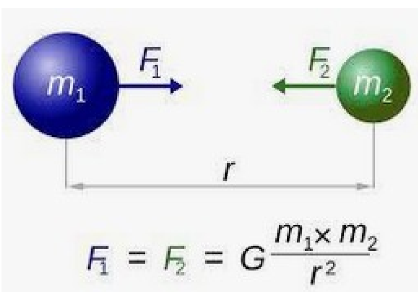
$$\Longrightarrow \quad v = \sqrt{Fr/m} = \sqrt{\frac{\left(8.22\times10^{-8}\ \mathrm{N}\right)\left(5.29\times10^{-11}\ \mathrm{m}\right)}{9.11\times10^{-31}\,\mathrm{kg}}} = 2.18\times10^{6}\ \mathrm{m/s}$$

Building on the previous example.

→ Compute the gravitational force between the proton and electron.
→ Compare with the electric force between them (find the ratio)

Mass electron = 9 10^{-31}kg
Charge electron = 1.9 10^{-19}C
radius of 5.29×10^{-11}m
G = 1.67 10^{-11} $m^3\ kg^{-1}\ s^{-2}$
Ke= 9 10^9 N m^2 /C^2

$$m_p = 1.673 \times 10^{-27}\,\mathrm{kg}$$



Newton's law of universal gravitation

$$F = G\frac{m_1 m_2}{r^2}$$

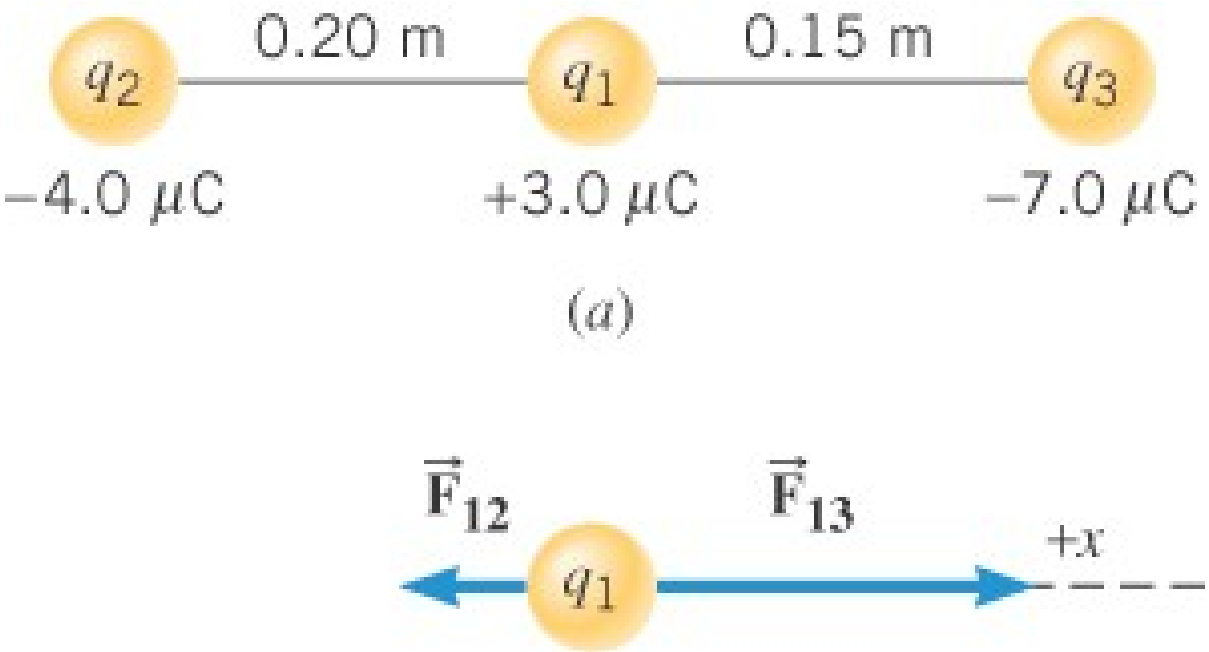
F = force
G = gravitational constant
m_1 = mass of object 1
m_2 = mass of object 2
r = distance between centers of the masses

The planets are neutral. So electric force is not considered.
At an atomic scale, we neglect the gravitational force.

TEXTBOOK 21.1 → to do

Example 4 Three Charges on a Line → superposition principle (adding vectors in 1D)

Determine the magnitude and direction of the net force on q_1.



(b) Free-body diagram for q_1

App adding vectorshttps://www.physicsclassroom.com/Physics-Interactives/Vectors-and-Projectiles/Vector-Addition/Vector-Addition-Interactive

(a)

(b) Free-body diagram for q_1

$$F_{12} = k\frac{|q_1||q_2|}{r^2} = \frac{\left(8.99\times10^{9}\ \mathrm{N\cdot m^2/C^2}\right)\left(3.0\times10^{-6}\mathrm{C}\right)\left(4.0\times10^{-6}\mathrm{C}\right)}{(0.20\mathrm{m})^2} = 2.7\mathrm{N}$$

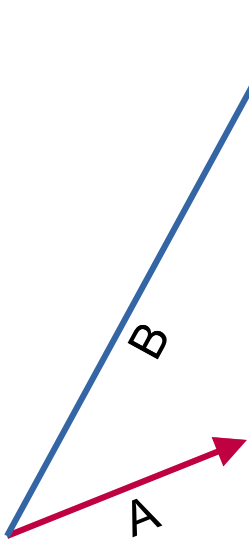
$$F_{13} = k\frac{|q_1||q_3|}{r^2} = \frac{\left(8.99\times10^{9}\ \mathrm{N\cdot m^2/C^2}\right)\left(3.0\times10^{-6}\mathrm{C}\right)\left(7.0\times10^{-6}\mathrm{C}\right)}{(0.15\mathrm{m})^2} = 8.4\mathrm{N}$$

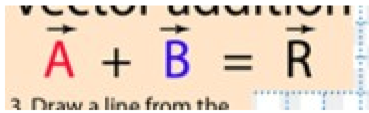
$$\vec{\mathbf{F}} = \vec{\mathbf{F}}_{12} + \vec{\mathbf{F}}_{13} = -2.7\mathrm{N} + 8.4\mathrm{N} = +5.7\mathrm{N}$$

Textbook 21.2, 21.3 → to do

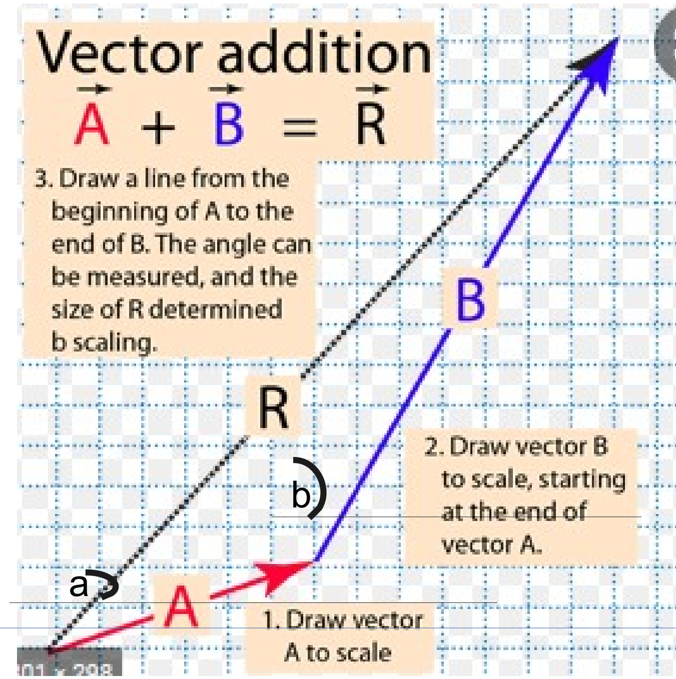
Superposition principle and Adding vectors in 2D

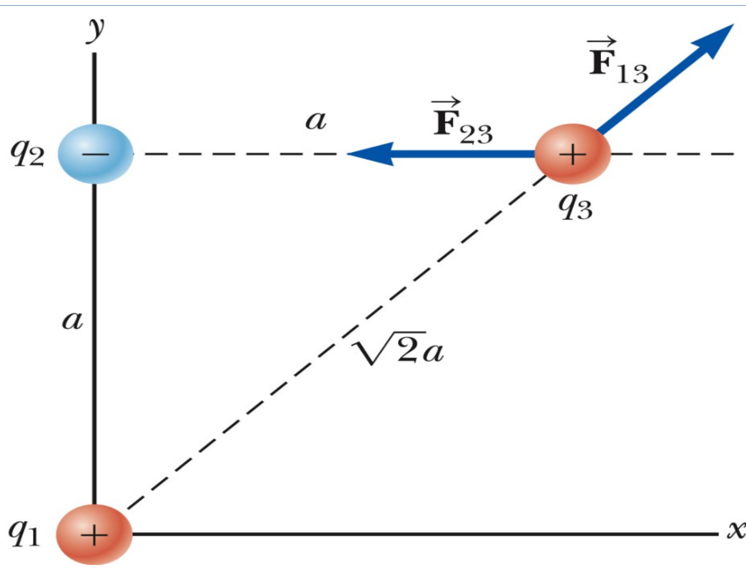
(review adding vectors. Simulation. Exploration physics emphasize polar graph – see canvas.)





	X	Y
A	|A|cos(a)	|A|sin(a)
B	|B|cos(b)	|B|sin(b)
R		





Example:

q1=4uC, q2=-4uC q3 = 1C

→ first draw the vectors.

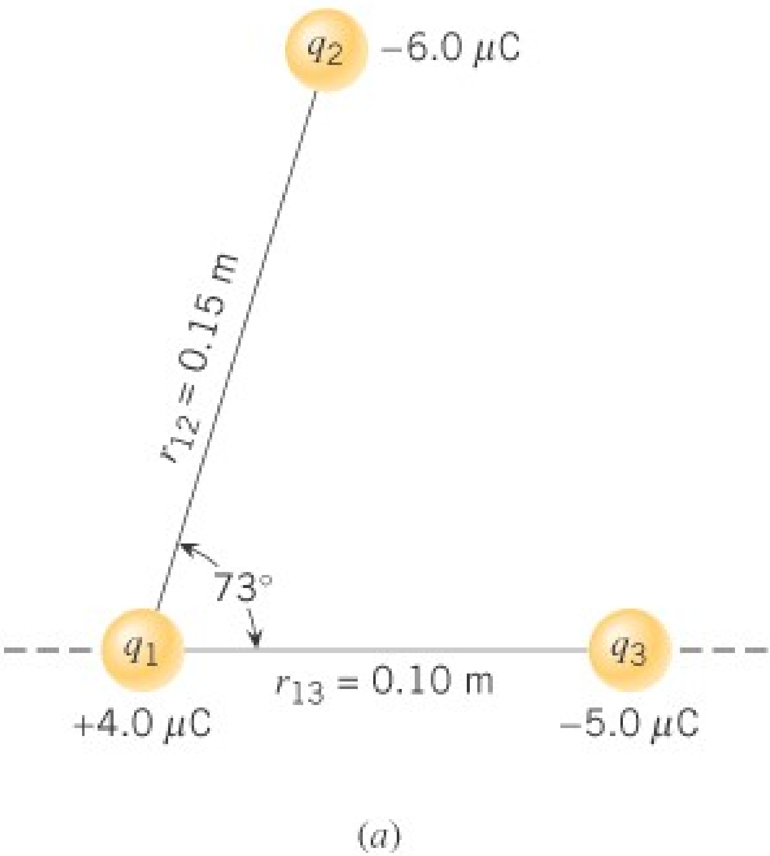
→ Find the magnitude of each force

→ add the components (cos or sin – use polar graph)

Note: the sum is the electric field If q3 is 1C

Q1=- q2 → The system (q1,q2) Is called an electric dipole.

Find the net force on q1



Step1: free-diagram for q1.
Draw Vector force.

Step2: find magnitudes F13 and F12

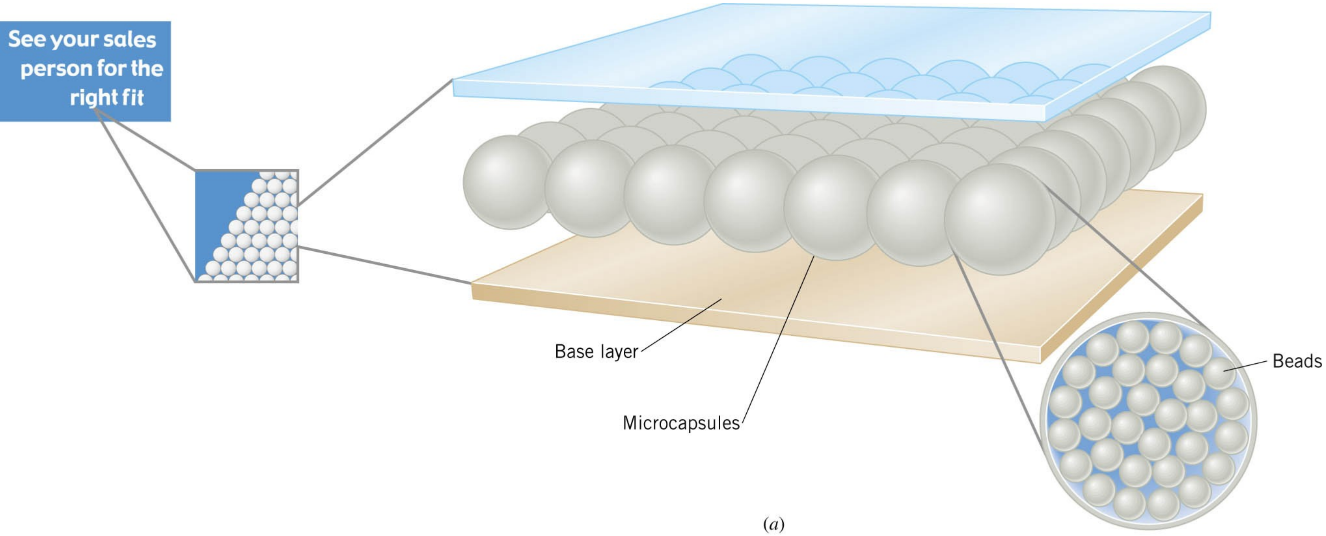
Step3 : Find Fx and Fy (components of Fnet .or sum) Use cosine and sine

Step4: find magnitude of Fnet using Pythagorean theorem
Find the direction of Fnet using Tangent. Use standard notation

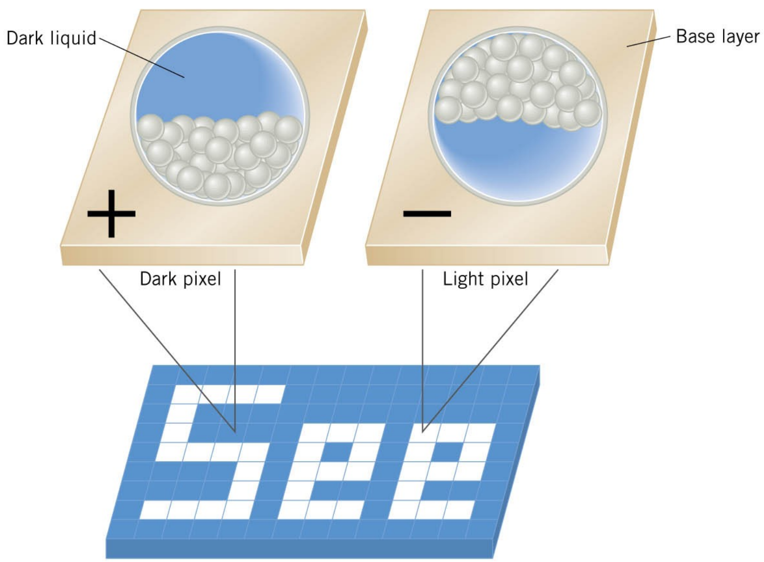
Figure 18.13*a* shows three point charges that lie in the *x*, *y* plane in a vacuum. Find the magnitude and direction of the net electrostatic force on q_1.

Use vector additions. Add the individual components.
Textbook 21.4

E- ink - E-reader

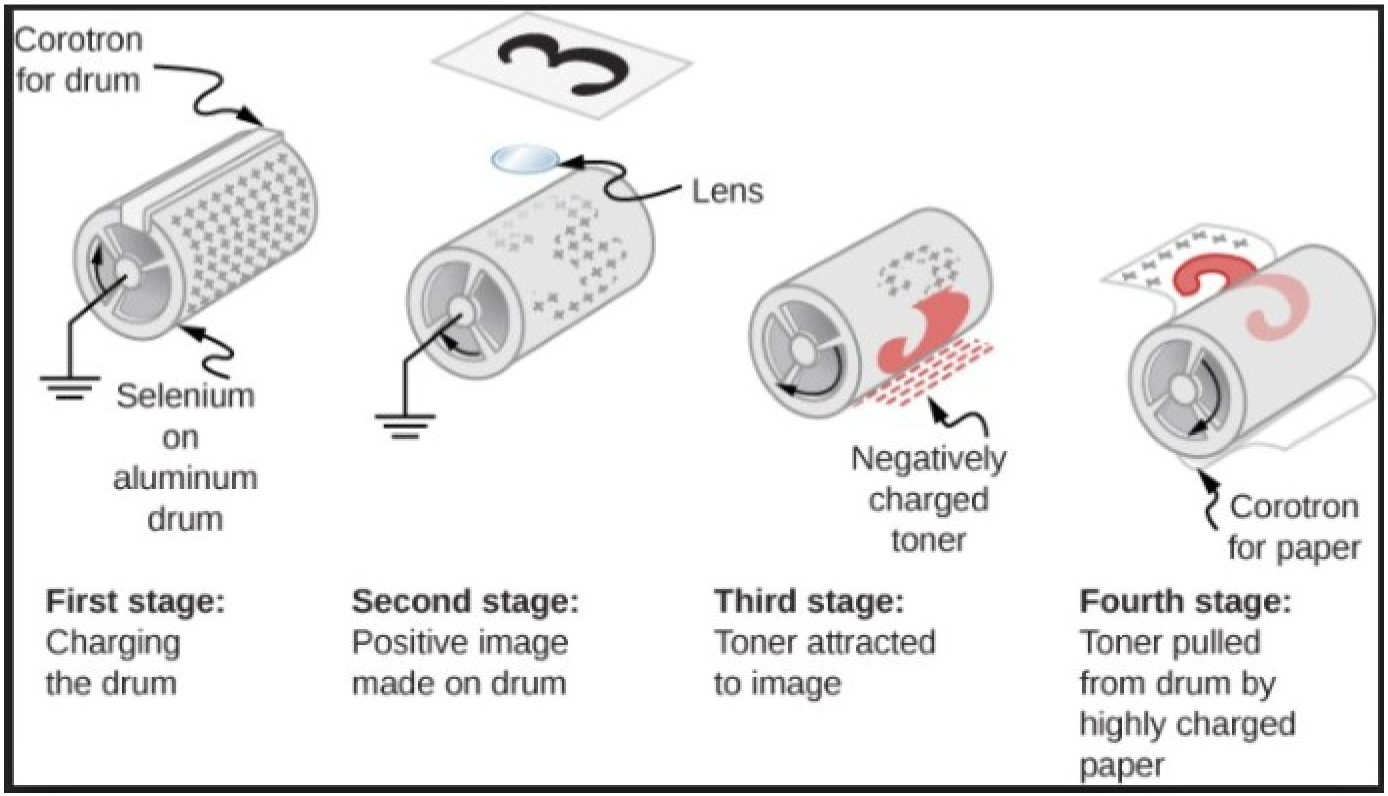


(*a*)



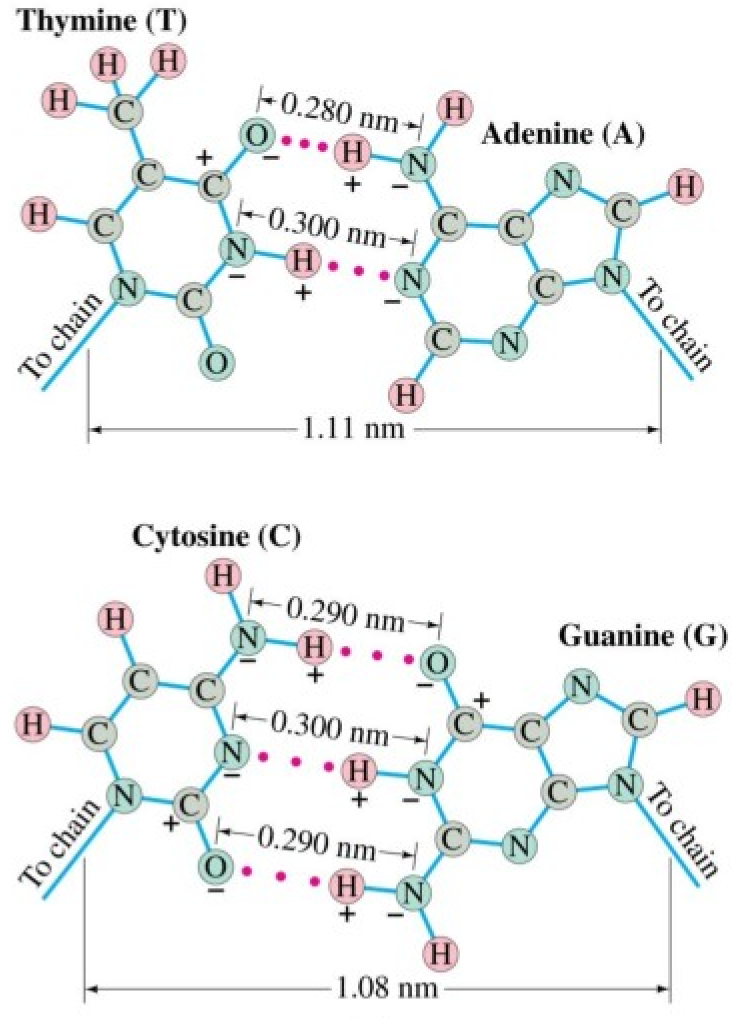
(*b*)

Copy machine



16-10 Electric Forces in Molecular Biology: DNA Structure and Replication

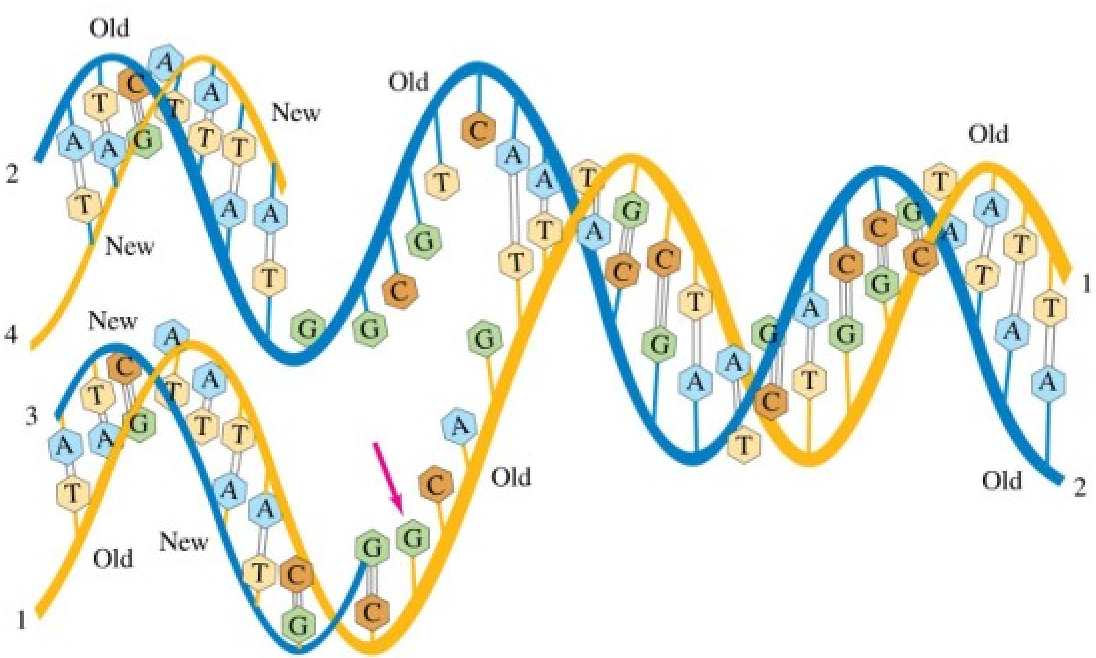
The A-T and G-C nucleotide bases attract each other through electrostatic forces.



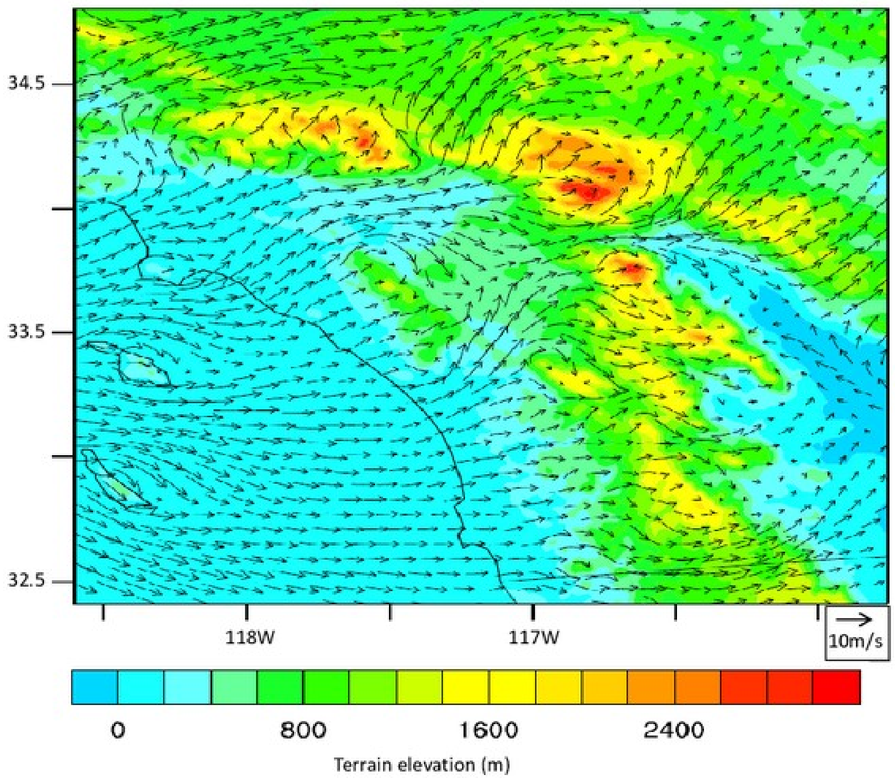
(b)

16-10 Electric Forces in Molecular Biology: DNA Structure and Replication

Replication: DNA is in a "soup" of A, C, G, and T in the cell. During random collisions, A and T will be attracted to each other, as will G and C; other combinations will not.

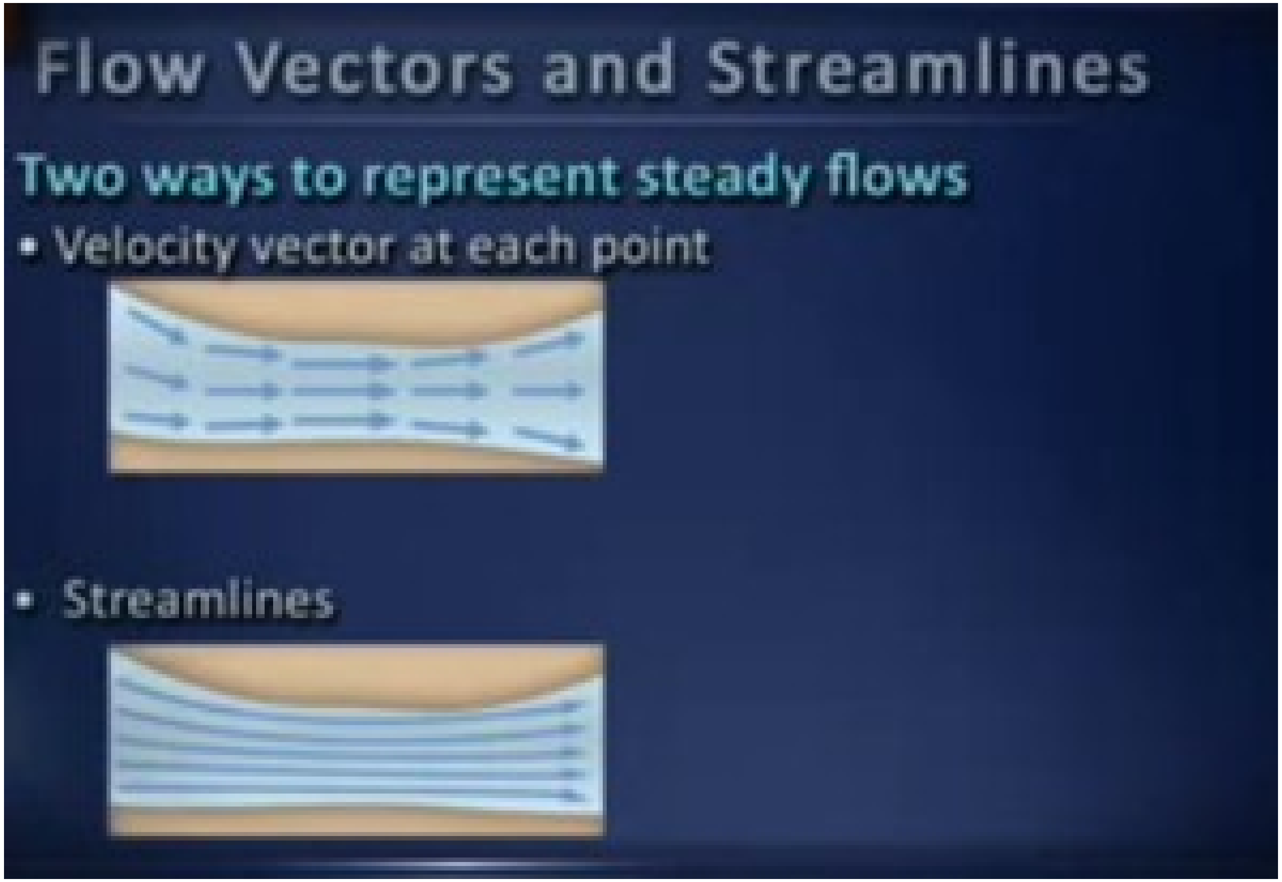


Vector fields →
any point in space is associated with a vector (magnitude and unit).

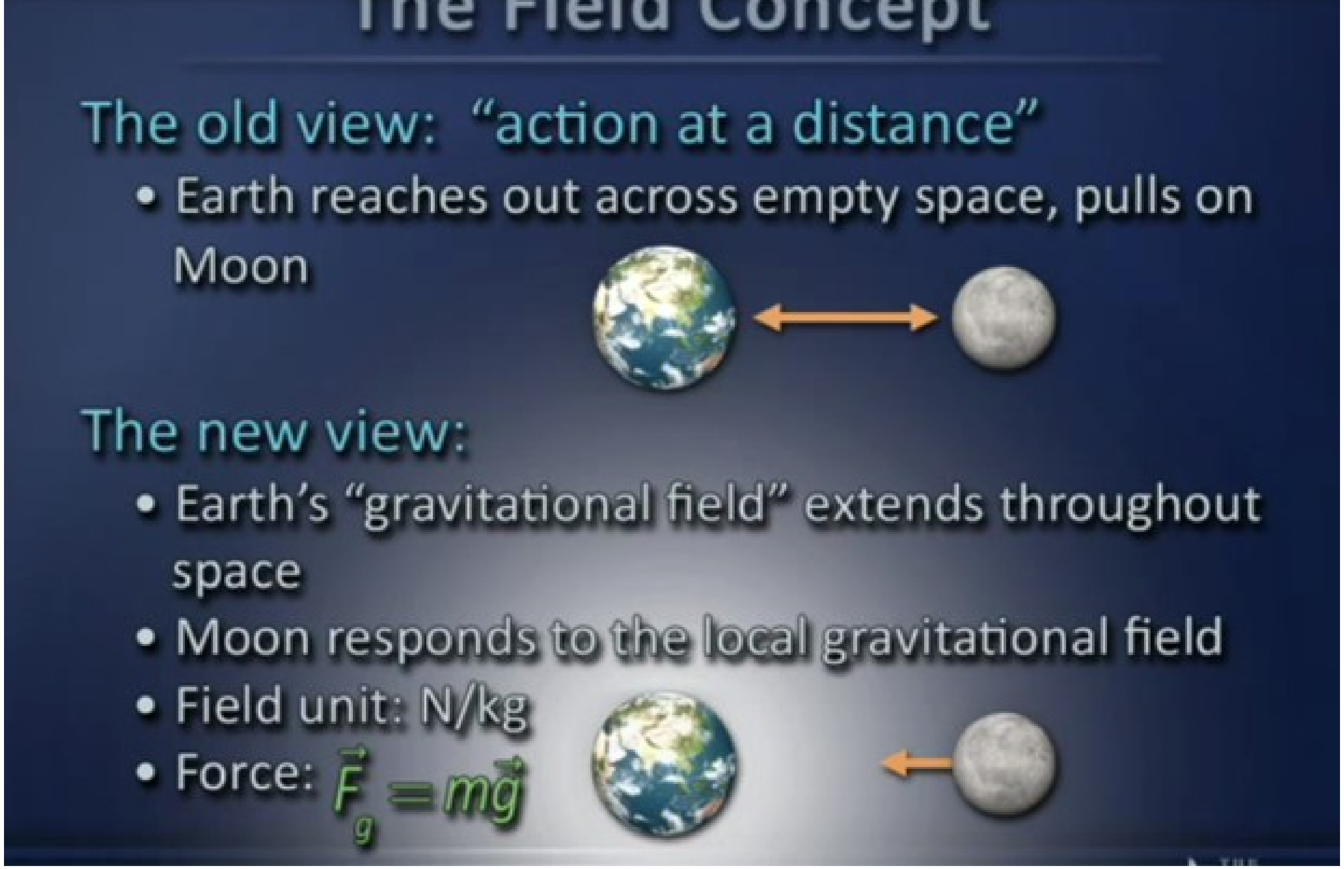


Wind vector field at 40 m above surface layer in southern California coastal area. Data are averaged from 20:00 to 02:00 UTC. 16-20 July 2005.

We can use arrows (big arrow = large wind). Or we can use line.
Large density of lines = large wind. Vector field versus lines



2 ways to represent a vector field:
→ arrows (larger = stronger field)
→ lines (more lines that is larger density of lines = stronger field)



The same way, a charge q in an electric field responds to it.
There is a force acting on the charge.

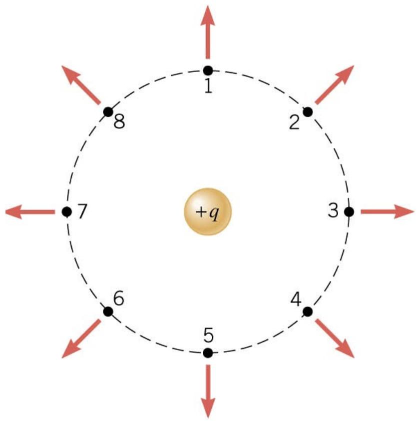
F = q E   F and E are vectors.
E is the electric field at the location of the charge.

$$\vec{F} = \vec{E} q_o$$

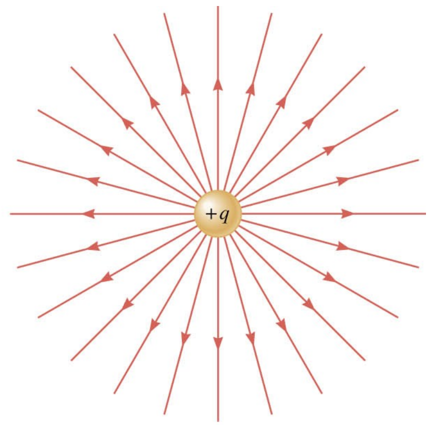
E has a magnitude, a direction, a unit.

Works only for a point charge. For a charged object,the E can vary point to point.

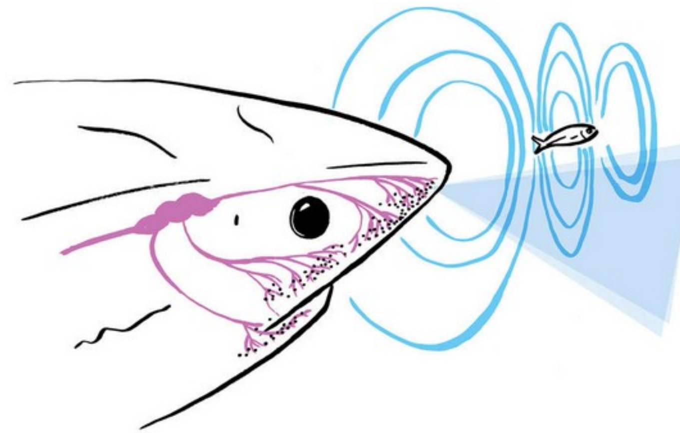
Electric field lines or ***lines of force*** provide a map of the electric field in the space surrounding electric charges.

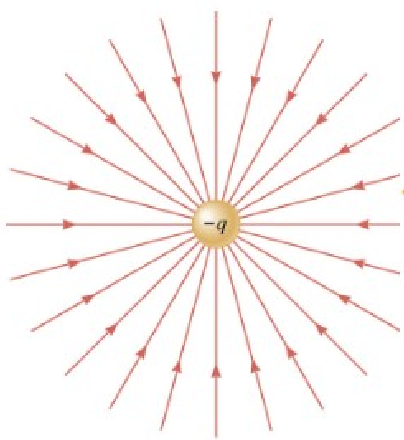


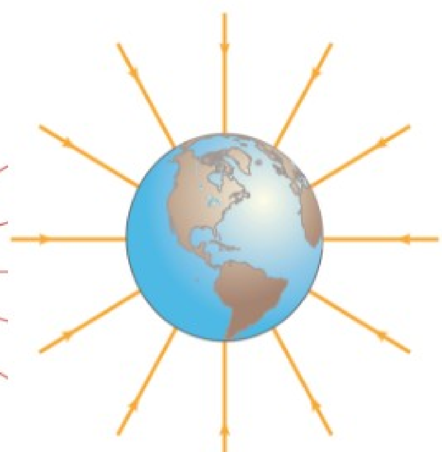
(a)



(b)

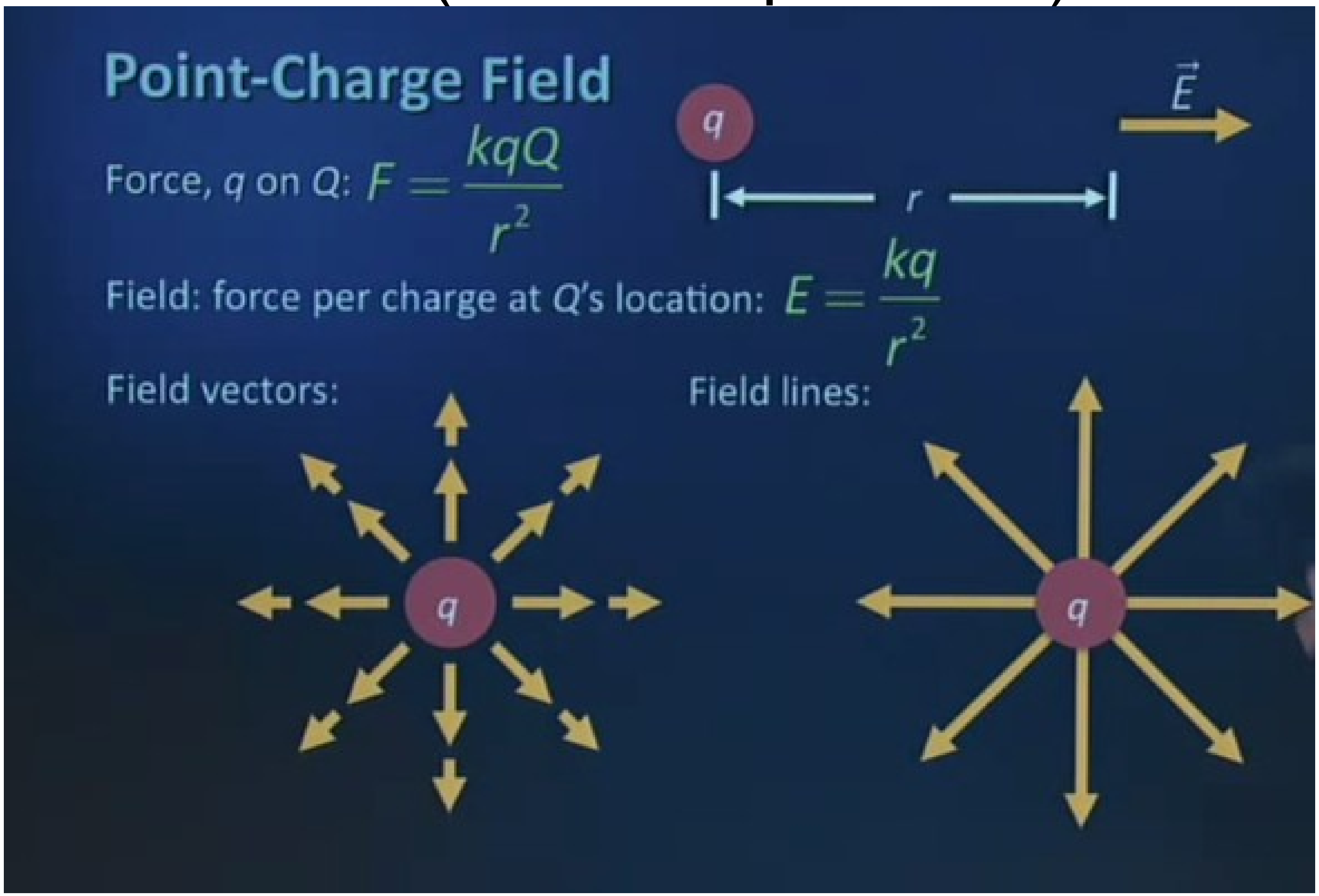






Electric field lines are always directed away from positive charges and toward negative charges.

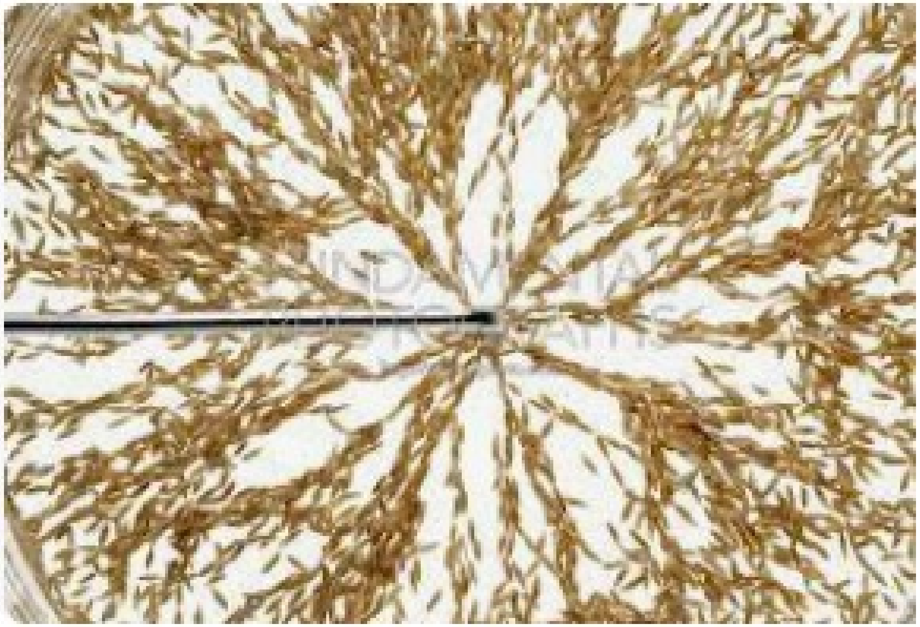
Electric field of a point charge
(inverse square law)



The electric field exists whether there is a point charge in it or not.

Textbook 21.6

Electric field of a point charge



Using grass seed in oil to
Visualize the electric field.
Inside the inner charged ring,
There is no electric field
(faraday cage).
Between the rings, the electric field
Is radial. Its a cylindrical capacitor
(like coaxial cable)

18.6 *The Electric Field*

DEFINITION OF ELECTRIC FIELD

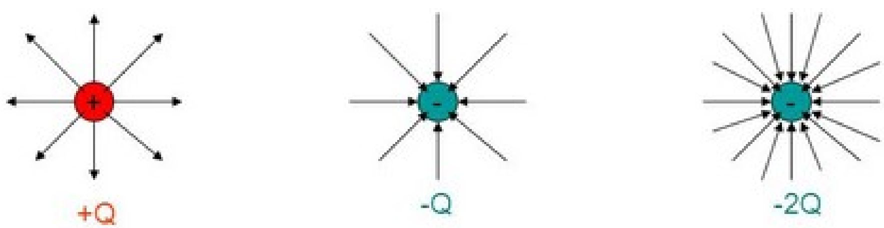
The electric field that exists at a point is the electrostatic force experienced by a small test charge placed at that point divided by the charge itself:

$$\vec{E} = \frac{\vec{F}}{q_o}$$

SI Units of Electric Field: newton per coulomb (N/C)

Definition:
- Electric field lines indicate the direction of the force due to the given field on a positive charge, i.e. electric force on a positive charge is tangent to these lines
- Number of these lines is proportional to the magnitude of the charge

Properties:
- Electric field lines start on positive charges or came from infinity, they end on negative charges or end at infinity
- Density of these lines is proportional to the magnitude of the field



- -The lines for a group of point charges must begin on + charge and end on – charge
- - The number of lines drawn leaving a + charge or ending a – charge is proportional to the magnitude of the charge
- - No two field lines can cross each other

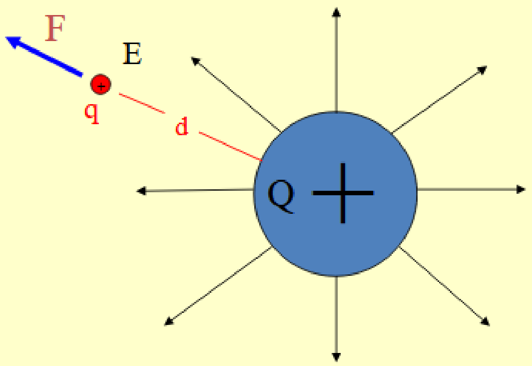
Video → lines of force, vector field and electric fields
Vision by Michael Faraday. He imagined field force
Also App/ Electric field PHet

Force on a charge due to an Electric Field

If a charge, ***q***, is placed at point "**x**" in the field where the Electric Field strength is ***E***, it will experience a force ***F***.

$$F = qE$$

$$E = \frac{F}{q}$$



$$|E| = \frac{k\,|q_1|}{r^2}$$

$$F = qE$$

E= Electric Field (N/C)

k= Coulombs Constant

(9 x 10^9 Nm^2/c^2)

q_1 = Charge of object 1

r = radius or distance

A positive charge moves along the electric field.
A negative charge moves against.

https://phet.colorado.edu/en/simulations/charges-and-fields

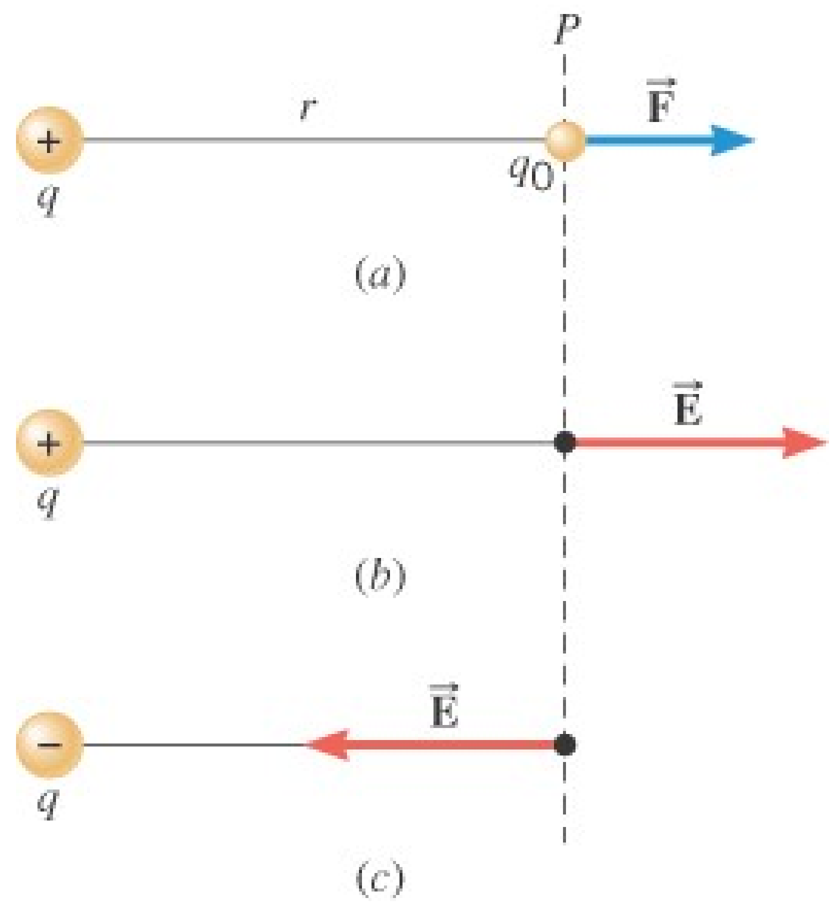
- Which of the following statements is FALSE?

  A. Electric charge is quantized.
  B. A negative charge placed in an electric field experiences a force in the direction of the field.
  C. The Coulomb force between two charges can be attractive or repulsive.
  D. A neutral atom becomes a positively charged ion when an electron is removed from it.

Example 10 The Electric Field of a Point Charge

The isolated point charge of q=+15μC is in a vacuum. The test charge is 0.20m to the right and has a charge q_o=+15μC.

Determine the electric field at point P.



$$\vec{\mathbf{E}} = \frac{\vec{\mathbf{F}}}{q_o}$$

$$F = k\frac{|q_1||q_2|}{r^2}$$

$$F = k\frac{|q||q_o|}{r^2}$$

$$= \frac{(8.99\times10^9\ \mathrm{N\cdot m^2/C^2})(0.80\times10^{-6}\,\mathrm{C})(15\times10^{-6}\,\mathrm{C})}{(0.20\mathrm{m})^2} = 2.7\mathrm{N}$$

$$E = \frac{F}{|q_o|} = \frac{2.7\ \mathrm{N}}{0.80\times10^{-6}\,\mathrm{C}} = 3.4\times10^6\ \mathrm{N/C}$$

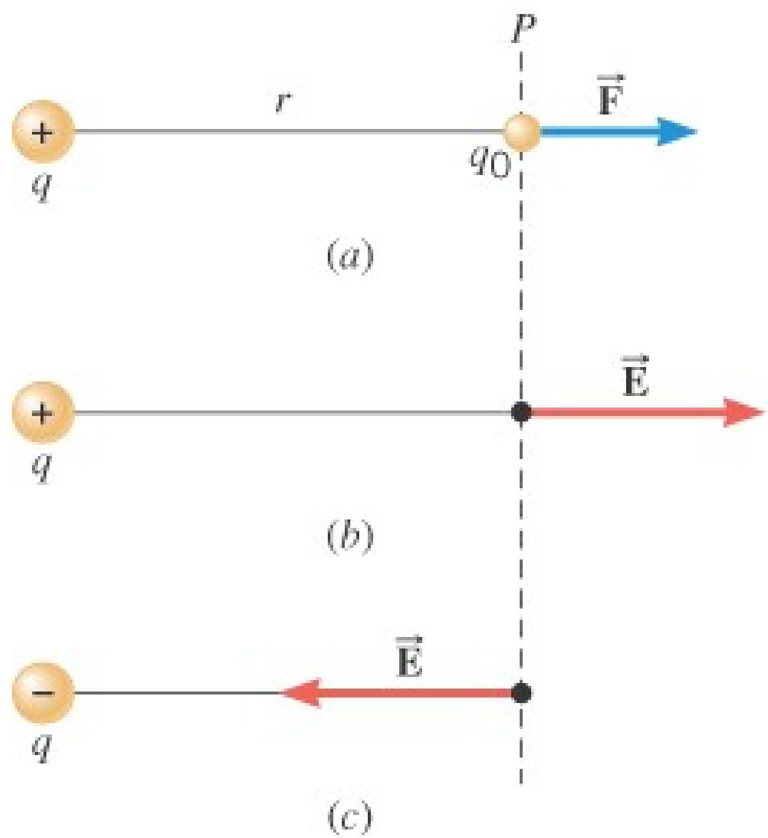
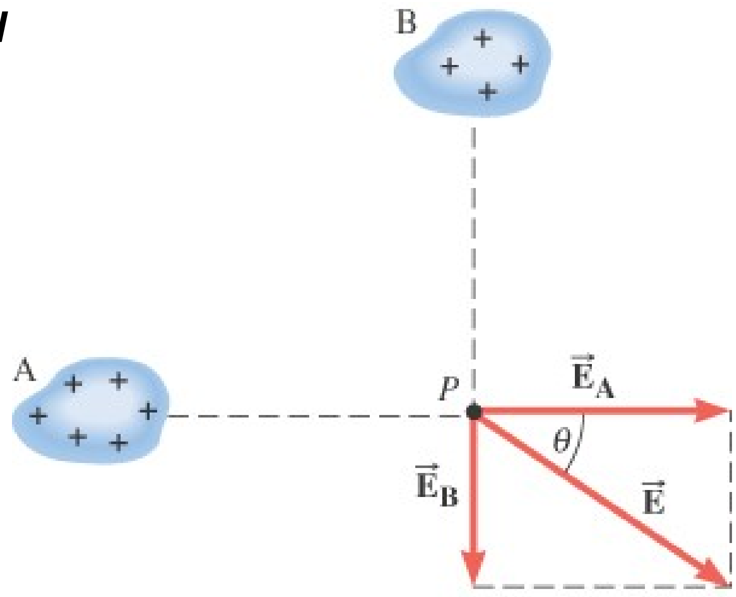


Table 1 Some Electric Fields[a]

Field	Value (N/C)
At the surface of a uranium nucleus	3×10^{21}
Within a hydrogen atom, at the electron orbit	5×10^{11}
Electric breakdown occurs in air	3×10^{6}
At the charged drum of a photocopier	10^{5}
The electron beam accelerator in a TV set	10^{5}
Near a charged plastic comb	10^{3}
In the lower atmosphere	10^{2}
Inside the copper wire of household circuits	10^{-2}

[a] Approximate values.

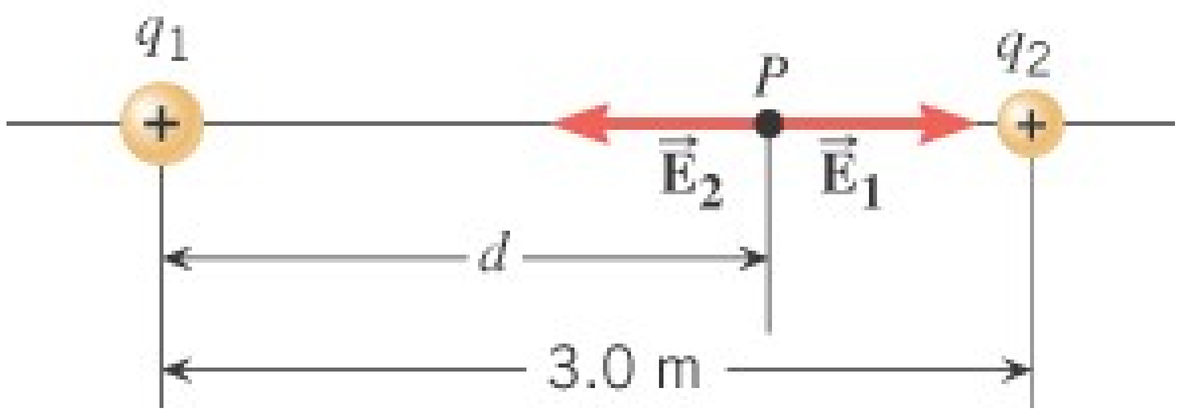
SUPERPOSITION PRINCIPLE



Electric fields from different sources add as vectors.

Example 11 The Electric Fields from Separate Charges May Cancel

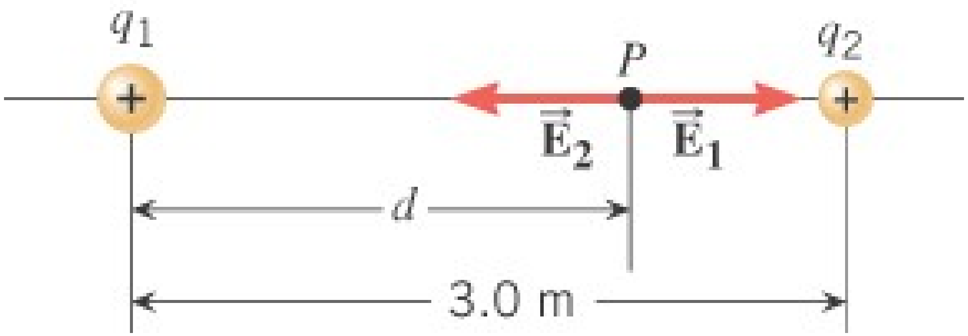
Two positive point charges, q_1=+16μC and q_2=+4.0μC are separated in a vacuum by a distance of 3.0m. Find the spot on the line between the charges where the net electric field is zero.



$$E = k\frac{|q|}{r^2}$$

Run simulation electric field of discrete distribution of charge.

https://phet.colorado.edu/sims/html/charges-and-fields/latest/charges-and-fields_all.html



$$E = k\frac{|q|}{r^2}$$

$$E_1 = E_2$$

$$k\frac{\left(16\times10^{-6}\,\mathrm{C}\right)}{d^2} = k\frac{\left(4.0\times10^{-6}\,\mathrm{C}\right)}{\left(3.0\mathrm{m}-d\right)^2}$$

$$2.0\left(3.0\mathrm{m}-d\right)^2 = d^2$$

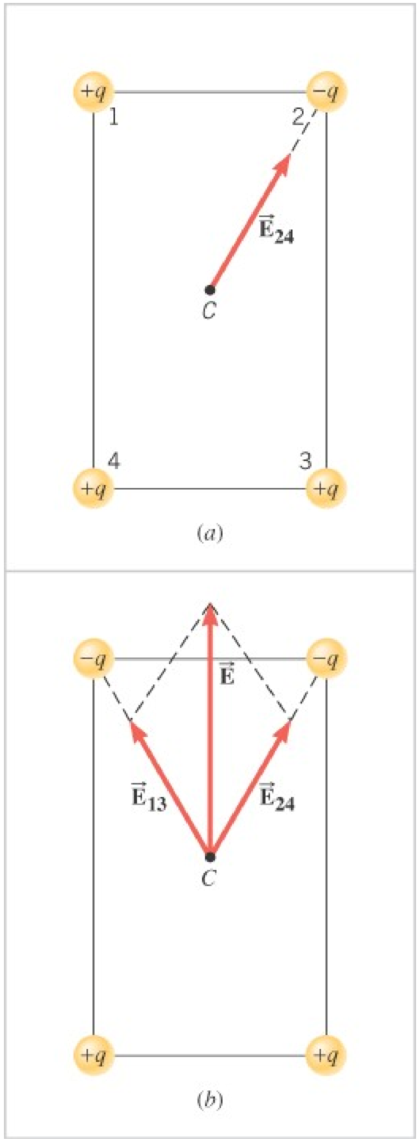
$$d = +2.0\ \mathrm{m}$$

Conceptual Example 12 Symmetry and the Electric Field

Point charges are fixes to the corners of a rectangle in two different ways. The charges have the same magnitudes but different signs.

Consider the net electric field at the center of the rectangle in each case. Which field is stronger?

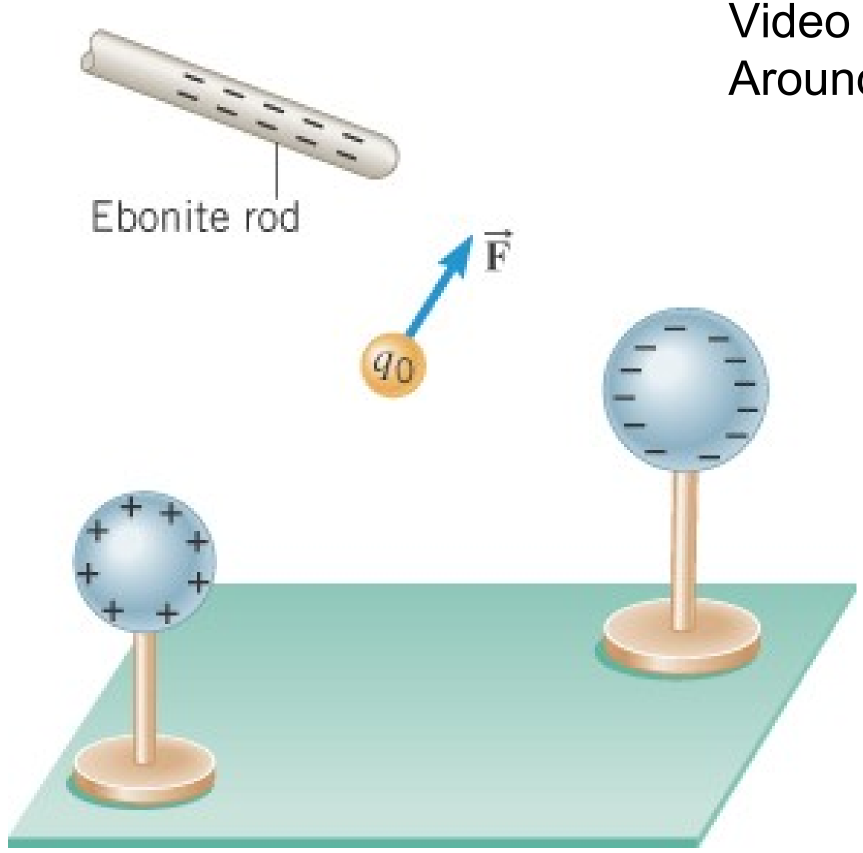
-

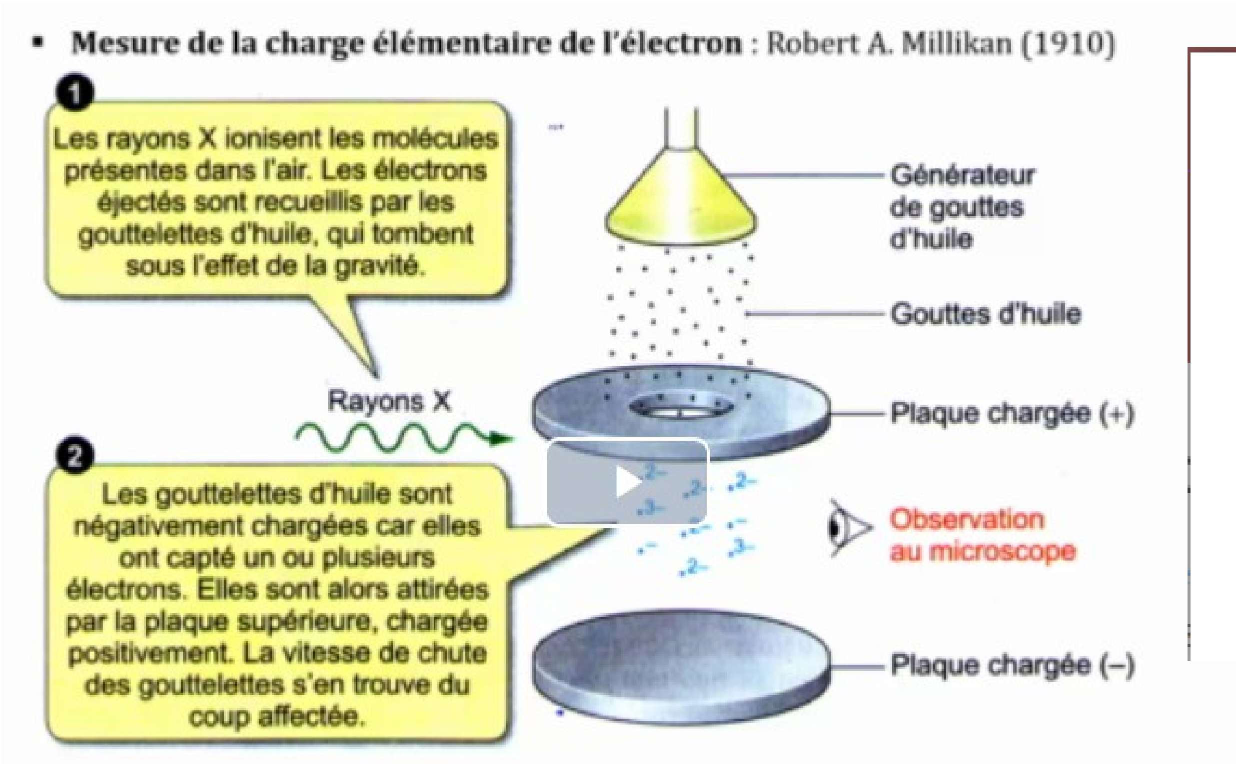


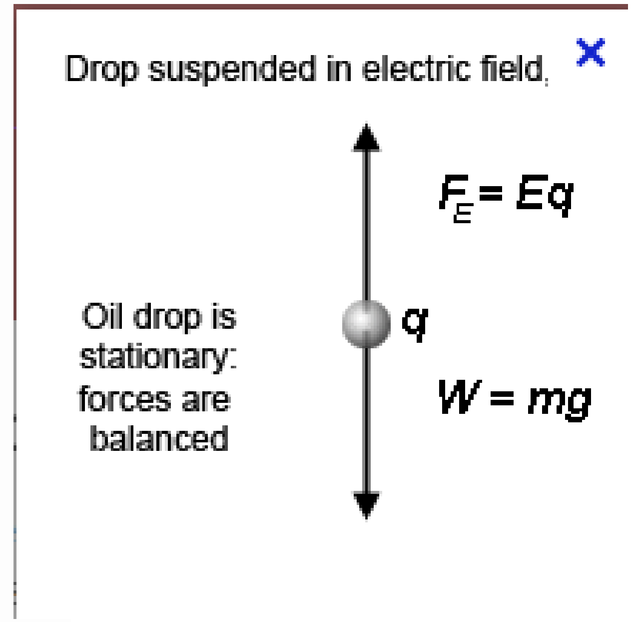
The positive charge experiences a force which is the vector sum of the forces exerted by the charges on the rod and the two spheres.

This ***test charge*** should have a small magnitude so it doesn't affect the other charge.

Video electric field lines
Around distribution of charges.





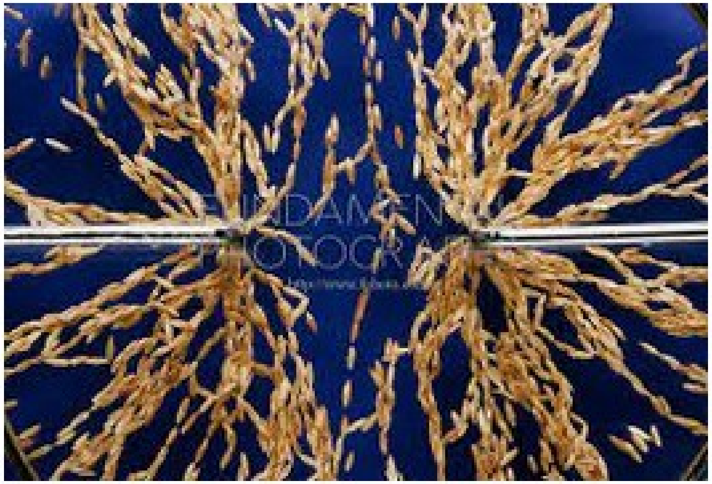


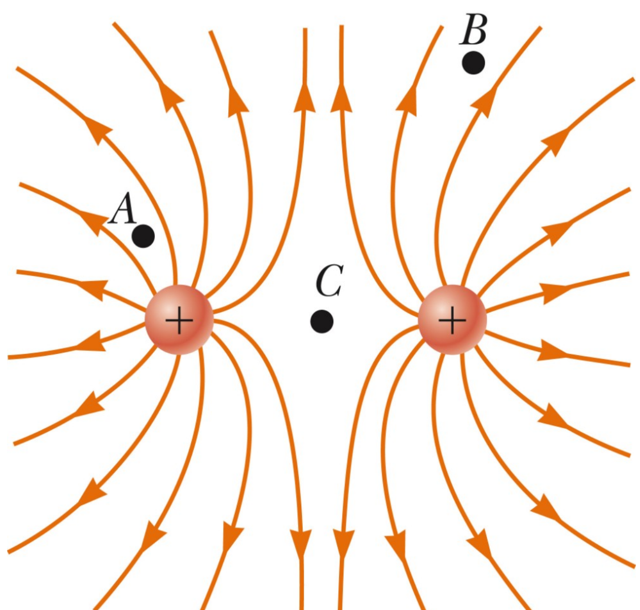
Experiment of Millikan

Millikan oil-drop experiment, first direct and compelling measurement of the electric charge of a single electron. ... Millikan was able to measure both the amount of electric force and magnitude of electric field on the tiny charge of an isolated oil droplet and from the data determine the magnitude of the charge itself.

In 1910 Robert Millikan succeeded in precisely determining the magnitude of the electron's charge.

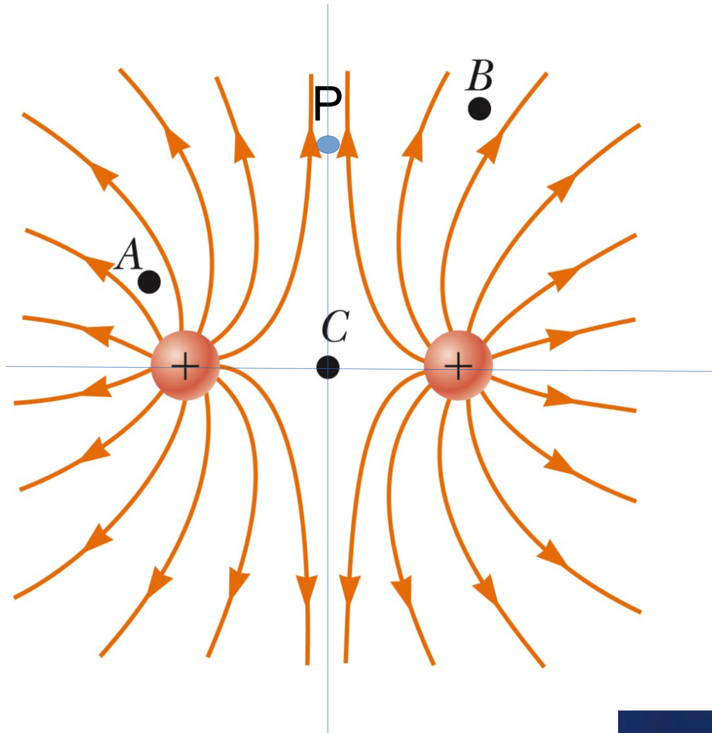
Other charges distribution





Show simulations MIT
With Van graaf

Video Walter Lewin



One charge q is at (-a,0) and one
Charge q at (a,0)

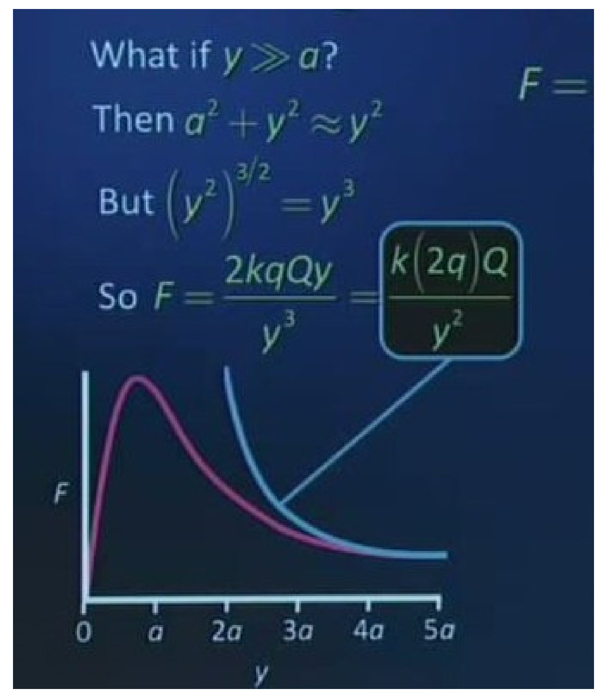
1) What is the electric field at C (0,0)?
Show with the simulation

2) The point P is at the location (0,y)
What is the electric field at P?
Magnitude and direction as a function
Of y, a , q.

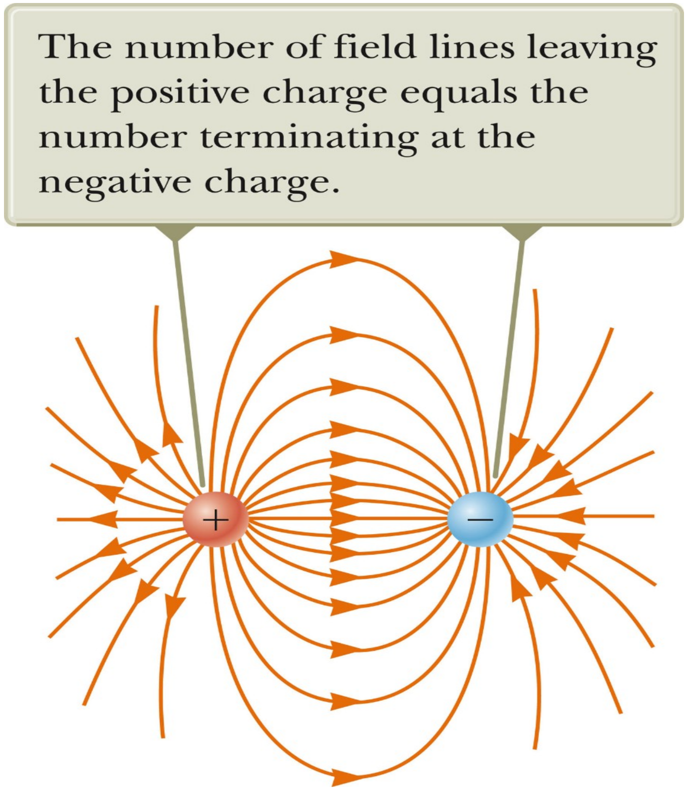
3) Show that E is (with Q=1C)
Graph E as a function of y

$$F=\frac{2kqQy}{\left(a^2+y^2\right)^{3/2}}$$

4) Show that for y>>a
Everything happens like we have
Only a 2q charge (point charge)



DIPOLE – Very important



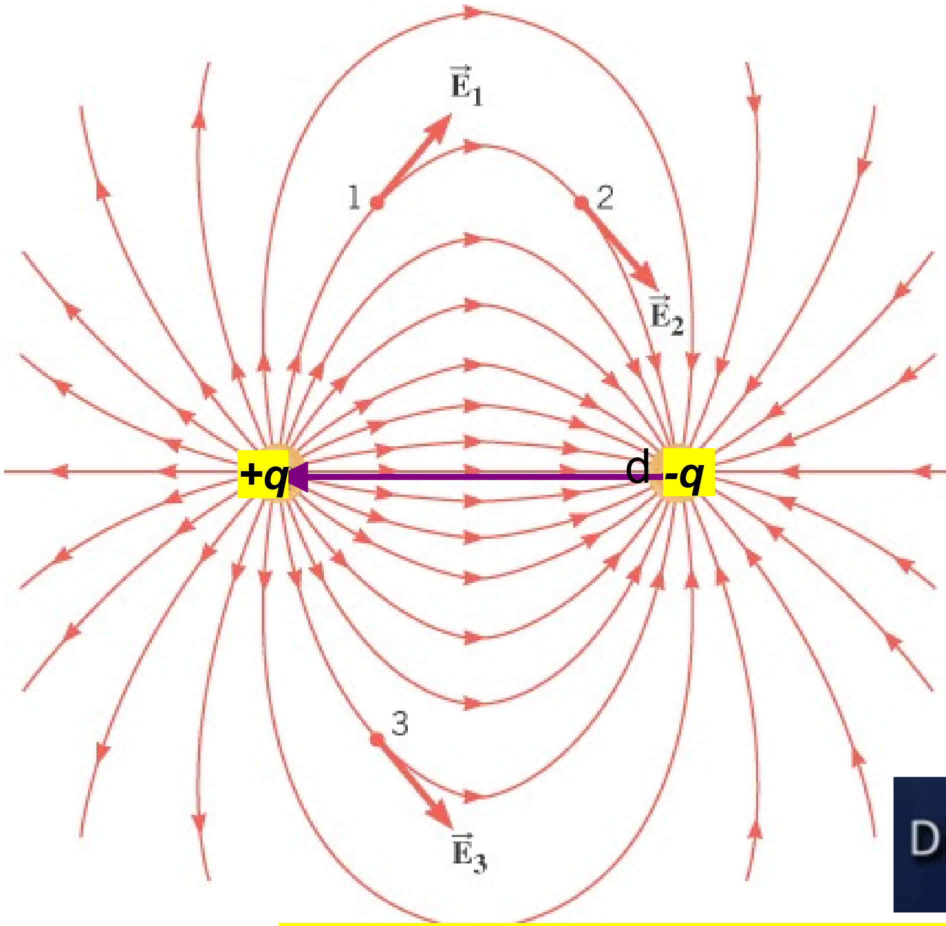
Demo grass seed (WL)

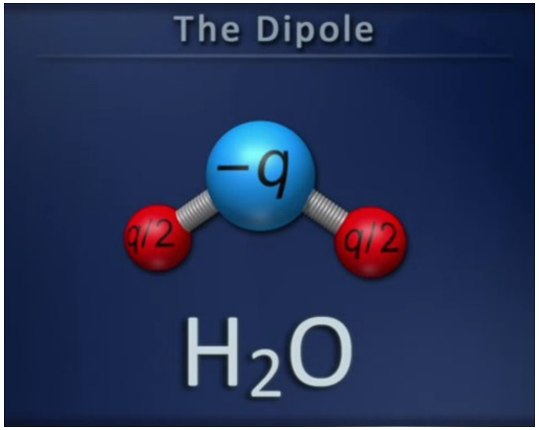
The number of lines leaving a positive charge or entering a negative charge is proportional to the magnitude of the charge.

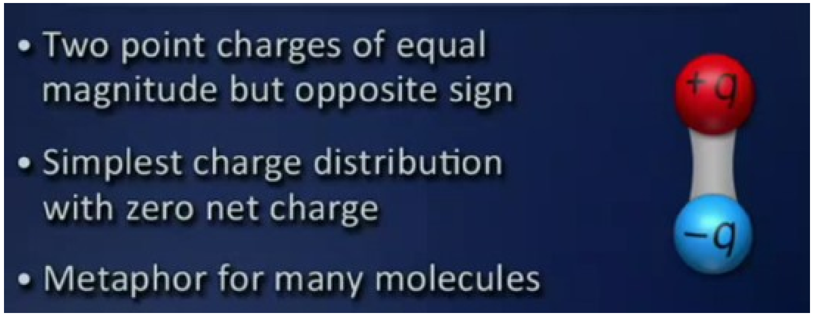
Electric dipole: Very important in Biology

2 charges with the same magnitude but opposite charge

The electric field goes from -   to +   . Same number of lines for q and -q







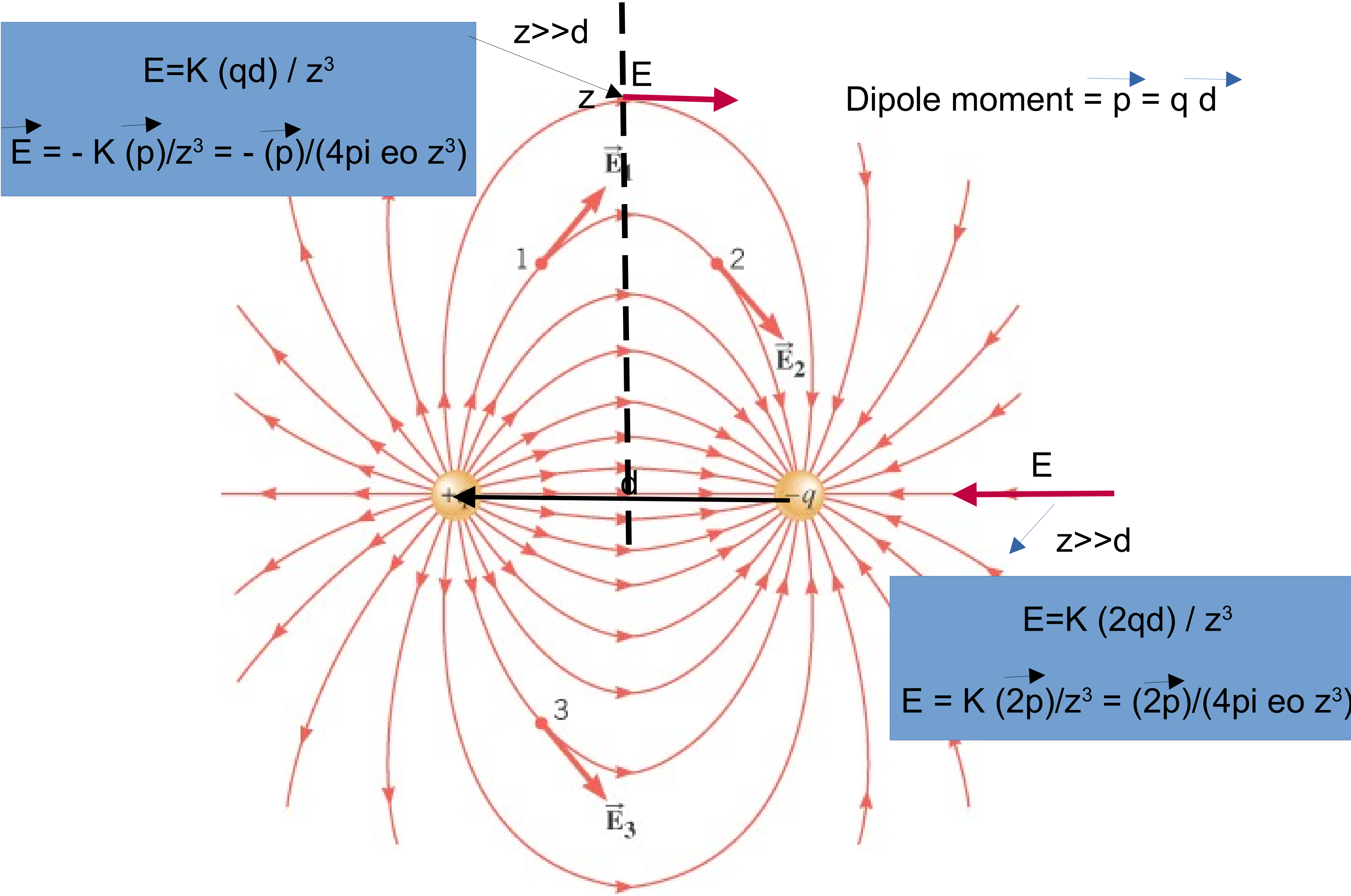
Show the vector p called electric dipole moment vector

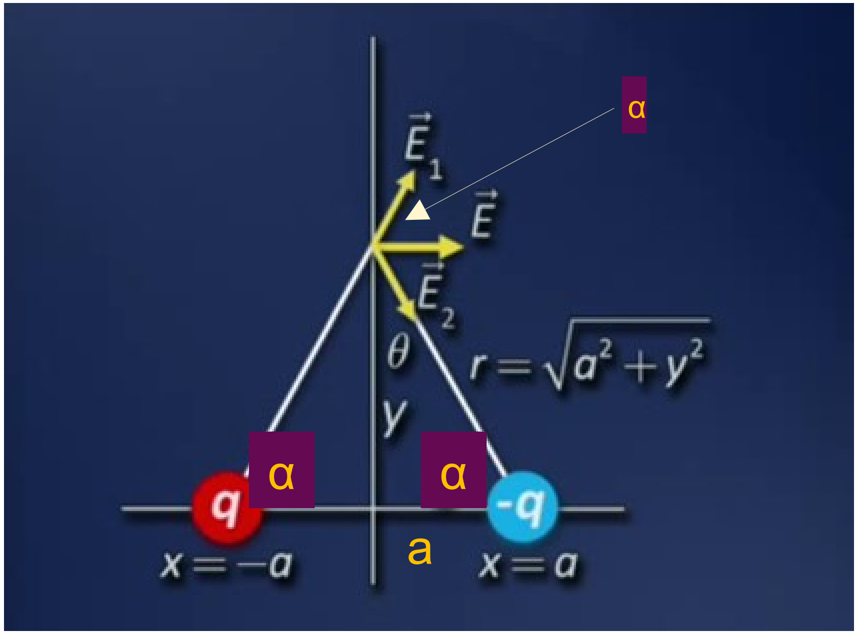
. Its a vector that goes from -q to +q. Unit is C. m

magnitude is |p| = q d and d is the distance between the charges.

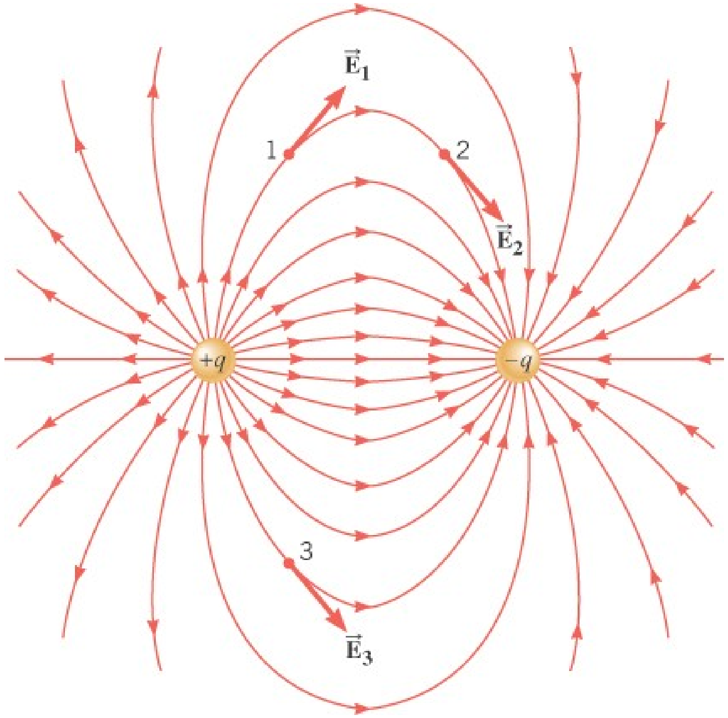
Discuss the different positions we can find the E.

See experiments seeds in castor oil. Use power supply 5KV or electrostatic machine.


E=K (qd) / z3
E = - K (p)/z3 = - (p)/(4pi eo z3)
z>>d
E
z
Dipole moment = p = q d
E1
1
2
E2
d
+q
-q
E
z>>d
E=K (2qd) / z3
E = K (2p)/z3 = (2p)/(4pi eo z3)
3
E3

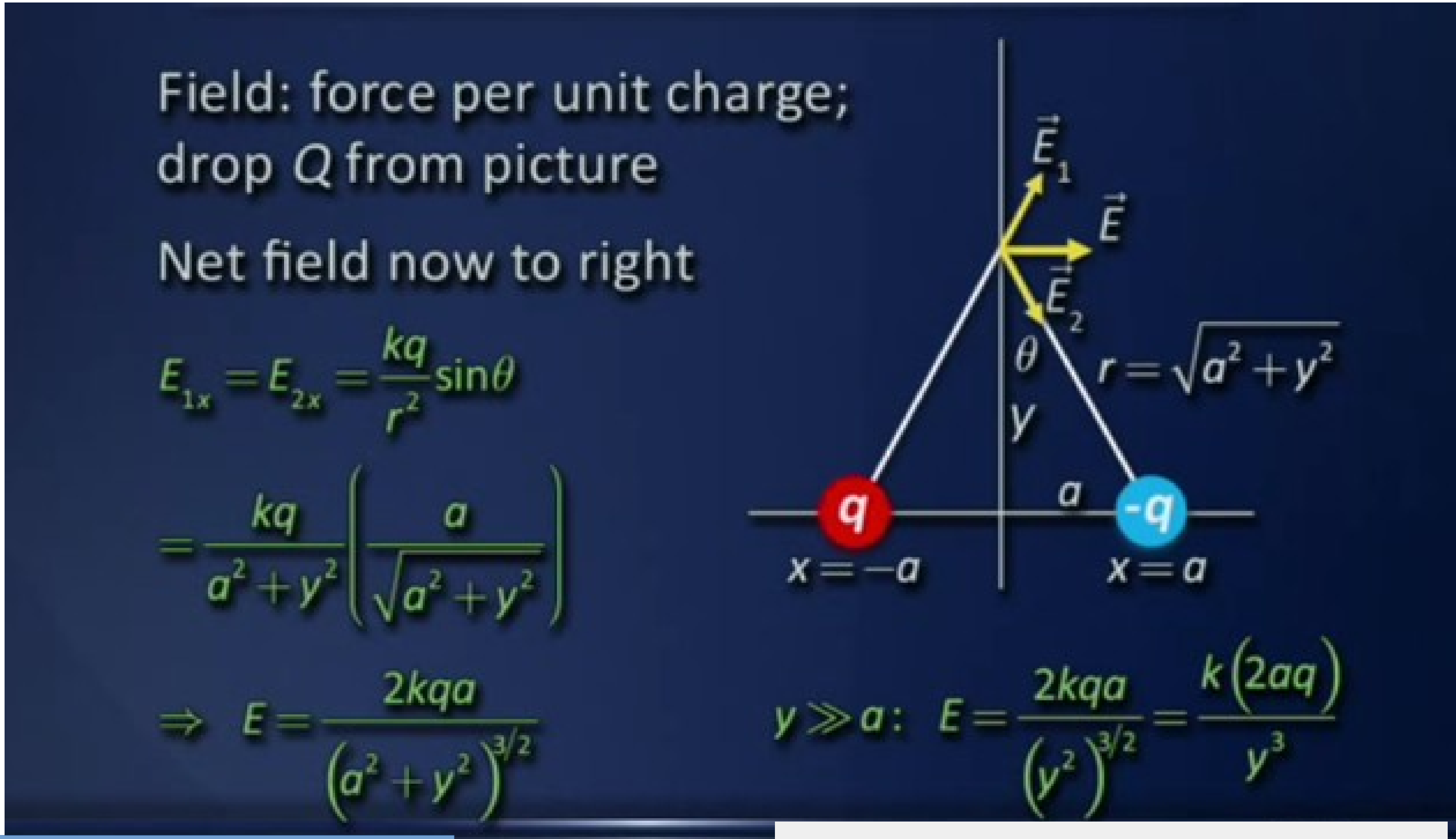


Find E at the distance y (as shown here)
What happens when y>>a



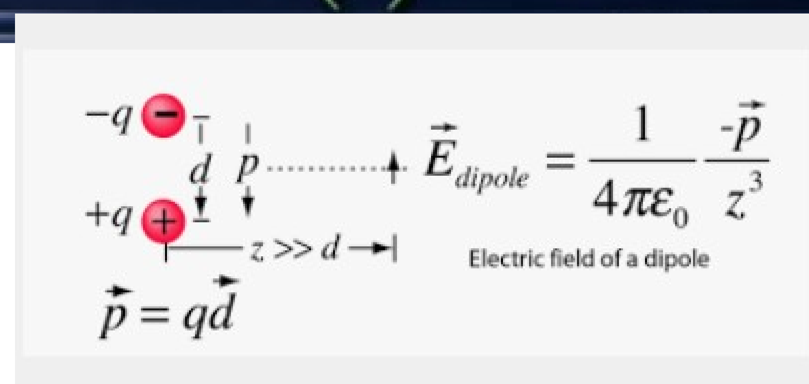
Discuss how the y-components cancel

Exnet = 2 x E cos (α) with E = E1= E2
cos(α) = a / sqrt (a^2 + y^2)



$\vec{E} = -K\,(\vec{p})/z^3 = -(\vec{p})/(4\pi\, \varepsilon_0\, z^3)$

Note: sin(theta) = cosine (alpha) → same thing.
E1x=E2x = E sin(theta) = E cos(alpha)
Also at large distance E (magnitude) decreases with distance^3



https://www.physics.udel.edu/~watson/phys208/binomial.html

Example: Electric Field of a Dipole

- Start with

$$E = E_+ - E_- = \frac{1}{4\pi\varepsilon_0}\frac{q}{r_+^2} - \frac{1}{4\pi\varepsilon_0}\frac{q}{r_-^2}$$
$$= \frac{1}{4\pi\varepsilon_0}\frac{q}{(z-d/2)^2} - \frac{1}{4\pi\varepsilon_0}\frac{q}{(z+d/2)^2}$$
$$= \frac{q}{4\pi\varepsilon_0 z^2}\left[\left(1-\frac{d}{2z}\right)^{-2} - \left(1+\frac{d}{2z}\right)^{-2}\right]$$

Taylor series
d/z <<1

- If d << z, then,

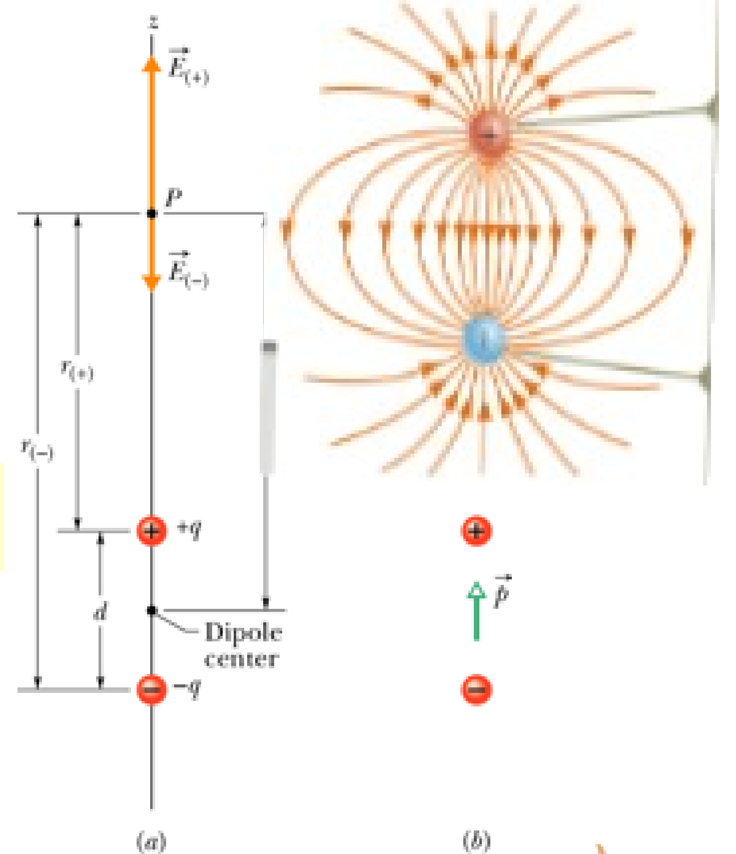
$$\left[\left(1-\frac{d}{2z}\right)^{-2} - \left(1+\frac{d}{2z}\right)^{-2}\right] = \left[\left(1+\frac{2d}{2z(1!)}+\ldots\right) - \left(1-\frac{2d}{2z(1!)}+\ldots\right)\right] \approx \frac{2d}{z}$$

- So

$$E = \frac{q}{4\pi\varepsilon_0 z^2}\frac{2d}{z} = \frac{1}{2\pi\varepsilon_0}\frac{qd}{z^3}$$
$$p = qd$$

$\vec{E} = K\,(2\vec{p})/z^3 = (2\vec{p})/(4\pi\, e_0\, z^3)$

- ✓ $E \sim 1/z^3$
- ✓ E =>0 as d =>0
- ✓ Valid for "far field"



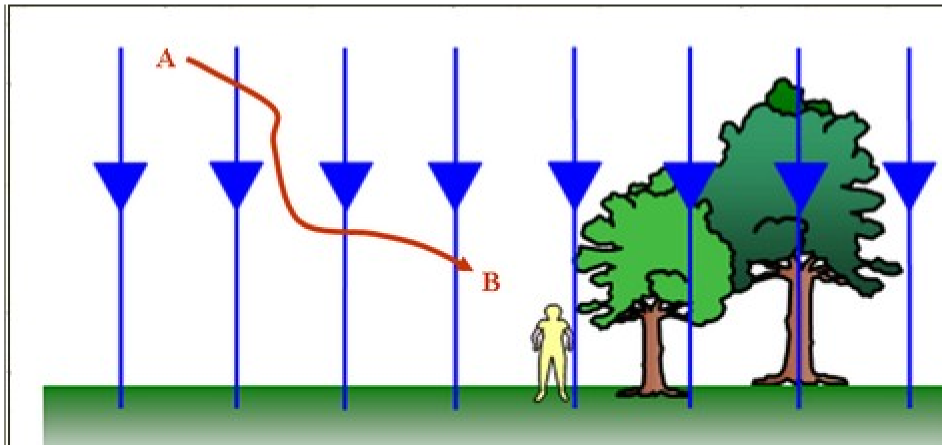
Remember Ke= 1/ (4pi e0)
So E = 2 K qd / z^3 = 2 K p /z^3 if z>> d
Magnitude here

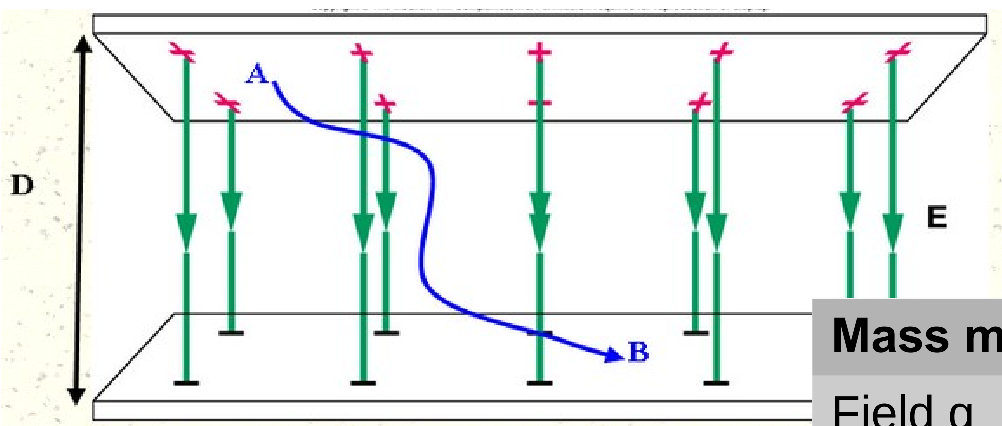
See example 21.14
The electric field decreases with distance^3
Same as a small loop of current and B.

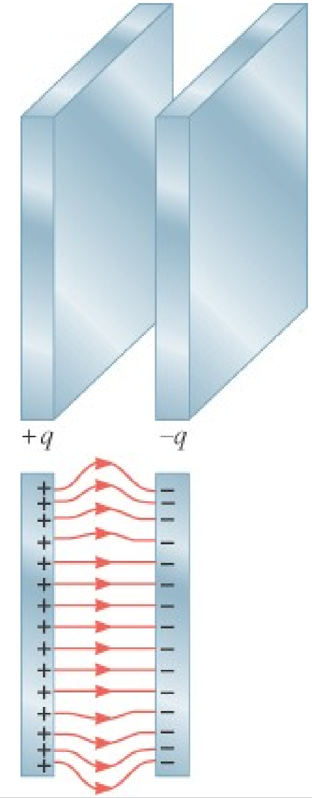
- Far from a dipole, you measure an electric field strength of 800 N/C. If you double your distance from the dipole, what will the electric field strength be at your new location?

   A. 400 N/C
   B. 200 N/C
   C. 100 N/C
   D. 50 N/C

18.7 *Electric Field Lines*





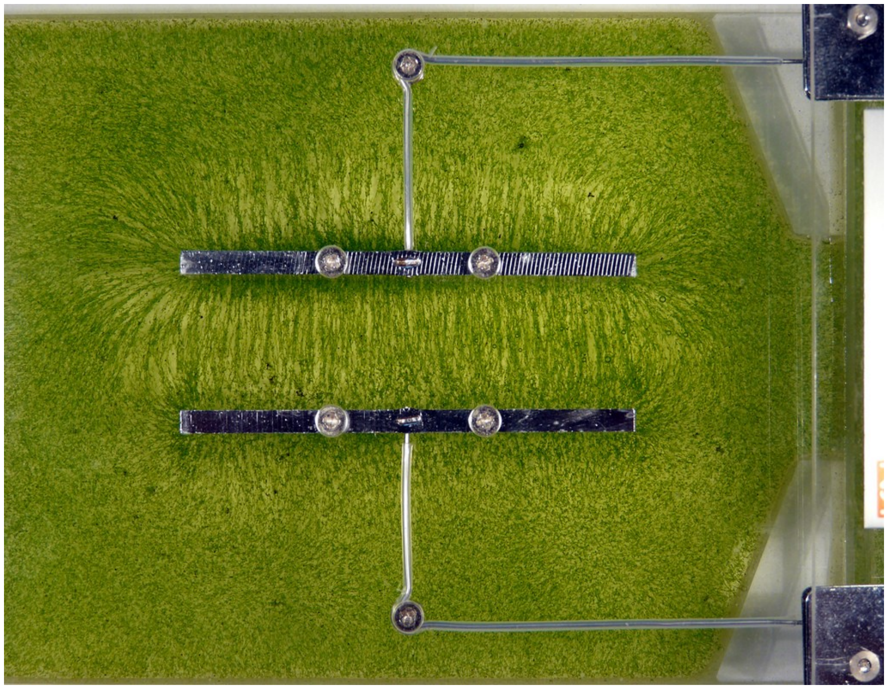


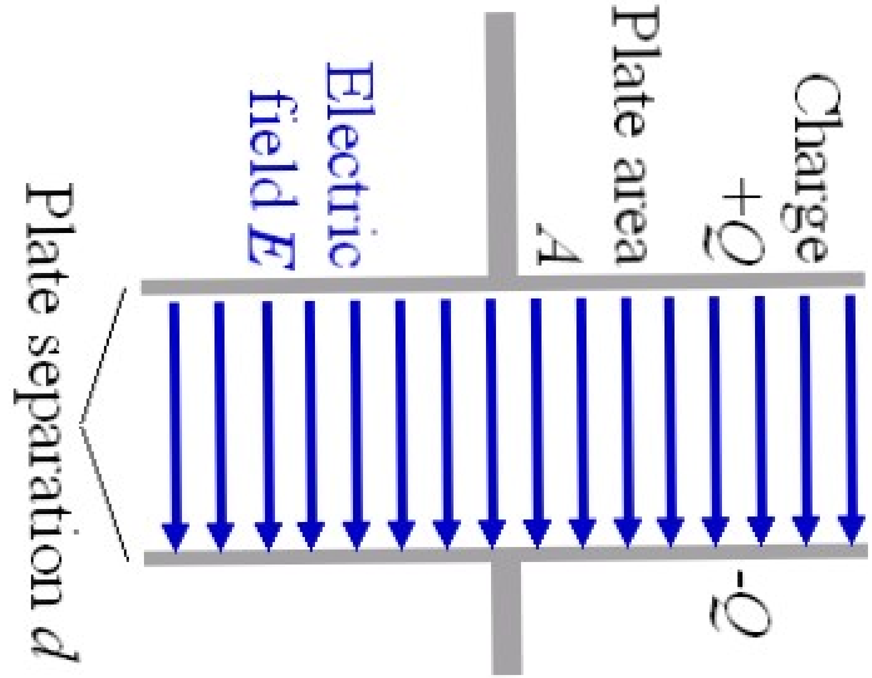
Mass m	**Charge q**
Field g	Field E
Weight = mg	Fe= qE
g is the force per unit mass	E is the force per unit charge

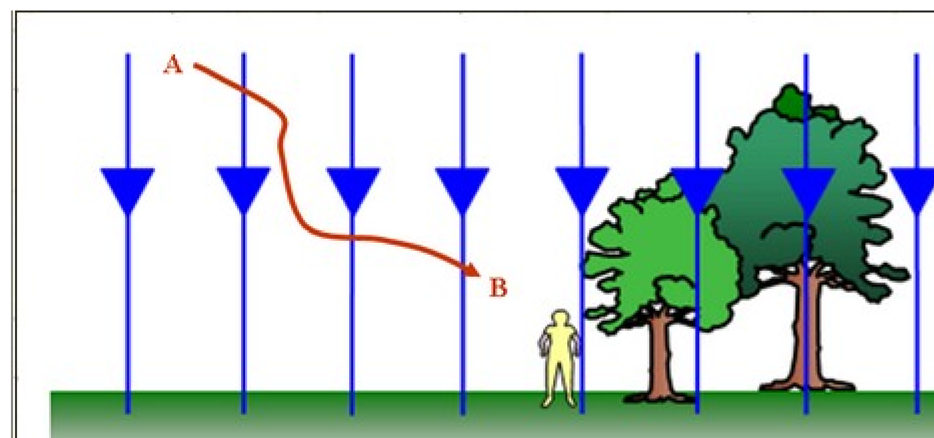
Electric field lines always begin on a positive charge and end on a negative charge and do not stop in midspace. The electric force is conservative.
The work done by us from B to A only depends on the initial and final positions. Not on the path.

Uniform field between 2 plates

→ capacitor
→ same direction and magnitude
→ like the field close to Earth surface







Uniform field between 2 plates

→ capacitor
→ same direction and magnitude
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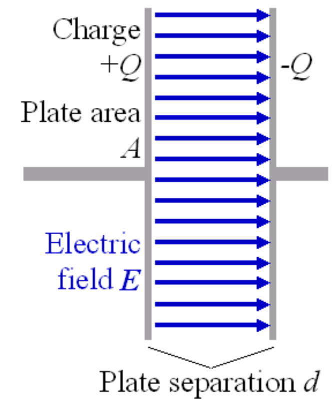
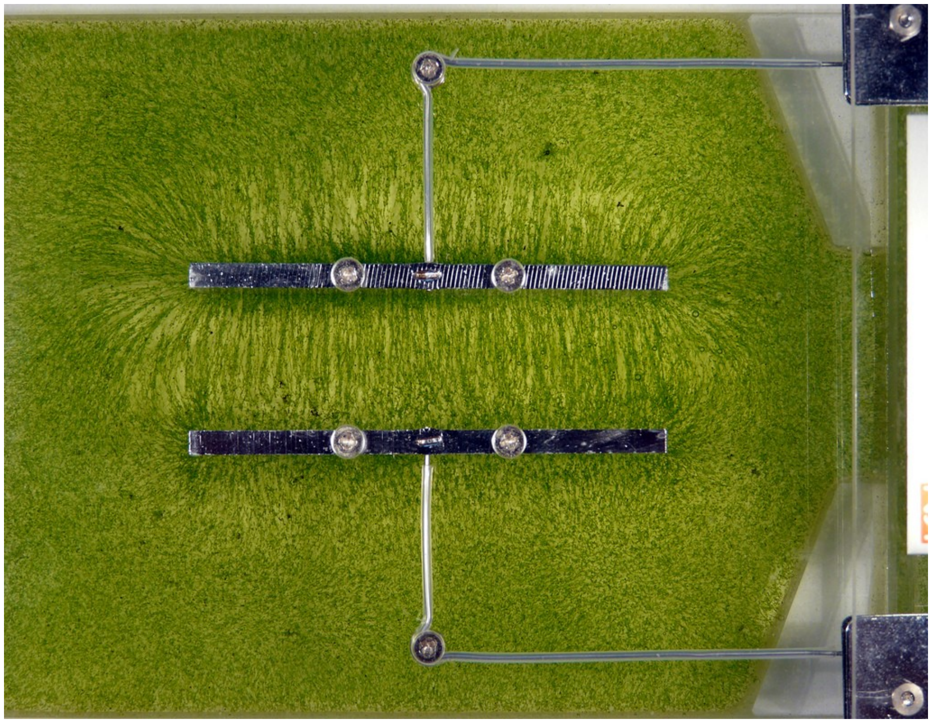
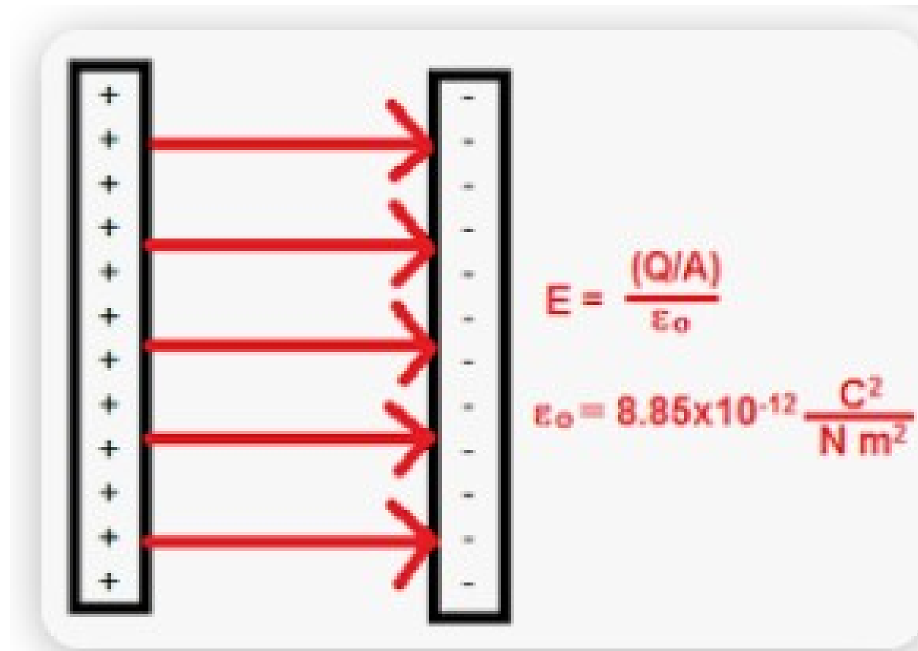
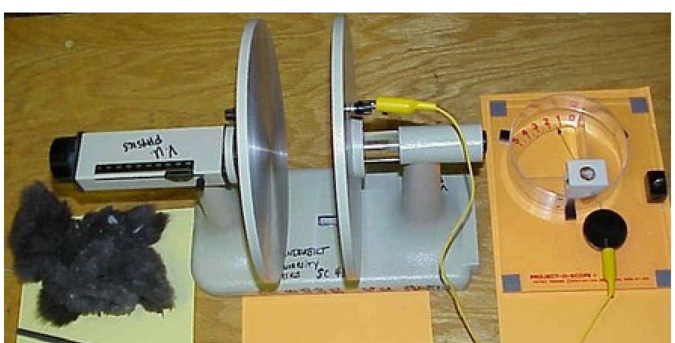








Video
Living cap WL

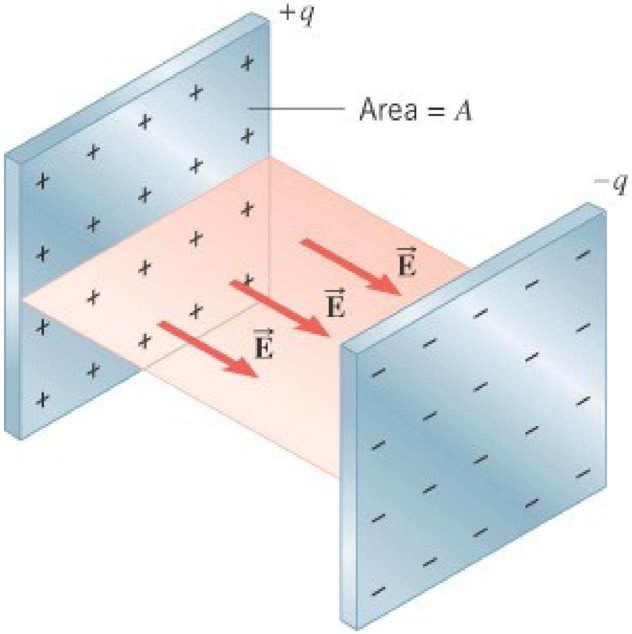
$e0=8.85 \times 10^{-12}$

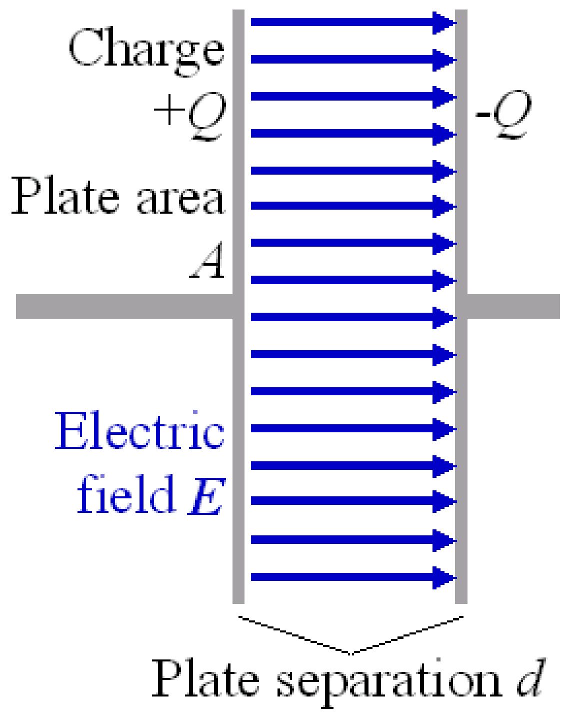
E = Q/(Ae0)   with σ = Q/A
(surface charge density).

3) Two parallel square metal plates that are 1.5 cm apart and 22 cm on each side carry equal but opposite charges uniformly spread out over their facing surfaces. How many excess electrons are on the negative surface if the electric field between the plates has a magnitude of 18,000 N/C? ($k = 1/4\pi\varepsilon_0 = 9.0 \times 10^9\ \mathrm{N \cdot m^2/C^2}$, $e = 1.6 \times 10^{-19}$ C)

Hint: first find Q . Use E = density/eo

Then Q = N e (charge is quantized).





Remember projectile motion??

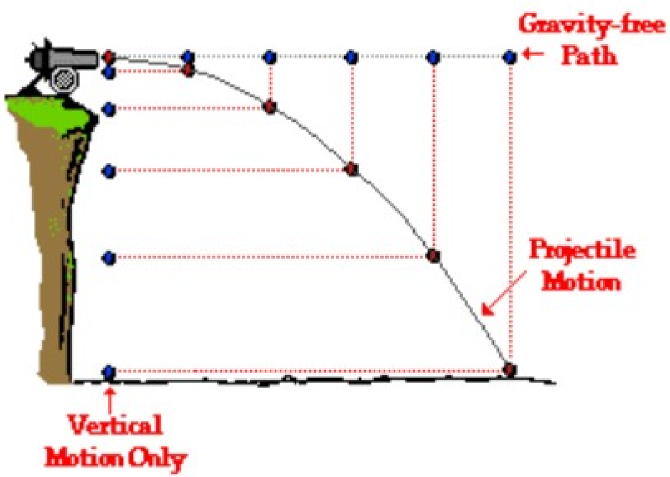
http://www.phy.hk/wiki/j/Eng/projectile/projectile_js.htm

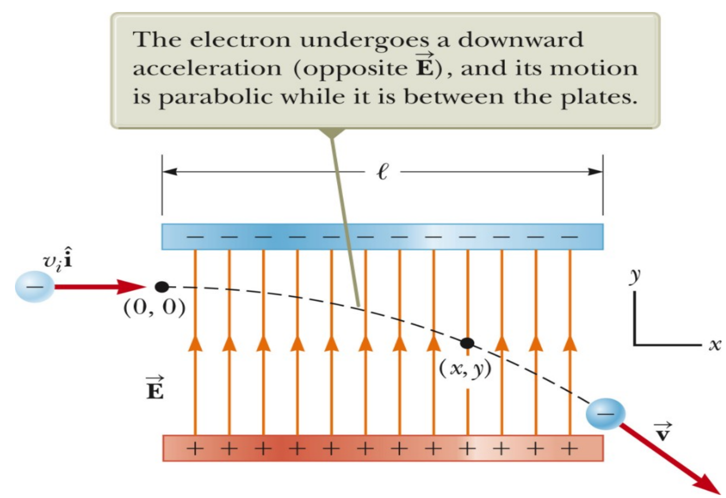
Kinematic Equations for Constant Acceleration in Two Dimensions

$v_x = v_{x0} + a_x t$	$v_y = v_{y0} + a_y t$
$x = x_0 + v_{x0} t + \frac{1}{2} a_x t^2$	$y = y_0 + v_{y0} t + \frac{1}{2} a_y t^2$
$v_x^2 = v_{x0}^2 + 2a_x(x - x_0)$	$v_y^2 = v_{y0}^2 + 2a_y(y - y_0)$

Kinematic Equations for Projectile Motion

$v_x = v_{x0}$	$v_y = v_{y0} - gt$
$x = x_0 + v_{x0} t$	$y = y_0 + v_{y0} t - \frac{1}{2} g t^2$
	$v_y^2 = v_{y0}^2 - 2g(y - y_0)$





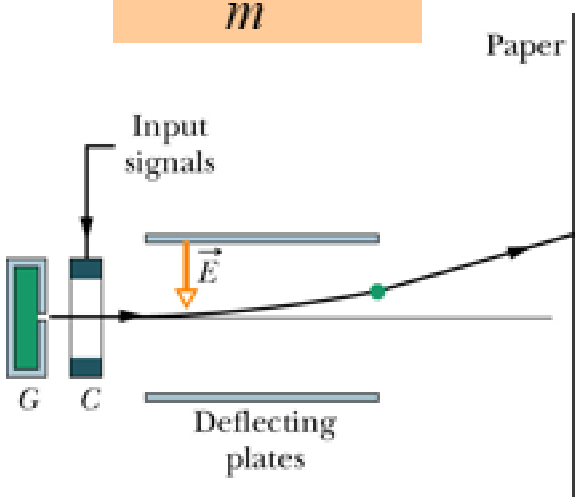
Motion of a Charged Particle in a Uniform Electric Field

See JJ Thomson experiment

$$\vec{F} = q\vec{E}$$

$$\vec{F} = q\vec{E} = m\vec{a}$$

$$\vec{a} = \frac{q\vec{E}}{m}$$

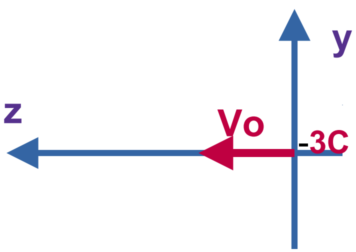


- If the electric field E is uniform (magnitude and direction), the electric force F on the particle is constant.
- If the particle has a positive charge, its acceleration a and electric force F are in the direction of the electric field E.
- If the particle has a negative charge, its acceleration a and electric force F are in the direction opposite the electric field E.

Do 21.7 in class

4. (moderate) A charged particle (-3.0C with a mass of 0.0002 kg) is injected into an E-field with an initial speed of 2000 m/s along the +z axis. The E-field is uniform in this region (500 N/C), and directed in the +y direction. Determine the acceleration components for all three directions (x,y, and z). Additionally, calculate the length of time needed to the particle to move $1x10^8$ m in the -y direction and the distance moved along the other two axes over that time frame. Assume that the initial position of the particle is at the origin of the axis system.

First draw the system (X,Y,Z). X is toward you. (use the 3 fingers)



Along X and Z there is no force → no acceleration
Along Y there is an acceleration = qE/mass.

Draw the motion of the charge (like projectile , parabola)

Use kinematics

What is the final velocity? Vy = (a)(t)   Vx=0   Vy=2000m/s
Use Pythagorean theorem to get V.

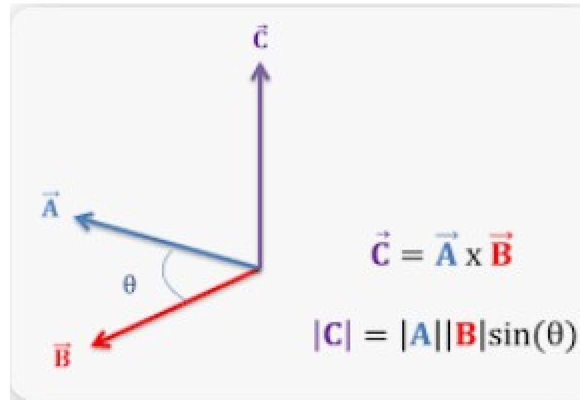
The Calculus part ..
To find the electric field of a continuous distribution of charges

Unit vector

$$\vec{F} = \frac{1}{4\pi\varepsilon_0}\frac{Qq_0}{r^2}\hat{r}$$

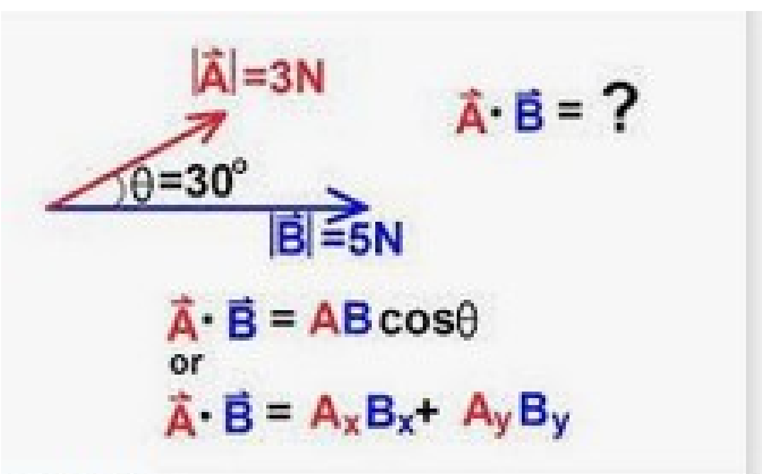
$$\vec{E} = \frac{\vec{F}}{q_0} = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}\hat{r}$$

Cross product



$$\det\begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix} = a_1\begin{vmatrix} b_2 & b_3 \\ c_2 & c_3 \end{vmatrix} - a_2\begin{vmatrix} b_1 & b_3 \\ c_1 & c_3 \end{vmatrix} + a_3\begin{vmatrix} b_1 & b_2 \\ c_1 & c_2 \end{vmatrix}$$

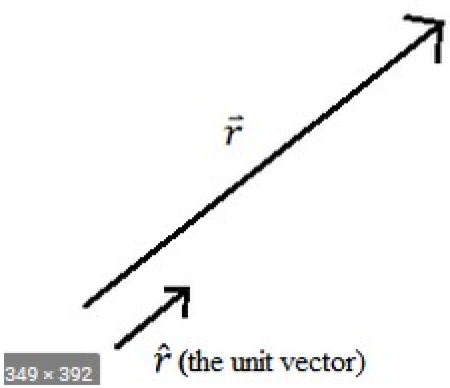
Dot product



$$\vec{r} = (3,4,10)$$

$$|\vec{r}| = \sqrt{3^2 + 4^2 + 10^2} = 11.18$$

$$\boxed{\hat{r} = \frac{\vec{r}}{|\vec{r}|} = \frac{(3,4,10)}{11.18} = (0.27, 0.35, 0.89)}$$



A unit vector is a vector with magnitude 1.

To find a unit vector, **u**, in the same direction of a vector, **v**, we divide the vector by its magnitude.

$$\vec{u} = \frac{\vec{v}}{\|\vec{v}\|} = \frac{1}{\|\vec{v}\|}\vec{v}$$

R = 11.18 r

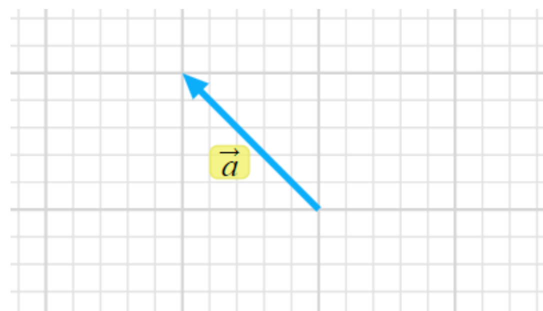
- A charge q_1 is located at (x, y)=(1 m, 0). What should you use for the unit vector $\vec{r}$ in Coulomb's law if you are calculating the force that q_1 exerts on charge q_2 located at the point (x, y)=(0, 1 m)?

A. $-\frac{\sqrt{2}}{2}\hat{i}-\frac{\sqrt{2}}{2}\hat{j}$

B. $\frac{\sqrt{2}}{2}\hat{i}+\frac{\sqrt{2}}{2}\hat{j}$

C. $-\frac{\sqrt{2}}{2}\hat{i}-\frac{\sqrt{2}}{2}\hat{j}$

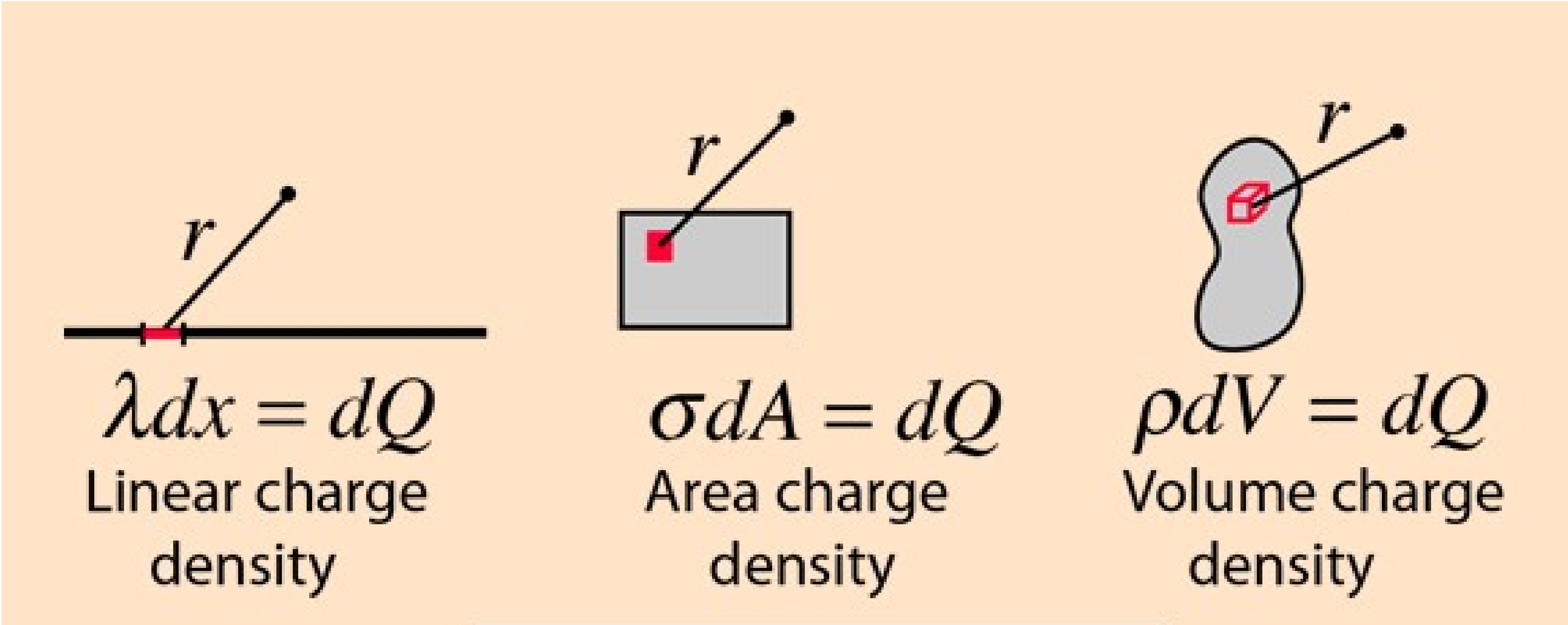
D. $-\frac{\sqrt{2}}{2}\hat{i}+\frac{\sqrt{2}}{2}\hat{j}$

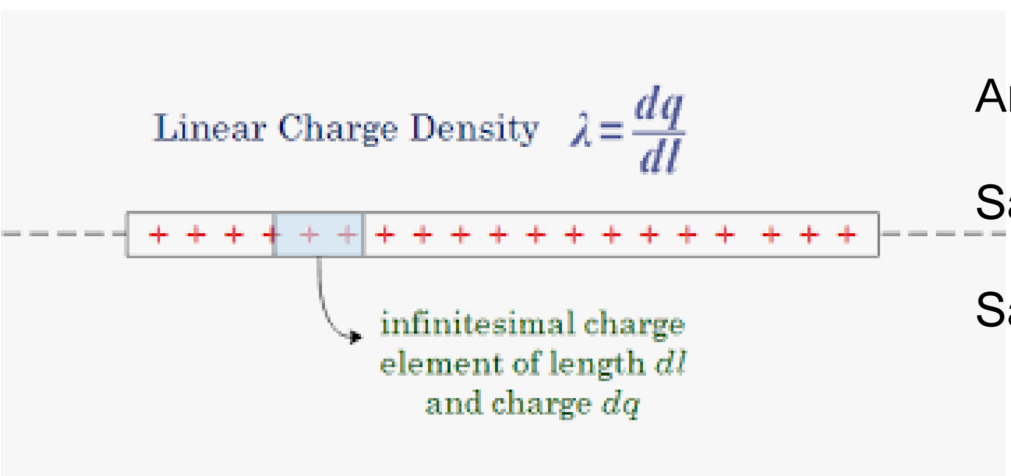


$$\hat{r} = \frac{\vec{r}}{|\vec{r}|}$$

https://phet.colorado.edu/sims/html/vector-addition/latest/vector-addition_en.html

Charge density → charge per unit element of length/area/volume
→ calculus III

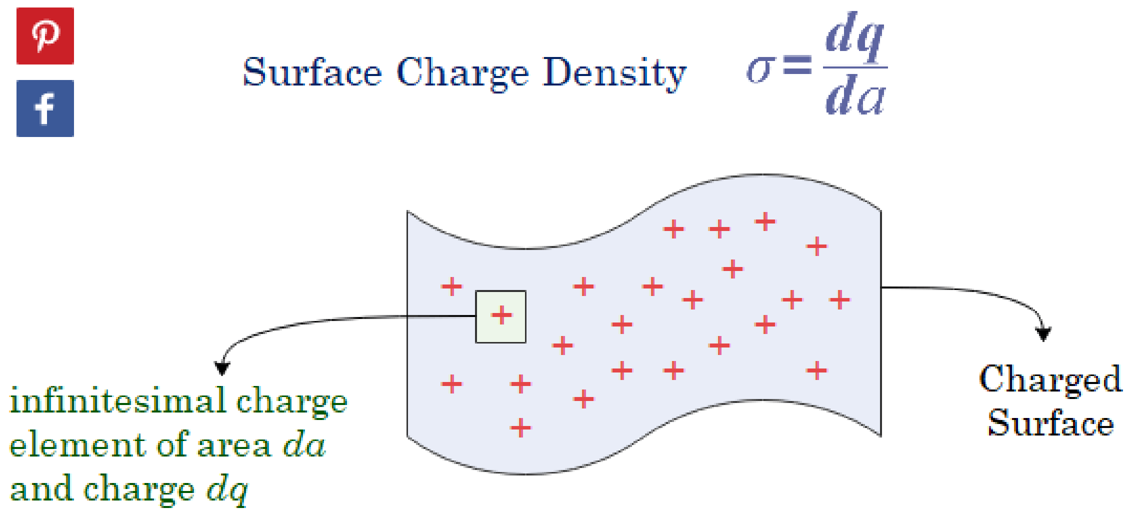


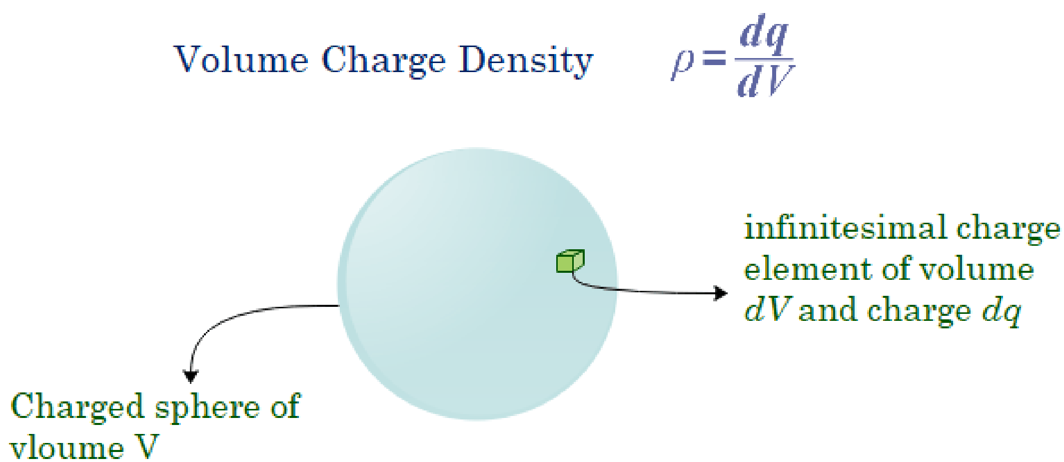


And λ= Qtotal / Ltotal

Same for area σ = Qtotal / Area total

Same for volume ρ = Qtotal/ Volume total

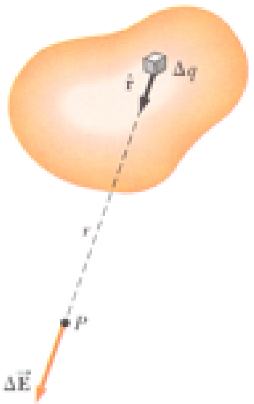




https://physicscatalyst.com/article/charge-density-formulas-and-solved-example/

Electric Field of a Continuous Charge Distribution

$$\Delta \vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{\Delta q}{r^2}\hat{r}$$



$$\vec{E} \approx \frac{1}{4\pi\varepsilon_0}\sum_i \frac{\Delta q_i}{r_i^2}\hat{r}_i$$

$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\lim_{\Delta q\to 0}\sum_i \frac{\Delta q_i}{r_i^2}\hat{r}_i = \frac{1}{4\pi\varepsilon_0}\int \frac{dq}{r^2}\hat{r}$$

- Find an expression for dq:
  - $dq = \lambda dl$ for a line distribution
  - $dq = \sigma dA$ for a surface distribution
  - $dq = \rho dV$ for a volume distribution
- Represent field contributions at P due to point charges dq located in the distribution. Use symmetry,

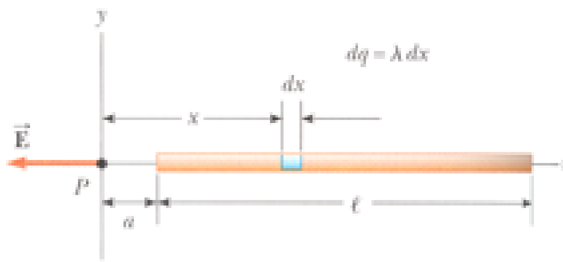
$$d\vec{E} = \frac{dq}{4\pi\varepsilon_0 r^2}\hat{r}$$

- Add up (integrate the contributions) over the whole distribution, varying the displacement as needed,

$$\vec{E} = \int dE$$

Example: Electric Field Due to a Charged Rod

- A rod of length l has a uniform positive charge per unit length λ and a total charge Q. Calculate the electric field at a point P that is located along the long axis of the rod and a distance a from one end.



- Start with

$$dq = \lambda dx$$

- then,

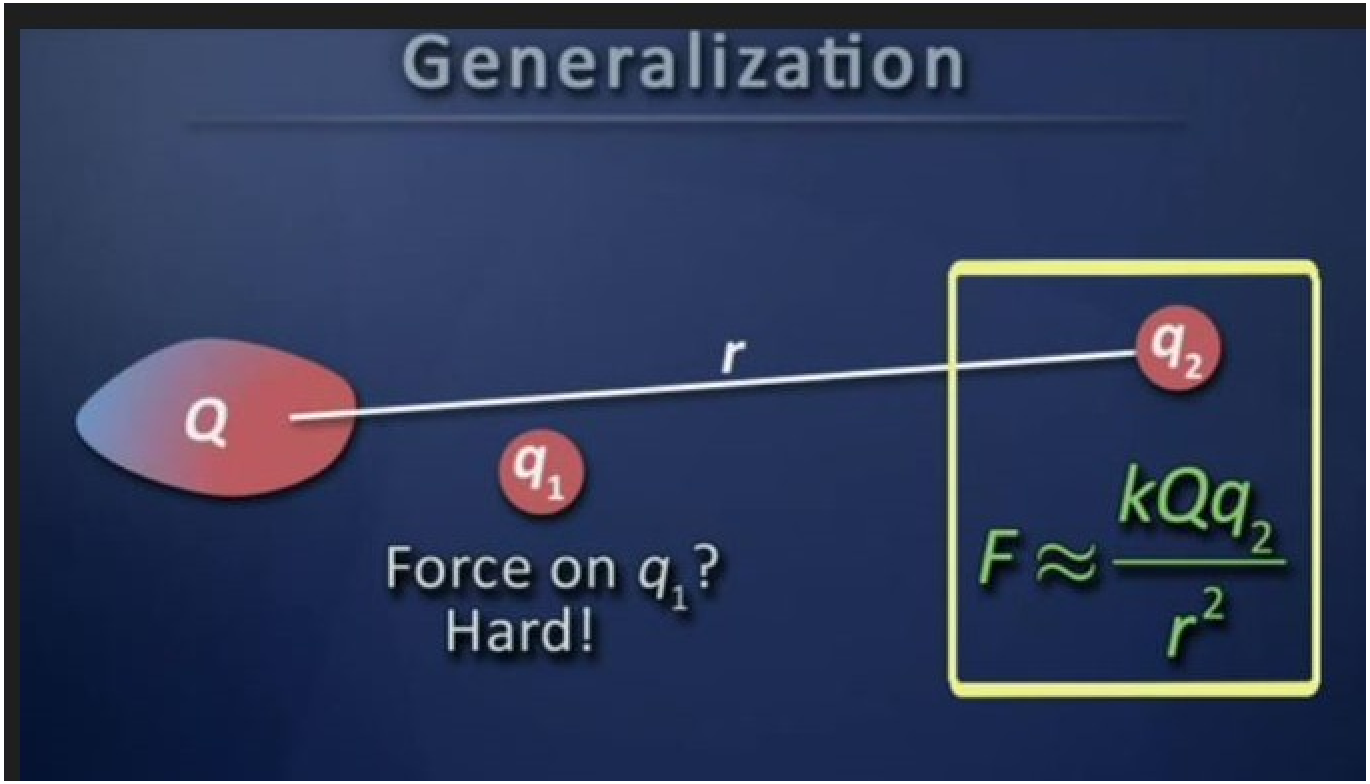
$$dE = \frac{1}{4\pi\varepsilon_0}\frac{dq}{x^2} = \frac{1}{4\pi\varepsilon_0}\frac{\lambda dx}{x^2}$$

- So

$$E = \int_a^{l+a} \frac{\lambda}{4\pi\varepsilon_0}\frac{dx}{x^2} = \frac{\lambda}{4\pi\varepsilon_0}\int_a^{l+a}\frac{dx}{x^2} = \frac{\lambda}{4\pi\varepsilon_0}\left[-\frac{1}{x}\right]_a^{l+a}$$

$$E = \frac{1}{4\pi\varepsilon_0}\frac{Q}{l}\left(\frac{1}{a} - \frac{1}{l+a}\right) = \frac{Q}{4\pi\varepsilon_0 a(l+a)}$$

- Finalize
  - l => 0 ?
  - a >> l?



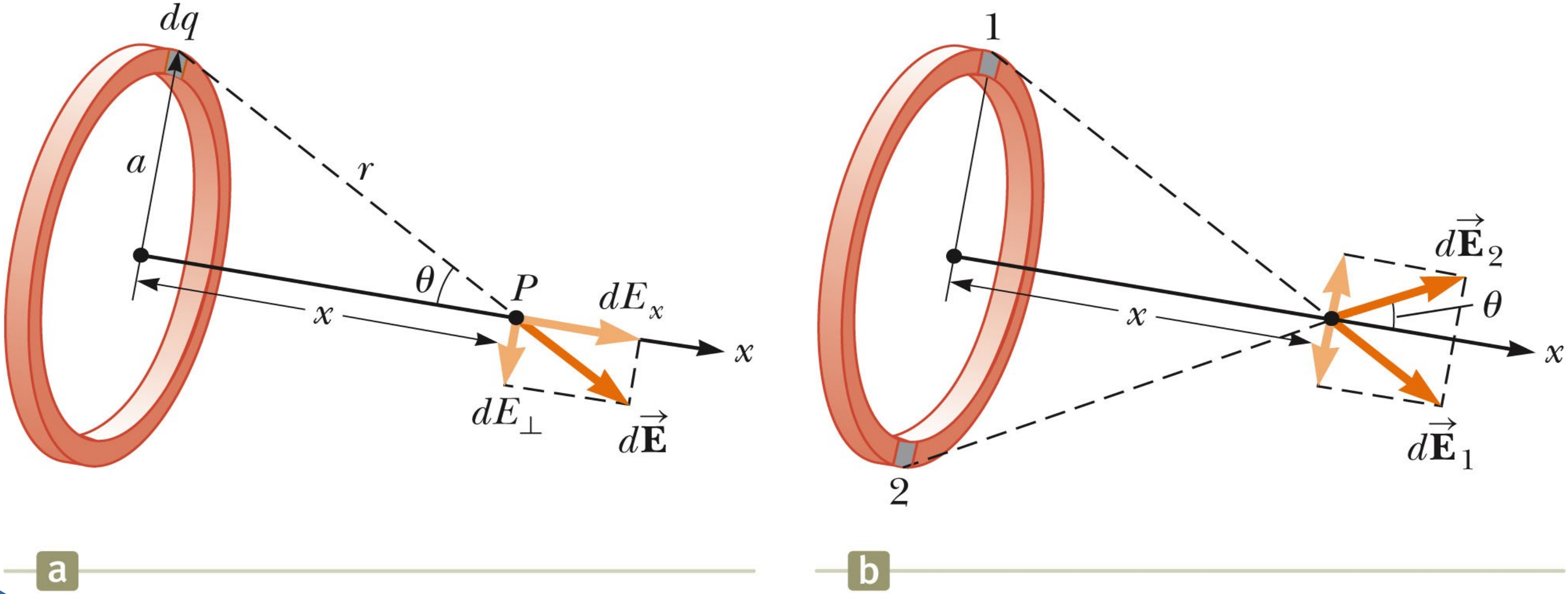
At a large distance, a “weird” distribution of charge → one point charge
At infinity (like the previous wire)

6. A ring of charge → $\overrightarrow{\text{Enet}}$ = vector sum ($\overrightarrow{\text{dE}}$)

Ex = Sum(dEx) scalar sum

Ey = Sun(dEy) = 0 scalar sum

$\overrightarrow{\text{Enet}}$ = Sum(dEx) $\vec{\text{x}}$



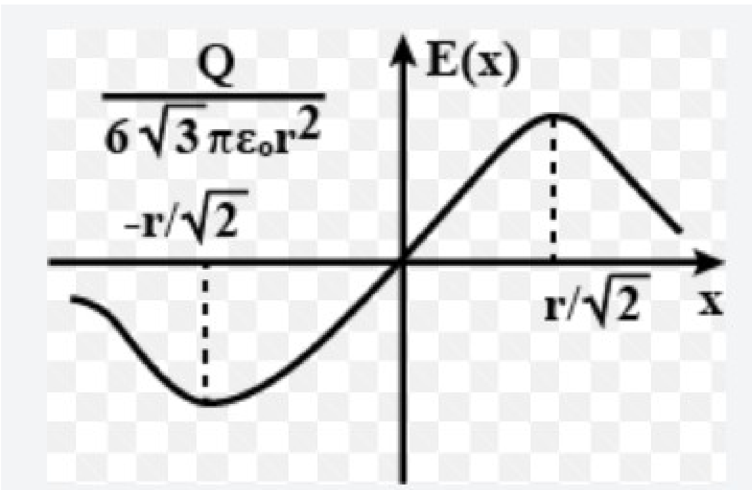
$\overrightarrow{\text{Enet}}$ = $\dfrac{kxQ}{\left(x^2+a^2\right)^{3/2}}\ \vec{x}$

If x>>a the electric field is like the one
From a point charge. Large distance.

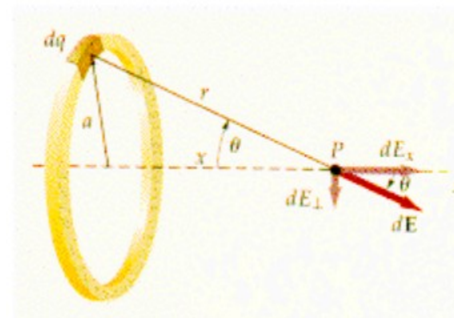
This is an easy one,
Its a 1D distribution.

http://www.physics.udel.edu/~watson/phys208/exercises/kevan/efield1.html

Electric field of a ring as a function of distance
Use wolfram alpha to see the shape of the graph
(plot x / (x^2+1)^1.5 with a = 1. radius = 1m)



To find the Max of E →
Take the derivative =0



Draw the electric field along the axis
As vectors. The length is proportional to
The magnitude E.

https://byjus.com/question-answer/1-what-is-the-electric-field-vs-radius-graph-in-a-ring/

30) A very long wire carries a uniform linear charge density of 7.0 nC/m. What is the electric field strength 16.0 m from the center of the wire at a point on the wire's perpendicular bisector? ($\varepsilon_0 = 8.85 \times 10^{-12}\ \mathrm{C^2/N \cdot m^2}$)

Hint: first draw. Suppose the wire is along the x-axis. You want to find E at
A distance d (16m) from the wire (along bisector). You see the symmetry.
There is no x-component (the x-components cancel each other – symmetry)
So dE = dEy (magnitude) = K dq / (x^2 + d^2) sin(a)
With sqrt((x^2 + d^2) is the distance between dx and the point P (where
E is computed). When you square the distance, the square root is eliminated
sin(a) = d/sqrt((x^2 + d^2)   do drawing.

You have dq = (charge density) . dx

So dE = K (charge density) (d) dx / (sqrt((x^2 + d^2))^1.5   note: 1 + 0.5 = 1.5

integral of 1/(x^2+16^2)^1.5 between - infinity to + infinity

Then integrate over x.
To find the integral, use wolfram alpha with d = 16m

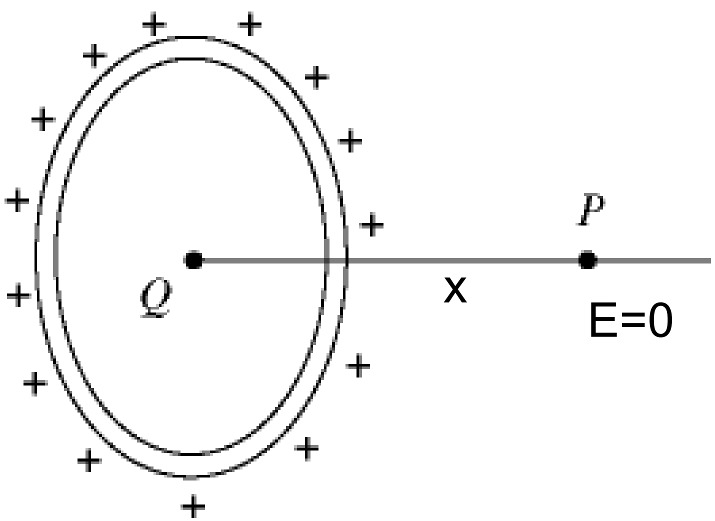
Note: sqrt((x^2 + d^2)) (x^2 + d^2) = (x^2 + d^2)^1.5

https://www.youtube.com/watch?v=R7yZ9XcrPrU   step by step tutorial

q_{ring}

a

33) In the figure, a ring 0.71 m in radius carries a charge of + 580 nC uniformly distributed over it. A point charge Q is placed at the center of the ring. The electric field is equal to zero at field point P, which is on the axis of the ring, and 0.73 m from its center. ($\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N \cdot m^2$) The point charge Q is closest to



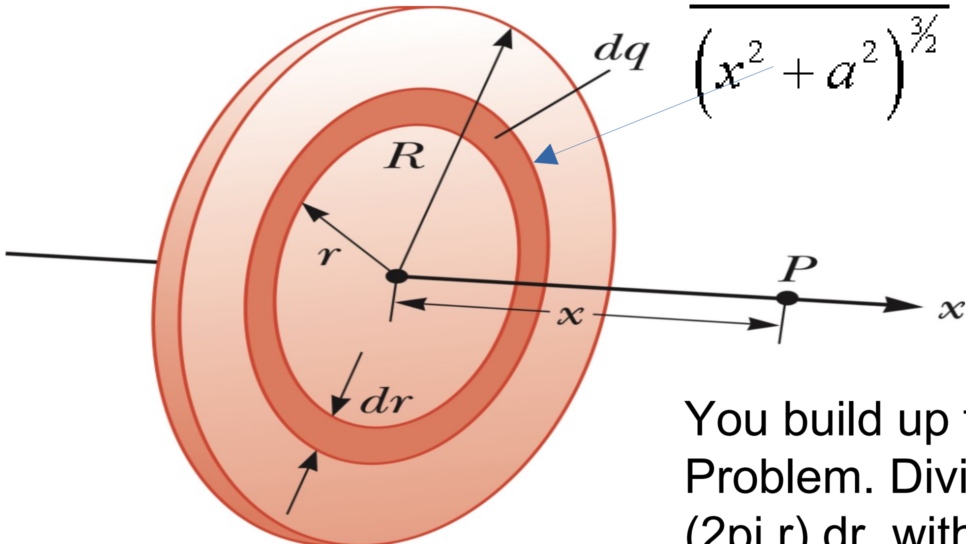
From previous slide we have for a charged ring and use the superposition principle

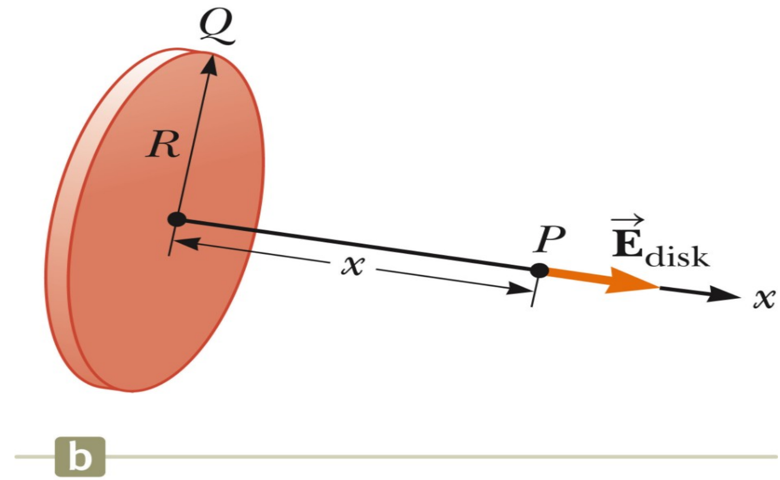
$$\frac{kx\ q_{ring}}{\left(x^2 + a^2\right)^{3/2}}$$

Keep nC. The unknown is Q the point charge

7. A charge disk

$$\frac{kxQ}{\left(x^2+a^2\right)^{3/2}}$$





You build up from previous Problem. Divide the area in small ring of areas (2pi r) dr with r between 0 and a .

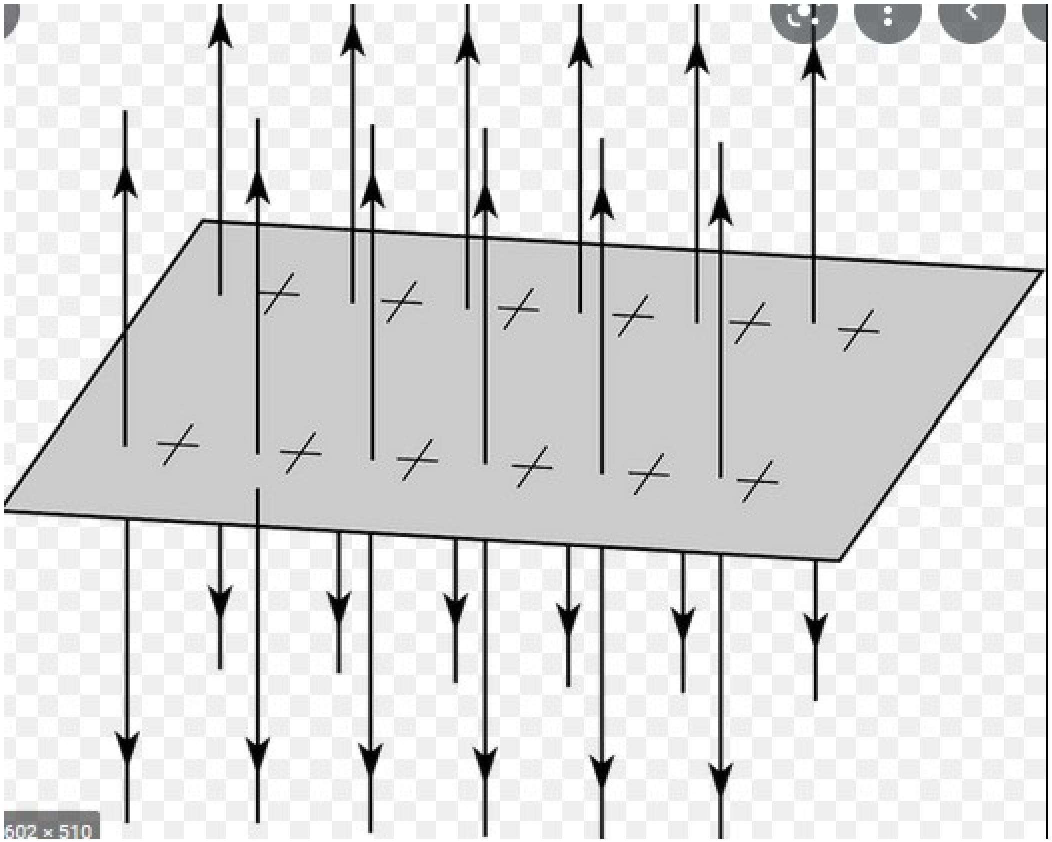
$$E_x = 2\pi k\sigma\left(1-\frac{x}{\left(x^2+R^2\right)^{1/2}}\right)$$

Also show that if R>>x or x=0
we get E = 2pi K sigma with K = 1/(2pie0)

$$E = \frac{\sigma}{2\varepsilon_0}$$

http://www.physics.udel.edu/~watson/phys208/exercises/kevan/efield1.html

https://www.youtube.com/watch?v=QnlKA0-EFGk (coding python)



We can apply the result to a
Very large sheet/

$$E = \frac{\sigma}{2\varepsilon_0}$$

$$\sigma = \frac{\text{Charge}}{\text{Unit area}}$$

$$Q = \sigma A$$

$$\varepsilon_0 = 8.85\times10^{-12}\,\frac{C^2}{Nm^2}$$

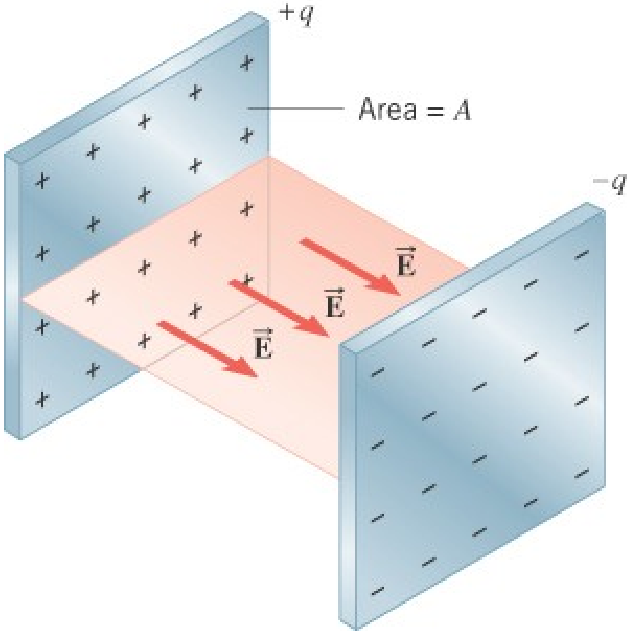
If the sheet is really large relative to the distance where we measure E,
E is uniform , does not depend on the distance (Walter Lewin demo)

DEMO – for large charged sheet → constant
Electric field close to the sheet. (d << size of sheet)

E = Q/(2Ae0)

$e0 = 8.85 \times 10^{-12}$

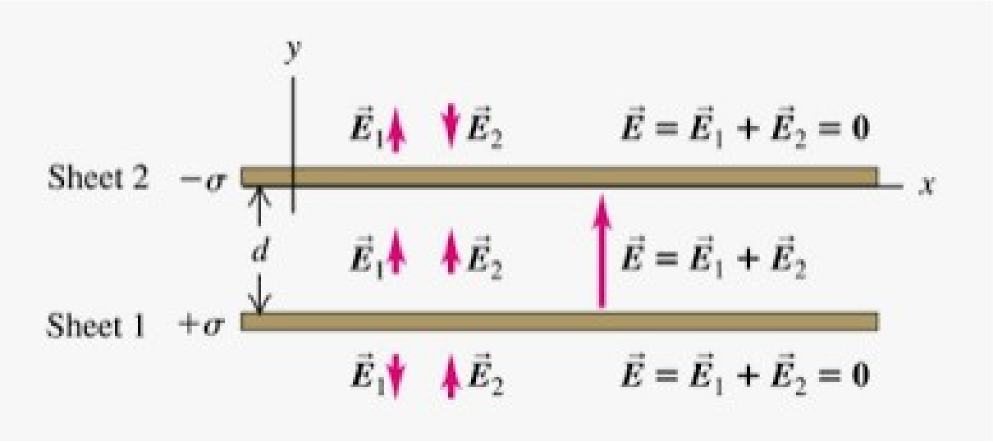
THE PARALLEL PLATE CAPACITOR



Parallel plate capacitor

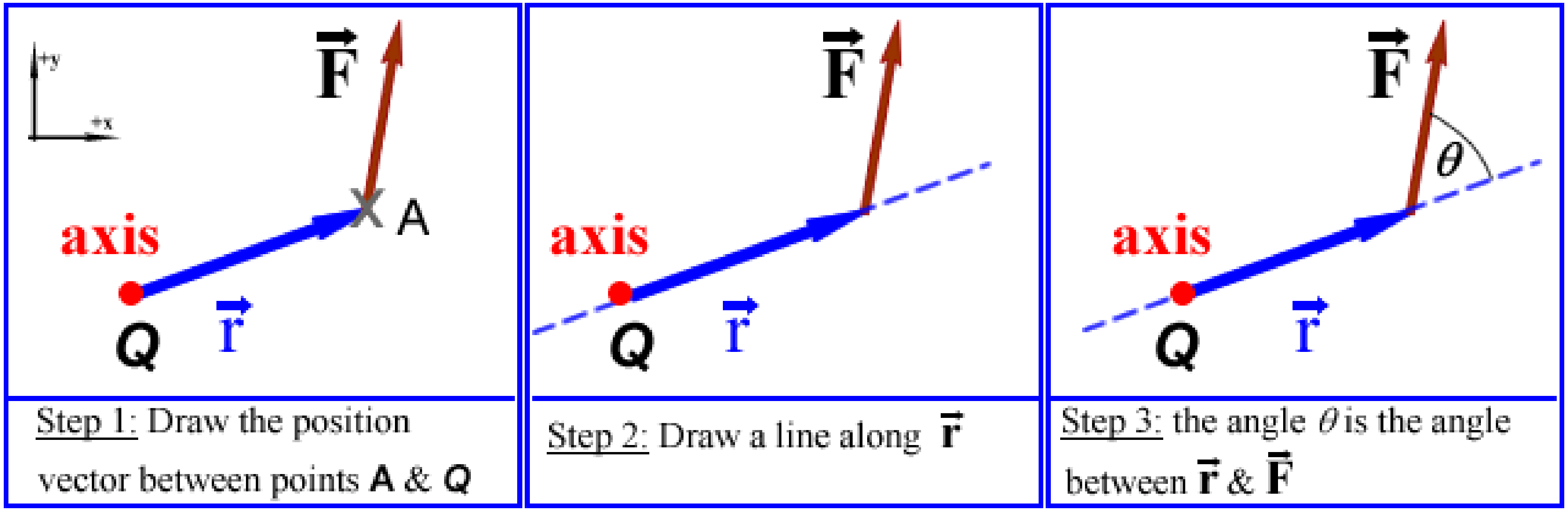
charge density

$$E = \frac{q}{\varepsilon_o A} = \frac{\sigma}{\varepsilon_o}$$

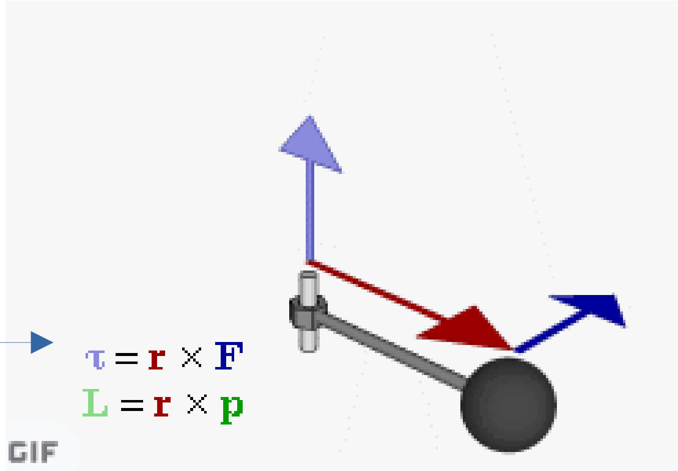


$$\varepsilon_o = 8.85 \times 10^{-12}\ \mathrm{C^2/(N \cdot m^2)}$$

Review Phys1: torque



$$|\vec{\tau}| = rF\sin\theta$$



A Dipole in an Electric Field

- Start with

$$\tau = Fx\sin\theta + F(d-x)\sin\theta = Fd\sin\theta$$

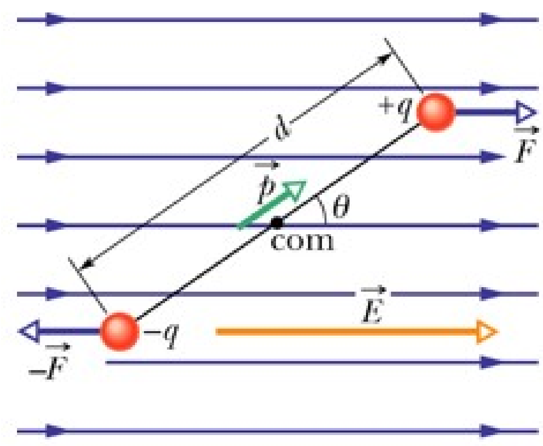
- Then

$$F = qE \quad \text{and} \quad p = qd$$

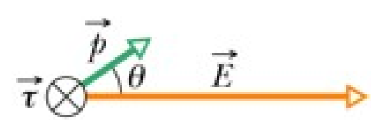
- So

$$|\tau| = pE\sin\theta$$

$$\boxed{\vec{\tau} = \vec{p}\times\vec{E}}$$



(a)



(b)

Discuss how t

p is a vector a

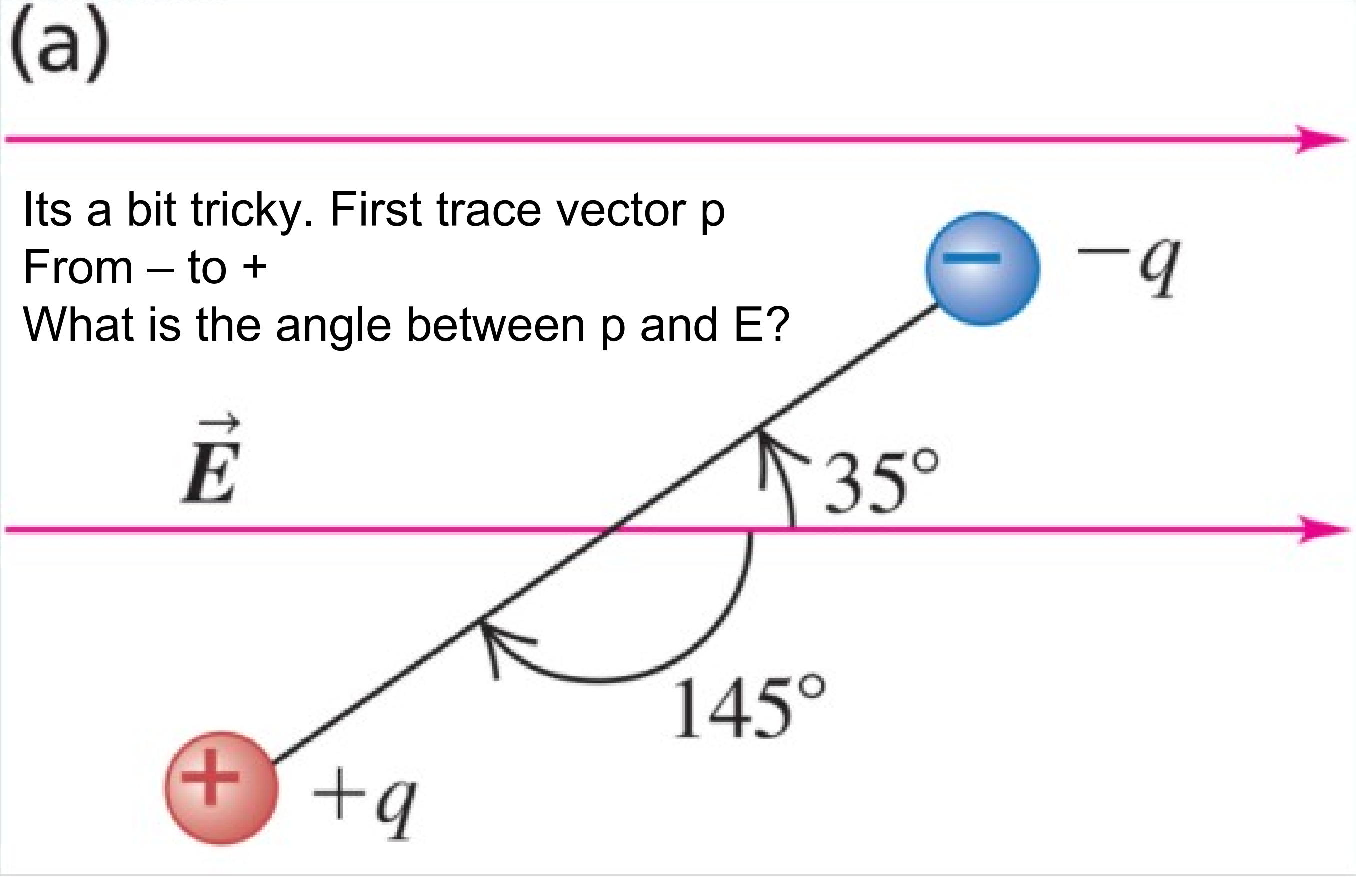
The unit is C.m The moment of a water molecule is 6.13 10-30 Cm

http://hyperphysics.phy-astr.gsu.edu/hbase/electric/dipolecon.html#c1

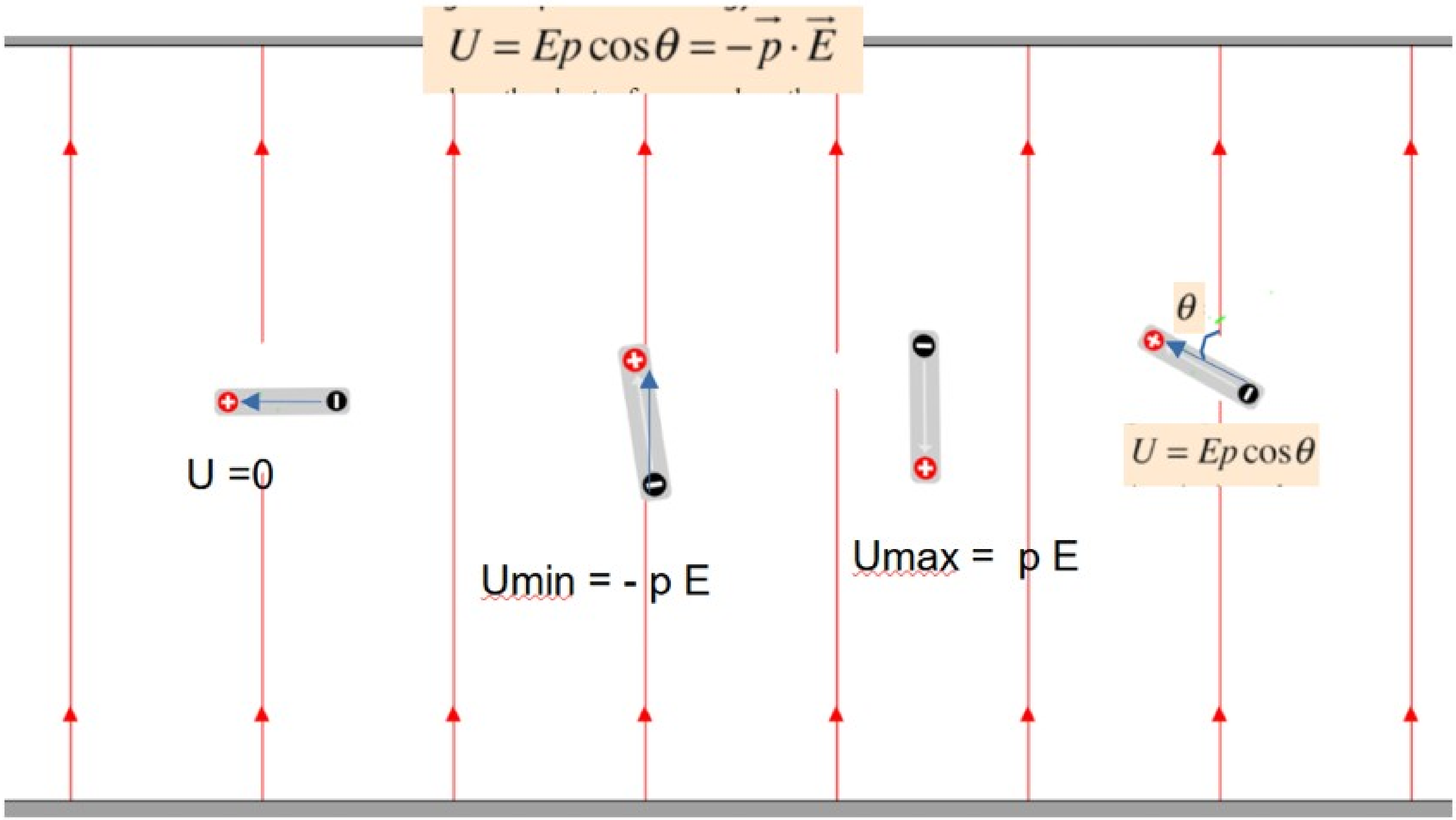
Figure 21.32a (next page) shows an electric dipole in a uniform electric field of magnitude 5.0×10^5 N/C directed parallel to the plane of the figure. The charges are $\pm 1.6 \times 10^{-19}$ C; both lie in the plane and are sep 0.125 nm $= 0.125 \times 10^{-9}$ m. Find (a) the net force exerted by the field on the dipole; (b) the magnitude and the electric dipole moment; (c) the magnitude and direction of the torque; (d) the potential energy of the system position shown.

The potential energy is – p.E (dot product between E and p)

FIGURE 21.32



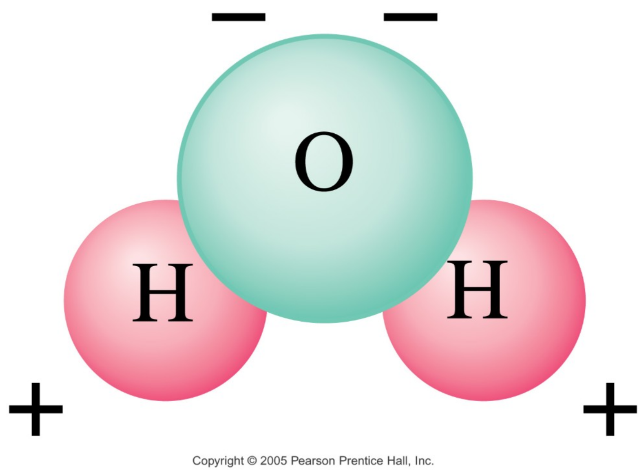
Its a bit tricky. First trace vector p
From – to +
What is the angle between p and E?



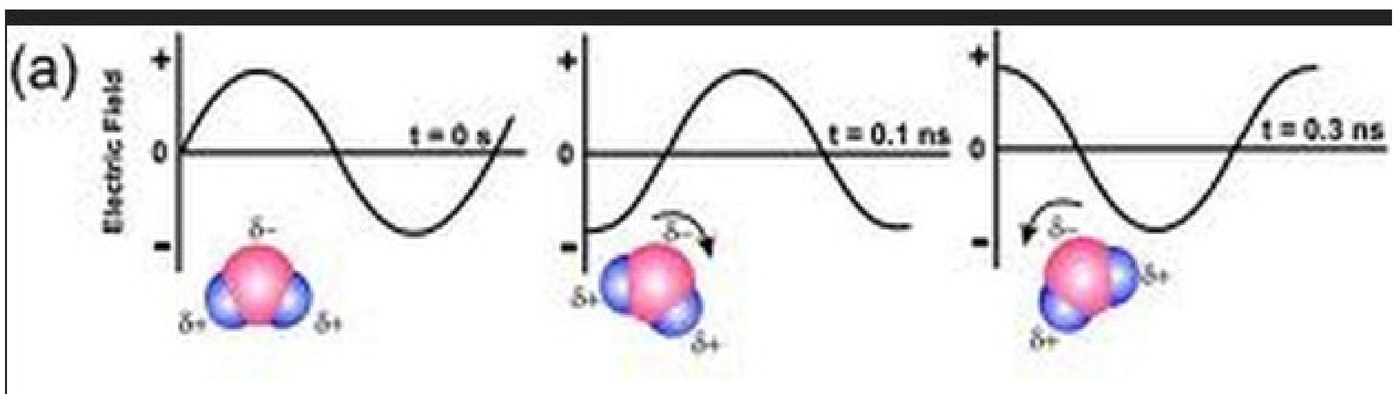
The blue vector is the vector p (from – to +). The red vectors are E (uniform)

https://sciencesims.com/sims/dipole-in-efield/

A water molecules in an electric field will shake if the electric field is alternating. Like the electric field in a microwave field.



https://phet.colorado.edu/en/simulations/microwaves



PE of an electric dipole
A Dipole in an Electric Field

- Start with

$$dW = \tau d\theta$$

- Since

$$U_f - U_i = \int_{\theta_i}^{\theta_f} \tau d\theta = \int_{\theta_i}^{\theta_f} pE\sin\theta d\theta = pE\int_{\theta_i}^{\theta_f} \sin\theta d\theta$$

$$= pE[-\cos\theta]_{\theta_i}^{\theta_f} = pE(\cos\theta_i - \cos\theta_f)$$

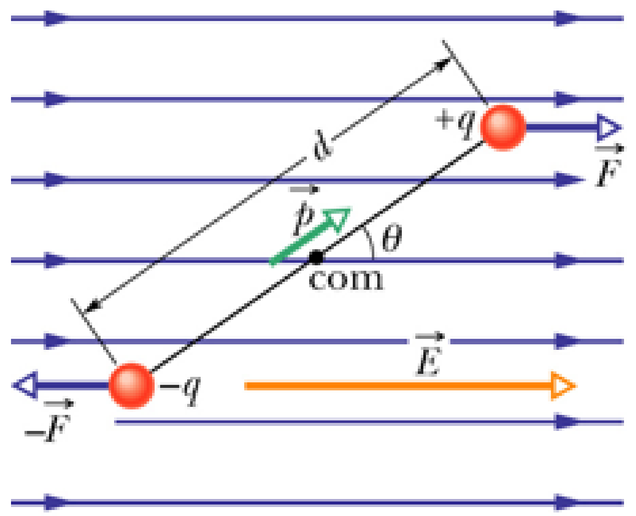
- Choose

$$\boxed{U_i = 0 \text{ at } \theta_i = 90^\circ}$$

- So

$$\boxed{U = -pE\cos\theta}$$

$$\boxed{U = -\vec{p}\cdot\vec{E}}$$

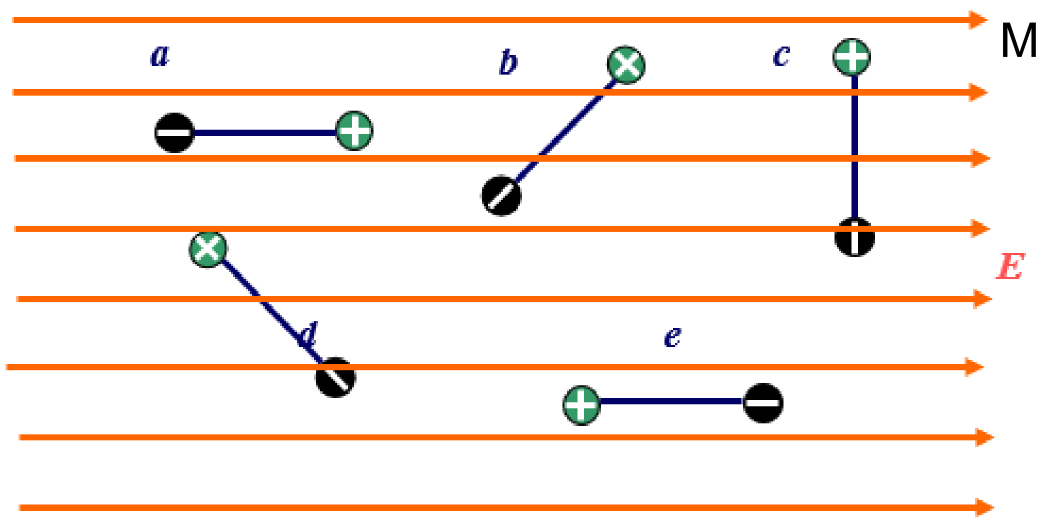


Discussion on direction. When is U minimum (U=-1), how E is relative to p

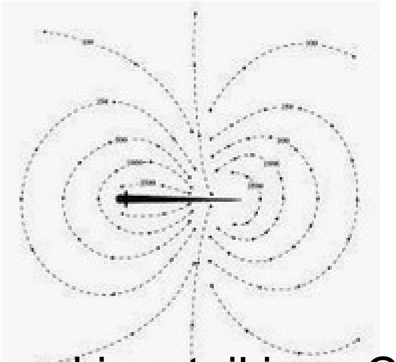
http://hyperphysics.phy-astr.gsu.edu/hbase/electric/dipolecon.html#c1

4. In which configuration, the potential energy of the dipole is the greatest?

Draw the vector p.
P has to be aligned with E for Minimum energy.



Do example 21.13



african knifefish- poor vision. Head is -, tail is +. Can sense a prey when E is disturbed.

Electrolocation

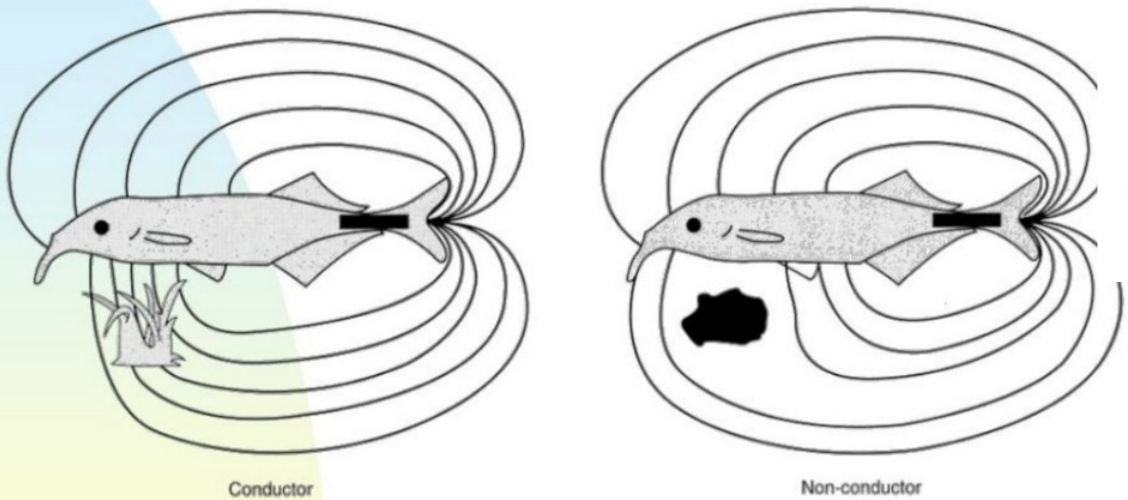
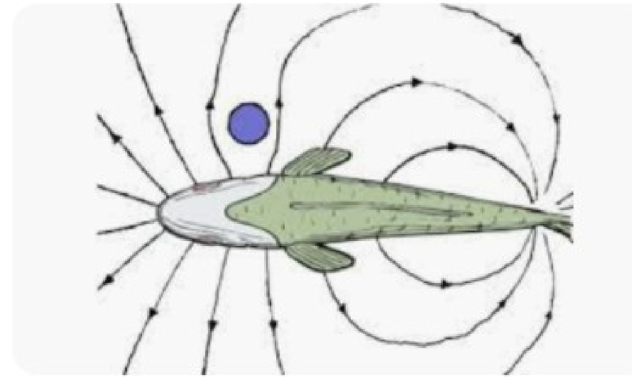
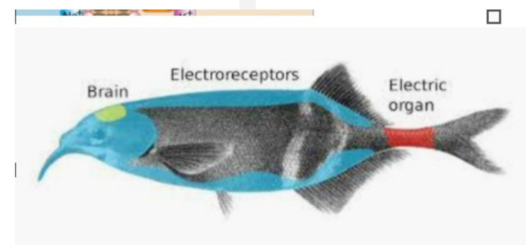


FIGURE 10.32 Electroreception. The electrical field generated by this fish is distorted by nearby objects. A good conductor, such as another living organism, draws the lines of force together. A nonconductor, such as a rock, spreads them out. Using electroreceptors distributed over its body surface, the fish senses the changes in the electrical field to "picture" its environment. (From von der Emde 1999.)

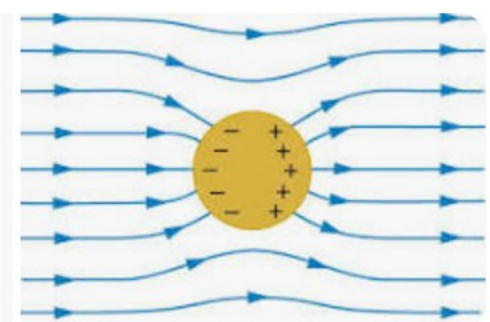


ResearchGate
electric field using the fish body ...

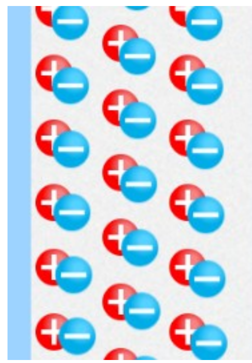


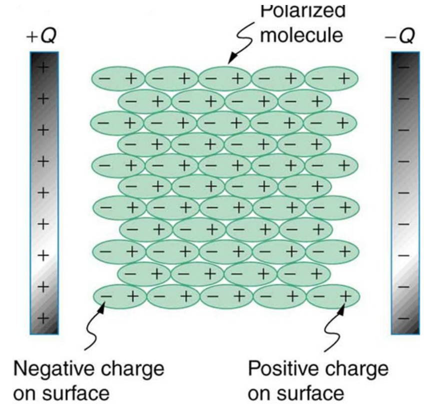
See the app with the balloon
Insulator can be polarized
So the electric field weakens.
See picture in folder

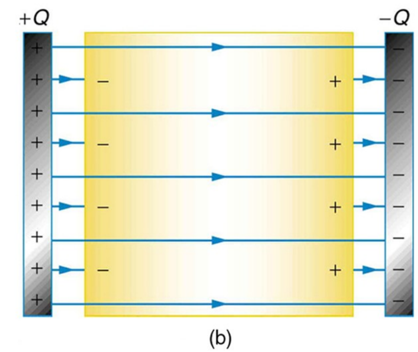
https://www.slideserve.com/devlin/electric-fish



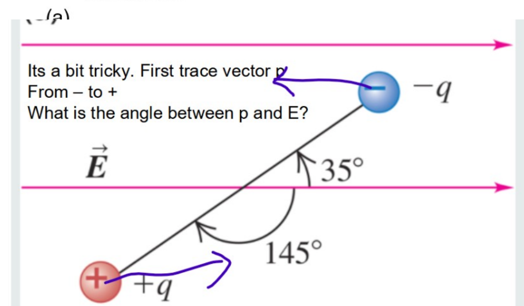
insulators inside an electric field become polarized. Like
1) Seeds in castor oil or dielectrics between the plates of a capacitor. The seeds are insulators and neutral. They become polarized and align with the electric field → torque. 2) A insulator inside an electric field weakens the field. Like dielectrics inside a capacitor or see the example of a dish.





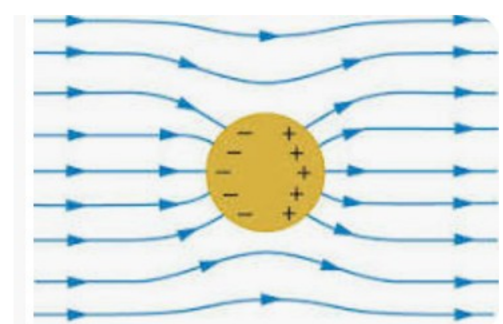


https://phet.colorado.edu/sims/html/balloons-and-static-electricity/latest/balloons-and-static-electricity_en.html

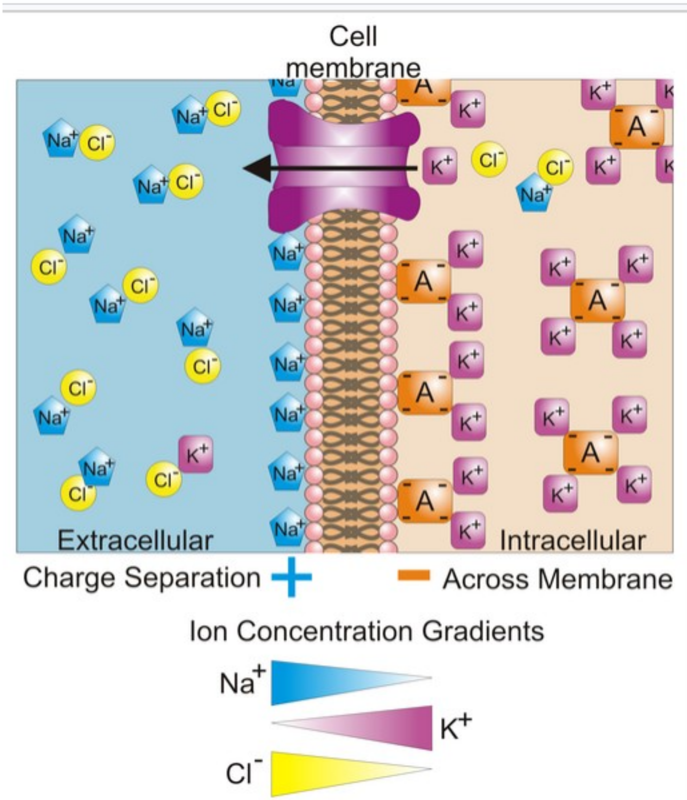


3) Water or chloric acid ($HClO_3$) are permanent dipoles They experience a torque inside an electric field.

4) a conductor inside a field will strengthens the field. Electrons move on the outside of the conductor and produce their own

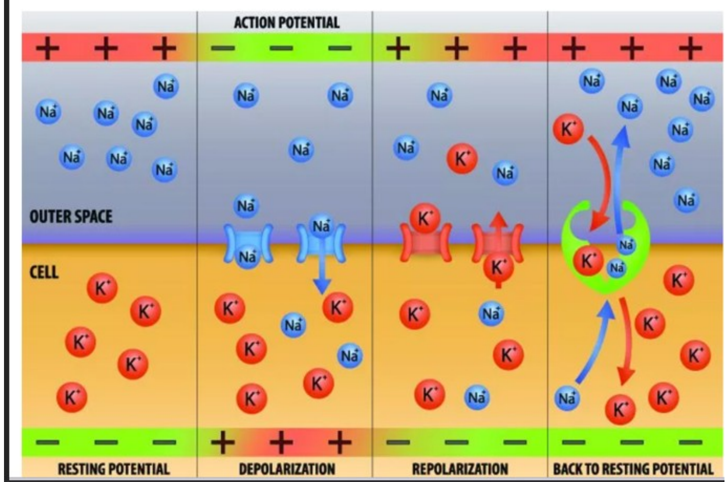


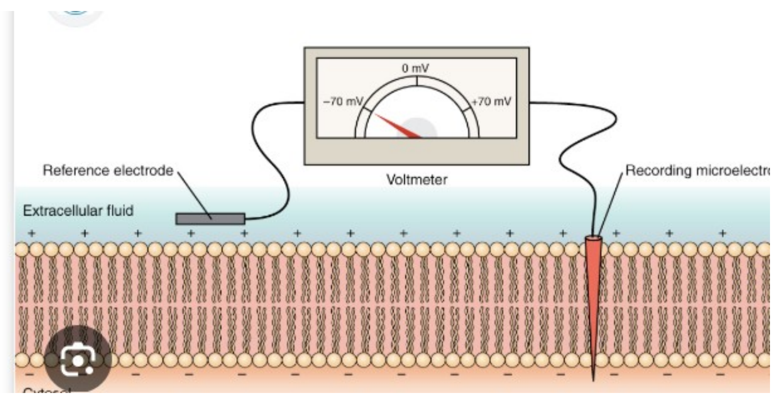
Electric field – Very important in Bio



Differences in the concentrations of ions on opposite sides of a cellular membrane lead to a voltage called the **membrane potential**. Typical values of membrane potential are in the range –70 mV to –40 mV. Many ions have a concentration gradient across the membrane, including potassium (K^+),

cellular membrane, with every action potential (impulse) similar in size. The response of a nerve or muscle cell to an action potential can vary according to how frequently and for what duration the action potentials are fired. An action potential requires an influx of positive ions to produce a specific change in voltage (threshold value). It occurs after a certain degree of internal cell membrane depolarization – a positive increase in electrical charge.





Sodium-potassium pump → to maintain
A negative potential across membranes of cells

https://www.youtube.com/watch?v=J0SMjoMJ03s

ATP as a versatile energy currency that drives various biological reactions and processes (maintaining voltage across cell membrane, synthesis of Proteins, muscle contraction, circulation of blood…, DNA stuff …)

To produce ATP. Electrons release energy through electron transport chain (ETC) in mitochondria. Electrons are transferred from one molecule to another (oxidation and reduction) , and they lose energy at each step (so higher energy orbital to lower). This energy is used to push protons (H+) against the electric field across the membrane of the mitochondria (against the potential gradient).
This is called a proton pump. Those high energy protons move back across the membrane but through a different path. This energy is used To spin a "rotor " in an enzyme (ATP synthase) . So mechanical energy. This rotor converts ADP to ATP. ATP is used to maintain a negative voltage across the membranes of neurones (-70mV). Sodium – potassium pump.

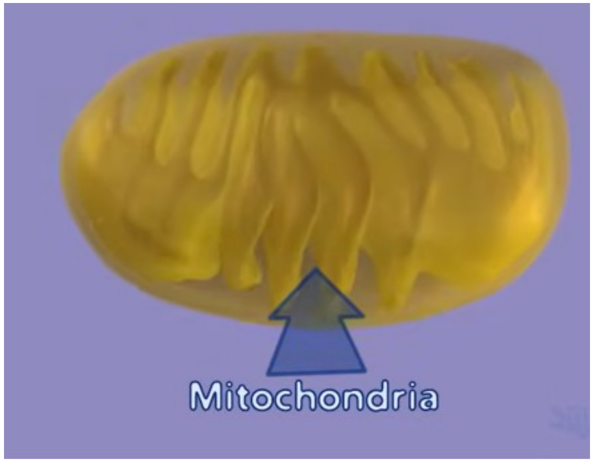
Hydrogen pump and making of ATP videos→

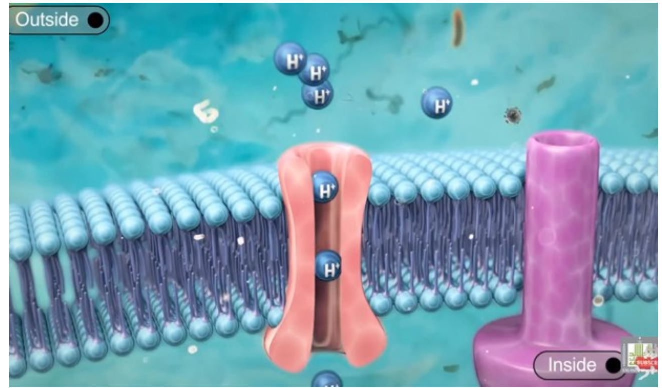
https://www.youtube.com/watch?v=_qkqHZMFwNs

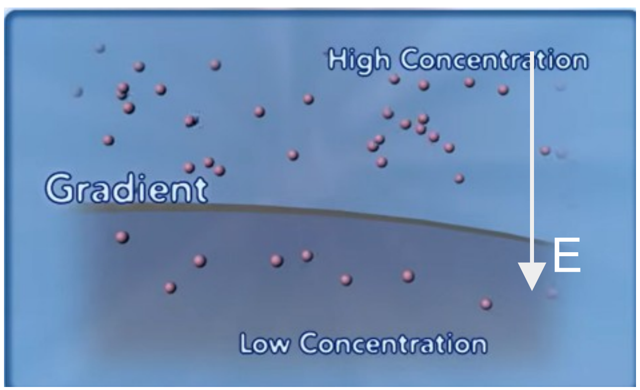
https://www.youtube.com/watch?v=3y1dO4nNaKY

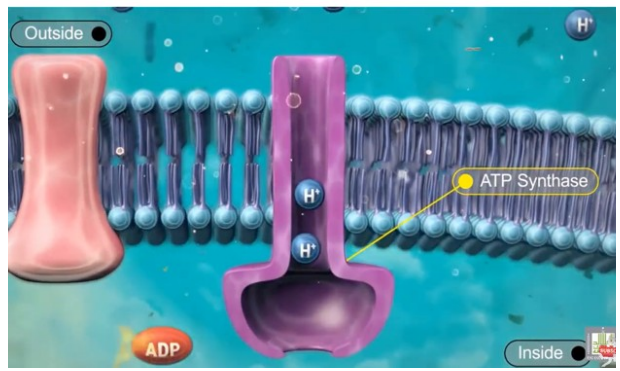
$NADH + H^+ + 1/2\ O_2 \rightarrow NAD^+ + H_2O$

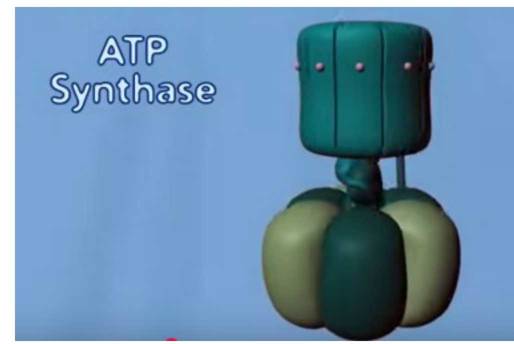
Electron transfer chain → electrons release energy.

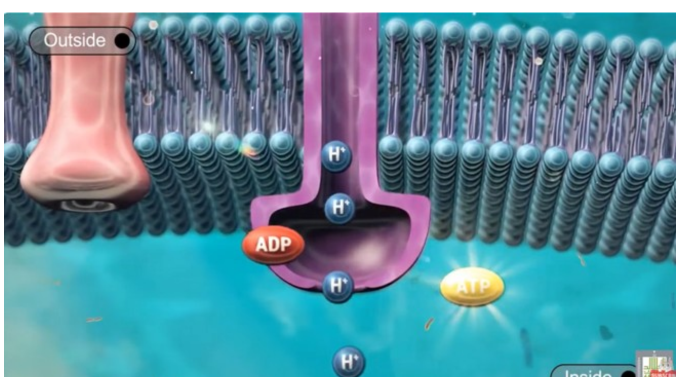


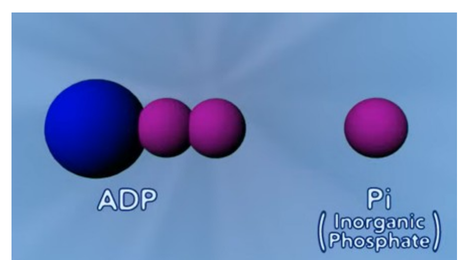






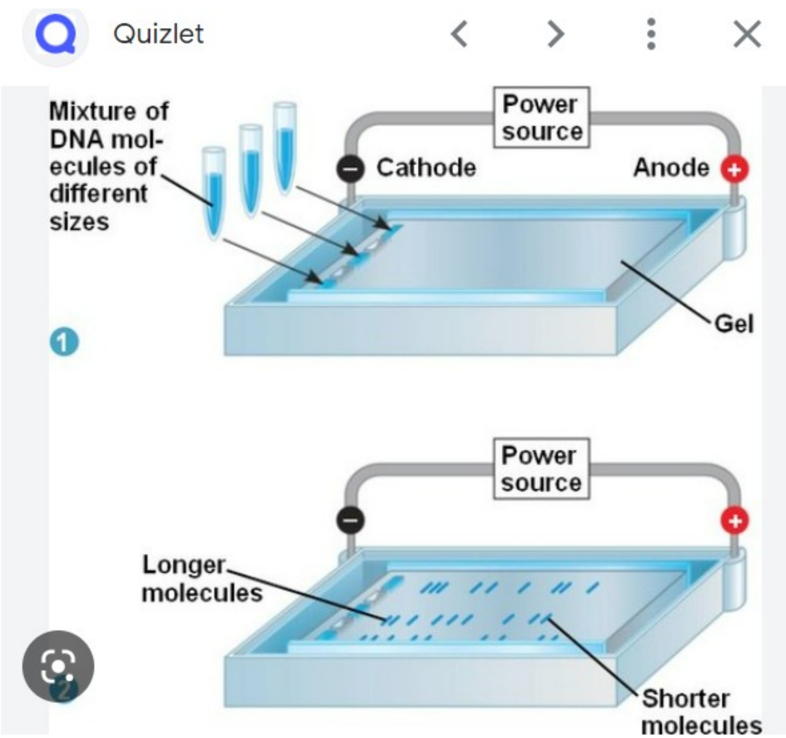




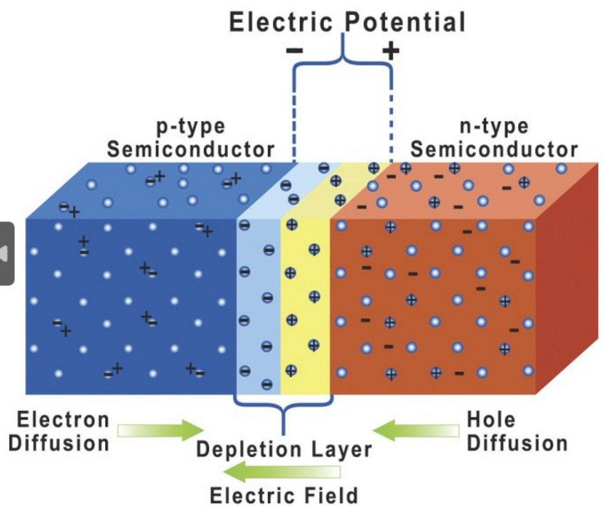


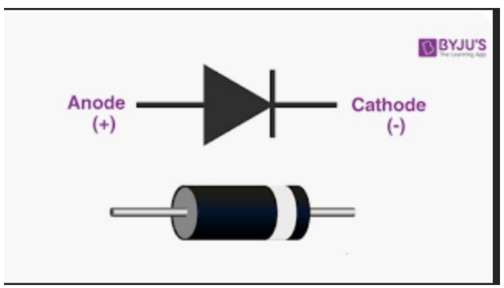
Application: DNA fingerprint

molecular biology



Gel Electrophoresis is a laboratory technique used to separate molecules based on their size and electrical charge. Imagine it like a race track for tiny particles! → the fragments are acted upon by 2 forces.
The electric force and the viscous force. The larger fragments will reach Their terminal speed later and lag behind. You can classify the fragments According to their mass
https://quizlet.com/204488499/biomed-eoc-review-gel-electrophoresis-flash-cards/





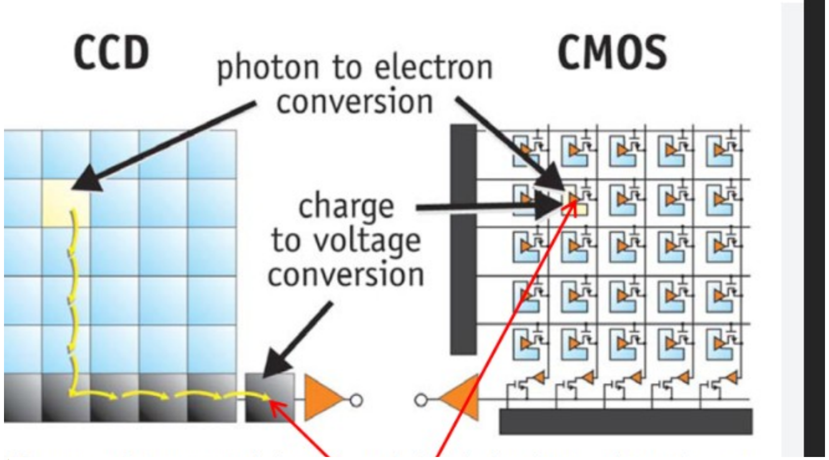
Integrated Circuits (ICs):

- Developed in the late 1950s, ICs combined multiple transistors, resistors, capacitors, and other electronic components onto a single silicon chip.
- This miniaturization allowed for exponentially denser and more powerful electronic devices.
- Without transistors, the miniaturization and complexity necessary for ICs wouldn't have been possible.

Silicon Revolution:

- This refers to the shift from germanium to silicon as the primary material for transistors and ICs in the late 1950s and early 1960s.
- Silicon offered several advantages over germanium, including lower cost, higher purity, and better performance at high temperatures.

integrated circuit (IC) revolution and the silicon revolution are based on the invention of transistors



1. **Photodiode and Charge Accumulation:**

- Each pixel contains a **photodiode**. When light hits the pixel, it excites electrons in the silicon, freeing them from their atoms.
- These free electrons move within the photodiode and get collected at a designated "potential well," essentially creating a pool of negative charge.
- The amount of accumulated charge is proportional to the intensity of light received by the pixel, meaning brighter areas hold more charge.

3. **Charge-to-Voltage Conversion:**

- This is where the magic happens! A special transistor called a **source follower** comes into play.
- The accumulated charge in the pixel well acts as the source for the transistor.
- By turning on the transistor, the charge flows through it, generating a **current** proportional to the amount of charge.
- This current flows through a **transistor amplifier**, boosting the signal strength.
- Finally, the amplified current is converted into a voltage using a **capacitor**.